United in Booms, Divided in Busts: Regional House Price Cycles and Monetary Policy

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Abstract

This paper shows that regional disparities in house price growth increase during house price busts. We link this to intensified house price belief extrapolation in busts and regional housing market heterogeneities. To explain this observation, we build a multi-region currency union model with a housing sector and extrapolative house price beliefs, introducing a novel method to handle such models efficiently. Analytically, we show that heterogeneities generate differences in house price growth, while extrapolation amplifies them. Quantitatively, the model matches the observed empirical patterns. Monetary policy targeting house prices stabilizes output and house prices, improving welfare relative to an inflation-targeting benchmark.

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I. Introduction

A substantial body of literature connects boom-bust cycles in house prices to over-optimism and over-pessimism regarding the future evolution of house prices.[1] The bust phases of these cycles are often associated with significant economic costs, making them a primary concern for policymakers, particularly central bankers.[2] However, housing markets are inherently regional, and consequently, housing cycles and their associated economic costs also exhibit pronounced regional variation. These regional divergences, especially during periods of economic downturn, therefore constitute an additional challenge for policymakers.[3] While a substantial literature examines housing cycles and their implications for monetary policy at the federal level, there is comparatively little focus on regional heterogeneities in house price dynamics.

This paper makes several contributions to the literature. Empirically, it documents a novel fact: regional disparities in house price growth are more pronounced during housing busts than during booms. We relate this pattern to extrapolative house price expectations updating and show that the degree of updating intensifies during busts.

To rationalize this finding, we develop a multi-region currency union model that incorporates regional heterogeneity in housing markets and extrapolative belief updating regarding house prices. We introduce a novel solution method that enables to efficiently solve structural models with extrapolative expectations. In our model, intensified extrapolation in busts amplifies house price responses, and thereby generates larger dispersion in house price growth originating from housing supply-side heterogeneities. We calibrate our model to U.S. metropolitan areas and demonstrate that it quantitatively matches the response of aggregate house prices and their regional dispersion in growth rates to monetary policy shocks in both booms and busts.

Finally, we illustrate that by leaning against aggregate house prices, the monetary authority can mitigate cross-regional disparities in house prices and economic activity, resulting in welfare improvements relative to standard inflation targeting.

We begin by presenting our central empirical finding that cross-regional dispersion in house price growth is more pronounced during busts, both unconditionally and conditionally on macroeconomic shocks. The cross-regional standard deviation of house price growth among major U.S.

[1] See, e.g., Case et al. (2012); Armona et al. (2019); Kuchler and Zafar (2019); Ma (2020); Kaplan et al. (2020)

[2] During the global financial crisis of 2008, 20% of household wealth (Dettling et al., 2018) and 8 million jobs were lost (Mian and Sufi, 2015). During this time, the Fed responded by drastically cutting interest rates.

[3] See, for instance, the Euro area crisis of 2009, which ultimately threatened the existence of the currency union.

cities is 50%-60% larger during busts than booms,[4] a pattern also present at U.S. state level and for the Euro Area. Conditional on monetary policy shocks, we observe a significant increase in cross-city standard deviations during busts, while the response is muted during booms. This result is robust to other types of shocks.

To rationalize the mechanisms underlying our empirical findings, we develop a multi-region currency union New Keynesian model. Building on Benigno (2004), we incorporate (i) regional heterogeneity in housing supply and (ii) extrapolative belief updating about house prices. Households form subjective expectations only about house prices, but their optimal choices—such as consumption—depend on expected future consumption, which in turn is shaped by future expected house prices. It therefore becomes necessary to characterize these expected endogenous choice variables as functions of prices that are perceived to be exogenous by households.

To address this, we propose a novel first-order solution method for general equilibrium models with time-consistent but non-rational expectations. Specifically, we show that a standard log-linearized Euler equation can be expressed in terms of current consumption, the expected future path of real interest rates, and the subjectively expected consumption at the limit. We refer to the latter as the *subjective expectations wealth effect*.[5]

We demonstrate that the *subjective expectations wealth effect* admits an explicit representation that depends solely on prices. Moreover, we show that knowledge of this term is sufficient to solve the model. Finally, the first-order approximation ensures compatibility with standard solution methods widely used in the DSGE literature, enabling the solution of large-scale models. In our quantitative analysis, we apply this approach to a currency union model comprising 20 regions.[6]

Equipped with our model, we demonstrate analytically that regional differences in housing markets, combined with house price belief extrapolation, can account for the asymmetry in house price growth dispersion observed across the house price cycle.[7] First, regional heterogeneity in housing markets creates regional differences in the responsiveness of house prices. Second, subjective expectations about house prices lead to amplification through extrapolation. Intuitively, when agents observe an increase in house prices, they anticipate further price increases in the

[4]Throughout the paper, a bust period is defined as one in which house prices declined on an aggregate or federal level in the previous period.

[5]Intuitively, when households extrapolate increases in house prices into the future, they perceive higher expected lifetime wealth, which raises expected consumption in the limit and, consequently, current consumption.

[6]Our solution method contrasts with that introduced by Winkler (2020), which, although based on first-order approximation, does not confine subjective expectations to specified asset prices and results in belief spillovers to all other prices. Conversely, Adam et al. (2022) solve a similar model using non-linear solutions.

[7]We focus on monetary policy shocks, but the general proposition extends to all shocks that affect housing demand.

future, leading to heightened demand for housing and, subsequently, driving prices even higher.

The interaction of these two forces generates the following dynamics: Regional heterogeneities in housing markets create differences in regional house price growth rates, which are amplified by extrapolation. This result allows us to align our theoretical model with the empirically observed patterns. Specifically, our model is capable of generating pronounced regional dispersion in house price growth during busts, provided extrapolation, the amplifying force, is stronger in those periods.

In a rational expectations version of the model, the ability of cross-regional differences in housing markets to produce differences in house prices is much weaker. Additionally, this version lacks extrapolative belief updating and, thus, cannot account for amplified cross-regional heterogeneity in busts in the absence of any additional frictions.

To test our theoretical predictions we move to the data. We estimate the extent to which U.S. households' house price expectations depend on prior expectations and past house price growth and find that the pass-through from past house price growth to expectations roughly doubles during busts compared to booms; that is, households update their beliefs more strongly during busts. Examining potential sources of these dynamics, we show that realized house price movements exhibit the same asymmetric characteristics as their expected counterparts, suggesting that the asymmetry in belief updating reflects the underlying asymmetry in actual house price dynamics.

We calibrate regional housing supply heterogeneities to the 20 largest U.S. metropolitan areas used in our empirical analysis and target the aggregate house price response to a monetary policy shock, independent of boom-bust asymmetries. When accounting for asymmetries in the belief updating process, the model accurately captures the magnitudes of aggregate house price changes in both booms and busts, even though these asymmetries were not targeted. The model also successfully replicates the regional house price growth dispersion observed during booms and busts in response to a monetary policy shock. In contrast, the rational expectations version of the model fails to match these dynamics, underestimating aggregate house price responses by a factor of three and generating no significant difference in regional house prices.

Finally, we investigate the extent to which leaning against house prices influences economic outcomes. Specifically, we assess the effects of including aggregate house prices in the Taylor rule compared to a baseline rule focused solely on inflation stabilization in response to a productivity shock. We find that house price targeting reduces volatility in both output and house prices, albeit at the cost of increased inflation volatility. Notably, it also reduces regional dispersion across all variables, synchronizing economic activity across regions. The underlying intuition is that by targeting house prices, the central bank limits the divergence in house price beliefs, mitigating the channel through which regional heterogeneities spill over to the real economy. We show

that house price targeting is beneficial under subjective expectations, expressing the utilitarian welfare as in Galí (2015).[8] Welfare improves by 9% in booms and by 13% in busts relative to the inflation-targeting baseline, with the larger bust gains reflecting stronger belief extrapolation in those periods.

Related literature. Our paper relates to a broad empirical literature emphasizing that house price beliefs deviate from rational expectations, highlighting momentum and revisions in belief formation (Case et al., 2012; Armona et al., 2019; Kuchler and Zafar, 2019; Ma, 2020). On the theoretical side, it connects to the behavioral macro-finance literature on deviations from rational expectations in asset price formation, including models of diagnostic expectations (Bordalo et al., 2018, 2021; L'Huillier et al., 2023; Bianchi et al., 2024a), learning and extrapolation (Adam et al., 2017; Winkler, 2020; Farhi and Werning, 2020; Fontanier, 2022), belief disagreement (Caballero and Simsek, 2019, 2020), and sentiments (Krishnamurthy and Li, 2020; Maxted, 2024). Regarding housing specifically, Glaeser and Nathanson (2017) and Schmitt and Westerhoff (2019) model extrapolative expectations in partial equilibrium, while Adam et al. (2012), Caines and Winkler (2021), and Adam et al. (2022) study these mechanisms in general equilibrium. Other studies explain house price behavior through optimism and pessimism, concave demand curves, or exogenous belief shifts (Burnside et al., 2016; Guren, 2018; Kaplan et al., 2020). We contribute by emphasizing the asymmetric evolution of house price beliefs across booms and busts and by proposing a novel solution method for models with capital gains extrapolation. We also contribute to the literature on regional housing supply heterogeneity. Mian et al. (2013), Mian and Sufi (2014), and Guren et al. (2021) explore housing wealth effects through supply elasticities, while Aastveit and Anundsen (2022) and Aastveit et al. (2023) show that inelastic regions display stronger house price responses to monetary policy. We connect to Glaeser et al. (2008), who show that supply-inelastic regions face higher bubble probabilities, extending this work by linking supply-side heterogeneity to asymmetric belief formation in general equilibrium. Our analysis builds on the literature on cross-regional heterogeneities within currency unions, including optimal policy (Benigno, 2004; Galí and Monacelli, 2008; Kekre, 2022) and the role of heterogeneities in wage and price setting, hand-to-mouth shares, and mortgage markets (Calza et al., 2013; Slacalek et al., 2020; Bletzinger and von Thadden, 2021; Pica, 2021; Almgren et al., 2022; Corsetti et al., 2022). For the United States, Beraja et al. (2019), Chen (2019), and Gitti (2024) document regional variation in exposure to fiscal policy, housing, and migration. We contribute by linking supply-side housing constraints to asymmetric house price belief formation. Finally, we relate to the literature

[8] Specifically, the welfare loss is expressed in terms of the equivalent permanent decline in consumption, measured as a fraction of steady-state consumption.

on leaning against asset prices. Stein (2012) argues for such policies to mitigate over-borrowing externalities, while Svensson (2017) and Weidmann (2018) emphasize their costs and the superior role of macroprudential tools, and Gourio et al. (2018) highlight the trade-off between crisis risk and cyclical volatility. In a rational expectations framework, Bianchi et al. (2024b) find that monetary policy should not target house prices—consistent with our rational expectations findings. Under deviations from rational expectations, however, leaning against the wind becomes beneficial (Adam and Woodford, 2021; Caines and Winkler, 2021; Fontanier, 2022; Caballero and Simsek, 2019). We contribute by showing that house price targeting additionally mitigates regional dispersion, thereby helping harmonize an economically integrated area—a point we believe to be novel in the literature.

Outline. The remainder of the paper is structured as follows. Section (II) documents the empirical fact that regional house price growth is more dispersed in busts than in booms. Section (III) presents the model and solution method, and Section (IV) provides the analytical intuition. Section (V) discusses evidence of stronger extrapolation in busts and the model calibration. Section (VI) reports the quantitative findings, Section (VII) presents the policy analysis, and Section (VIII) concludes.

II. Regional house prices in booms and busts

In the following we document a novel empirical fact: regional disparities in house price growth are significantly larger during housing busts than during booms. This asymmetry in regional dynamics has, to our knowledge, not been previously emphasized in the literature.[9] We establish this finding using both unconditional moments—capturing overall cross-regional dispersion—and conditional moments, when controlling for specific shocks. For our analysis, we classify periods with positive aggregate house price growth ($\Delta q_{t-1}^{agg} > 0$) as booms and those with negative growth ($\Delta q_{t-1}^{agg} < 0$) as busts. This definition follows Glaeser et al. (2008); our findings are robust with respect to other definitions.

Cross-regional house price variation, unconditional. To investigate this pattern, we focus on regional house price growth data, specifically at the major city or state level, for the United States. For each point in time, we compute the cross-regional standard deviation of house price growth rates. We then split the sample into boom and bust episodes, where these episodes are defined based

[9] While regional heterogeneities and their effects on house prices are well studied (e.g., Saiz, 2010; Guren et al., 2021; Aastveit and Anundsen, 2022), asymmetries between booms and busts remain unexplored.

on the aggregate house price growth. Our preferred specification is based on U.S. major cities in the cross-section.[10] The data is given in monthly frequencies, allowing for sufficient observations of booms and busts. For the U.S. state level, we use quarterly data. Our sample includes all states except for Alaska and Hawaii but includes the District of Columbia. Both U.S. data sets span from 1990 to 2024. Table (1) reports the mean and median cross-regional standard deviations for house price booms and busts, as well as their share in busts relative to booms.

Table 1: U.S. cross-regional house price growth standard deviation (σ_c) in booms and busts

	cities		*states*	
	mean	*median*	*mean*	*median*
$\frac{Bust}{Boom}$	**1.60**	**1.64**	**1.23**	**1.26**
$\sigma_c : Boom$	0.47	0.41	0.87	0.80
$\sigma_c : Bust$	0.74	0.67	1.08	1.01
$p - Val.\ Bust > Boom$	0.000	0.000	0.007	0.007
Number regions	20		51	
Obs. boom	331		109	
Obs. bust	82		28	
Sample	$1990M1 - 2024M5$		$1990Q1 - 2024Q1$	

Notes: The Table reports the mean and median values, across time, of the estimates of cross-sectional standard deviations of house price growth rates within a given economic entity during house price booms and busts. $\frac{Bust}{Boom}$ denotes the regional standard deviation in busts relative to booms. The test for $Bust > Boom$ is based on a one-sided t-test.

Our findings indicate that the cross-regional standard deviations observed during busts are larger for both the mean and the median estimates than those observed during booms. Specifically, if one considers monthly data for U.S. cities, the cross-regional standard deviations during busts are 60% larger than during booms, judging by the relative shares. The ratio decreases to 23% – 30% for the U.S. states. We find that a t-test rejects the H_0 hypothesis of regional standard deviations being larger in booms for both regional specifications. Table (C.1) in the Appendix shows that the same holds for the euro area at the country level.

Cross-regional house price variation, conditional. While the previous analysis focused on unconditional variations in house price growth during booms and busts, we now move to a conditional approach. Using local projections, we estimate the response of cross-regional standard deviations of house price growth to a monetary policy shock. We utilize city-level data due to its sufficiently

[10] We use the Case-Shiller U.S. National Home Price Index for the city level. The 20 major U.S. cities included are Atlanta, Boston, Charlotte, Chicago, Cleveland, Dallas, Denver, Detroit, Las Vegas, Los Angeles, Miami, Minneapolis, New York, Phoenix, Portland, San Diego, San Francisco, Seattle, Tampa, and Washington DC.

large time series dimension. Equation (1) depicts the estimation equation. The left-hand-side variable (y_{t+h}) denotes the three-month moving average of the cross-regional standard deviation of house price growth at horizon h after the impact of the shock. α^h denotes a constant. The monetary policy shock, ϵ_t^{MP}, is the high frequency identified from Degasperi and Ricco (2021). The monetary policy shock further conditions on times of booms ($\Delta q_{t-1}^{agg} > 0$) and busts ($\Delta q_{t-1}^{agg} < 0$). We adopt this approach because monetary policy shocks are relatively small and do not influence the U.S. economy's transition between boom and bust phases.[11] We use the absolute values of the shocks as the cross-regional dispersion should be more responsive in busts independent of the sign of the shock. The controls, x_t, contain 6 lags of the left-hand side variable, the log of the house price index on a federal level, the log of industrial production, the log of the CPI, the Federal Funds Rate (FFR), and the shocks. The sample runs from 1991 to 2019.

$$y_{t+h} = \alpha^h + \mathbb{1}(\Delta q_{t-1}^{agg} > 0) \times \beta_1 |\epsilon_t^{MP}| + \mathbb{1}(\Delta q_{t-1}^{agg} < 0) \times \beta_2 |\epsilon_t^{MP}| + x_t + u_{t+h} \tag{1}$$

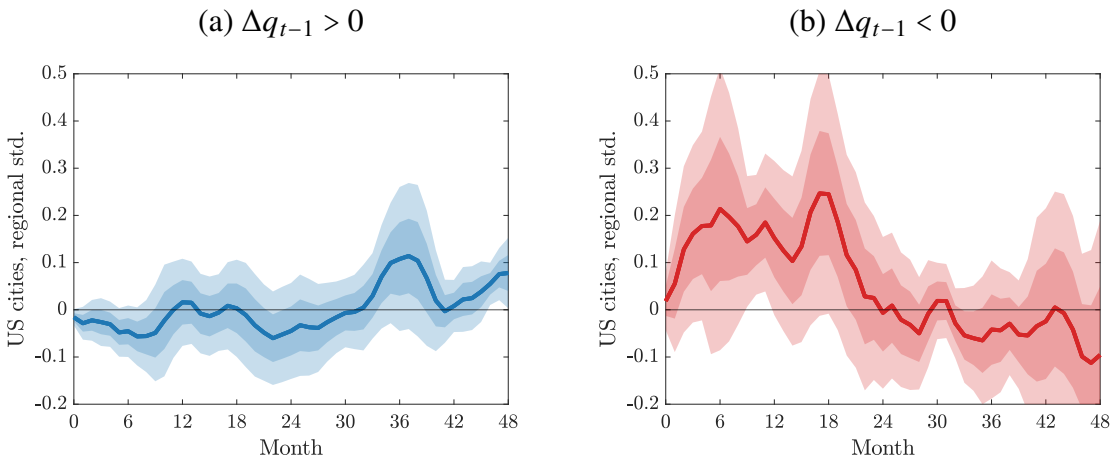
Figure (1) presents the results. We see no significant change in cross-regional standard deviations in booms; see panel (a). In times of busts, panel (b), we find that a monetary policy shock leads to a significant increase in the variation of cross-regional house price growth over the next two years. At the relative peaks, the bust response is approximately twice as high as the one in booms. Hence, conditional on monetary policy shocks, regional house price growth is more dispersed in busts relative to booms.

Robustness. Naturally, one may wonder whether house price booms and busts are simply expansions and recessions. Figure (C.1) shows that this is not the case. It plots the cross-city three-month moving average of house price growth. In particular, it shows that the standard deviations were already elevated before and remained elevated after the global financial crisis. We further observe an increase in the standard deviation *after* COVID, during 2022-2023. This period is associated with a house price bust but no recession. Further, we observe that the recessions brought by the Dotcom bust (2001) and the COVID-19 crisis (2020-23) were not associated with house price busts and did not experience elevated cross-regional heterogeneity in house price growth. Therefore, house price busts and recessions seem not to be perfectly correlated.

Concerning the definition of house price busts, we show, drawing on city-level data, that the results reported in Table (1) remain robust when adopting the boom/bust classification proposed by Albuquerque et al. (2025) (see Table (C.1)). Furthermore, since this definition is based on aggregate house prices, it raises the question of the extent to which regional house price cycles

[11] Regressing house price growth on the aggregate level on the monetary policy shock reveals that $R^2 = 0.0003$. This emphasizes that these shocks do not drive the cycle.

Figure 1: City-level std. house price growth response to a monetary policy shock, boom-bust



Notes: Responses to a monetary policy shock (25 bps on impact, absolute value) ; confidence intervals: 68% and 95% (Newey-West). The lhs variable is the three-month moving average of the cross-city standard deviation of house price growth.

co-move with the national cycle. To address this, we compute the frequency with which city-level and aggregate house price cycles are aligned. We find that cycle alignment occurs in approximately 85% of cases. Accounting for noise in aggregate and regional house price growth rates increases the share to 92%. Taken together, these findings suggest that local and aggregate housing cycles are synchronized the vast majority of the time.

We also study the response of regional house price growth dispersion to a main business cycle shock constructed by Angeletos et al. (2020). The shock is given in quarterly frequencies and we, therefore, move to state-level data to extend our cross-section. The main idea is that the main business cycle shock should be strong enough to affect the house price cycle.[12] Consequently, we study the response of the cross-state standard deviation of house price growth to an expansionary main business cycle shock. We observe that this shock indeed reduces house price growth heterogeneities; see Figure (C.2). As a sanity check, we also report the response of house price growth to the shock and observe that it is increasing. Our results are, therefore, in line with the findings reported above.

[12]Regressing house price growth on the shock gives an R^2 of 0.1, meaning that this shock can explain a non-trivial part of the variation in house price growth.

III. The Model

With these empirical impressions in mind, we now describe our multi-region currency union model, inspired by Benigno (2004), which we use to connect our empirical evidence and conduct our policy experiments. A notable difference between our model and Benigno's is the incorporation of incomplete bond markets across regions instead of having perfect consumption insurance. In addition, we incorporate a housing sector into the model, introducing two critical elements.

First, we account for subjective expectations in the formation of house prices. Previous research demonstrates that house price expectations deviate from rational expectations and play a crucial role in explaining the boom-bust dynamics in the housing market.[13] In our model, subjective expectations are applied exclusively to house prices, as the primary focus of this study is on housing market dynamics. Furthermore, the availability of forecast data is limited to a select number of variables, making it feasible to incorporate subjective expectations only for house prices where the empirical data can effectively inform the model.

Second, we allow for regional differences in housing markets. The only difference between regions is on the housing supply side, the distinction being the time required to construct houses, referred to as "time-to-build." This parameter enables us to model supply-side heterogeneities in a tractable manner.[14] Below, we also briefly discuss how housing demand-side heterogeneities may be modeled.

In the following, we focus on a specific region i. The remaining regions are modeled symmetrically.

III.A The Economy

Households. A representative domestic household in region i derives utility from consuming domestic and foreign varieties of goods from regions $j \neq i$, leisure, and housing. The preferences are as follows:

$$\mathbb{E}_0^{\mathcal{P}} \sum_{t=0}^{\infty} \beta^t \left(\frac{\xi_c c_{i,t}^{1-\sigma}}{1-\sigma} + \frac{\xi_{ih} h_{i,t}^{1-\nu}}{1-\nu} - \frac{\chi n_{i,t}^{1+\varphi}}{1+\varphi} \right)$$

$$c_{it} = \left[\lambda_{ii}^{\varsigma} c_{ii,t}^{1-\varsigma} + \sum_{j}^{N} \lambda_{ij}^{\varsigma} c_{ij,t}^{1-\varsigma}\right]^{\frac{1}{1-\varsigma}}, \quad c_{ii,t} = \gamma_i \left[\int c_{ii,t}(k)^{\frac{\epsilon-1}{\epsilon}} dk\right]^{\frac{\epsilon}{\epsilon-1}}$$

[13] See, for example, Case et al. (2012); Armona et al. (2019); Kuchler and Zafar (2019); Ma (2020).

[14] In the empirical analysis below we are more precise on why time-to-build seems a reasonable approximation for supply-side differences from an empirical point of view.

where $\mathbb{E}_0^{\mathcal{P}}$ denotes the subjective expectations operator discussed below. ξ_j and $j \in \{c, h, a, x\}$ denote model-exogenous shock terms, and $h_{i,t}$ and $n_{i,t}$ denote housing and hours worked, respectively. $\gamma_i \in (0, 1)$ denotes the measure of households in region i. Following Benigno (2004), γ_i is simultaneously the economic size of the region, that is, the mass of variety-producing firms. $c_{i,t}$ denotes the consumption goods basket of region i that is assembled from the varities of domestically produced goods $c_{ii,t}(j)$ and the varieties of imported goods $c_{ij,t}(k)$, which in turn are aggregates of the constant elasticities of substitution (CES) of the two groups of goods and their varieties. Specifically, $c_{ij,t}(k)$ denotes the consumption of good k consumed by region i and produced by region j. A preference bias for the goods produced in the respective region of residence ("home bias") arises if $\lambda_{ii}, \sum_{j \neq i}^{N} \lambda_{ij} \neq \gamma$.[15] The slope of the housing demand curve is governed by the parameter ν.

The household budget constraint is then given by

$$c_{i,t} + q_{i,t}(h_{i,t} - (1-\delta)h_{i,t-1}) + b_{i,t+1} + \frac{P_{ii,t}}{P_{i,t}} x_{i,t} = \\ w_{i,t} n_{i,t} + (1 + r_{i,t}) b_{i,t} + q_{i,t} \cdot H(x_{i,t-\tau_i}, \xi_x) - T_{i,t} + \Sigma_{i,t} + \mathfrak{b}_{i,t}$$

The budget constraint is expressed in units of the region i's final consumption basket, c_i. $\Sigma_{i,t}$ are profits from all domestic firms, which are owned evenly by all domestic households; $w_{i,t}$ is the real wage; $T_{i,t}$ represents government lump-sum taxes; and $b_{i,t}$ is a one-period nominal zero-coupon bond that is traded union-wide. $q_{i,t}$ denotes the real house price, and $x_{i,t}$ is the number of domestic consumption units dedicated to the production of new housing units.

To invest in housing production, households need to purchase domestically produced goods and transform them into housing investment units. Housing production is defined as $H(x_{i,t-\tau_i}, \xi_x) = \xi_x \frac{x_{i,t-\tau_i}^{\eta}}{\eta}, \eta \in (0, 1)$. τ_i denotes the number of quarters it takes to construct new housing units. Housing investment in period t will, therefore, lead to a return on this investment in period $t + \tau_i$. Regional heterogeneities on the housing supply side are modeled by allowing for cross-regional differences in time-to-build (τ_i).

In the steady state, we calibrate the steady-state values for $\xi_{hj}, \forall j$ s.t. $b_{j,ss} = 0, \forall j$. Hence, in the steady state, there are no net debtors or net creditor regions. This ensures that the only form of regional heterogeneity is situated on the housing side.

It is convenient to express bond holdings in units of region i's final basket. Households take as given the real interest rate, $r_{i,t}$, which is determined in equilibrium by the following Fisher-type

[15] Throughout the paper we maintain the assumptions that (i) the degree of home bias is symmetric, and (ii) the bias is such that households favor domestically produced products ($\lambda_{ii} \geq \gamma_i$).

equation: The value of bond holdings in units of numéraire is $B_{i,t} = P_{i,t} \cdot b_{i,t}$ and the nominal bond pays $i_{t-1} - \psi b_{i,t}$ units of the currency as interest.[16] The real interest rate is, thus, given by

$$1 + r_{i,t} = \frac{1 + i_{t-1} - \psi b_{i,t}}{1 + \pi_{i,t}}$$

where $\pi_{i,t} := P_{i,t}/P_{i,t-1} - 1$. Finally, $\mathfrak{b}_{i,t} := (\beta^{-1} - 1)\big(\gamma_i + \sum_{j \neq i}^{N} \gamma_j \frac{P_{j,t-1}}{P_{i,t-1}}\big)(1 + \pi_{i,t})^{-1} \bar{b}_i$, taken as exogenous by the household, captures the payment streams between i and j that guarantee that households are content with holding no bonds in the non-stochastic steady state with zero inflation and real exchange rate parity.[17]

Subjective expectations for house price setup. We model house price expectations following the capital gains extrapolation literature (e.g. Adam and Marcet, 2011; Adam et al., 2017). Section (V) provides empirical support for this choice; Adam et al. (2022) offers additional evidence. Households are endowed with a set of beliefs in the form of a probability measure over the full sequence of variables that they take as given, henceforth external variables $(\xi_t, r_{i,t}, w_{i,t}, \Sigma_{i,t}, T_{i,t}, \mathfrak{b}_{i,t}, \pi_{i,t}, (P_{ii,t}/P_{ij,t}), q_{i,t})_{t \geq 0}$.[18] We denote this measure as $\mathcal{P}$. Rational expectations are a special case of this setup in the form where households' beliefs agree with the objective or, equivalently, with the "true" or "equilibrium-implied" distribution of the external variables, $\mathcal{P} = \mathbb{P}$. Although households may hold expectations that are generally inconsistent with the equilibrium-implied (conditional) distribution of the external variables, it is worth emphasizing that, first, households have a time-consistent set of beliefs and, second, they behave optimally, given their beliefs. That is, households are *internally rational* in the sense of Adam and Marcet (2011). Moreover, the fact that all households are identical in their beliefs and preferences is not common knowledge among agents, so households cannot discover the misspecification of their beliefs, $\mathcal{P} \neq \mathbb{P}$, by inductively reasoning through the structure of the economy. Given the observed path of the external variables up to period t, households then use this information and $\mathcal{P}$ to form a

[16] The nominal interest rate is elastic in domestic households' aggregate bond holdings. We follow Schmitt-Grohé and Uribe (2003) to ensure stationarity of the first-order dynamics. In Appendix A, we provide a simple micro-foundation for debt-elastic interest rates.

[17] Given that bond holding entails a real cost in equilibrium, see footnote 16, introducing payments $\mathfrak{b}_{i,t}$ as a way to ensure that there are no bond holding costs in the non-stochastic steady state with zero inflation and real exchange rate parity (i.e., $1 + \pi_{i,t} = 1 + \pi_{j,t} = 1 + \pi_{ii,t} = 1 + \pi_{ij,t} = \frac{P_{ii,t}}{P_{ij,t}} = 1$). This ensures that this steady state is efficient, given that fiscal policy undoes the monopolistic competition distortion. $\mathfrak{b}_{i,t}$ may be interpreted as the real interest rate paid by a non-marketable nominal consol that perpetually pays the nominal rate $(\beta^{-1} - 1)(\gamma_i + \sum_{j \neq i}^{N} \gamma_j P_{j,t-1}/P_{i,t-1})$ of which the household is endowed with $\bar{b}_i$ units. The endowments of these consols ensure that nominal payments balance; that is, $\gamma_i \bar{b}_i + \sum_{j \neq i} \gamma_j \bar{b}_j = 0$, see Appendix A. Since we will linearize the model around a steady-state with zero bond holding, $\mathfrak{b}_{i,t}$ will be zero in equilibrium.

[18] ξ_t denotes the vector of exogenous shocks, which common across regions.

conditional expectation over the continuation sequence of the external variables, which we denote as $\mathbb{E}_t^{\mathcal{P}}$. As usual, we denote the conditional rational expectations operator by $\mathbb{E}_t$.

We assume that agents have rational expectations with respect to all external variables, except for house prices, $q_{i,t+s}$.[19] Households entertain the idea that house prices follow a simple state-space model:

$$\begin{aligned} \ln \frac{q_{i,t+1}}{q_{i,t}} &= \ln m_{i,t+1} + \ln e_{t+1} \\ \ln m_{i,t+1} &= \varrho \ln m_{i,t} + \ln v_{t+1}, \quad \varrho \in (0,1) \\ \begin{pmatrix} \ln e_t & \ln v_t \end{pmatrix}' &\sim \mathcal{N}\left(\begin{pmatrix} -\frac{\sigma_e^2}{2} & -\frac{\sigma_v^2}{2} \end{pmatrix}, \begin{pmatrix} \sigma_e^2 & 0 \\ 0 & \sigma_v^2 \end{pmatrix} \right) \end{aligned} \tag{2}$$

Hence, agents perceive house price growth rates as the sum of a transitory, and a persistent component. Crucially, $\ln e_t$ and $\ln v_t$ are not observable to the agents, rendering $\ln m_{i,t}$ unobservable. Agents apply the optimal Bayesian filter; that is, the Kalman filter, to arrive at the observable system.[20]

Lemma 1 (House price belief updating)**.** *Applying the Kalman filter to the state-space model and log-linearizing around the non-stochastic steady state gives*

$$\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+s} = \widehat{q}_{i,t} + \frac{1-\varrho^s}{1-\varrho} \varrho \widehat{\overline{m}}_{i,t} \tag{3}$$

$$\widehat{\overline{m}}_{i,t} = \varrho(1-g)\widehat{\overline{m}}_{i,t-1} + g\Delta\widehat{q}_{i,t-1}, \tag{4}$$

where $\ln \overline{m}_{i,t} := \mathbb{E}_t^{\mathcal{P}}(\ln m_{i,t})$ *is the posterior mean,* $g = \frac{\varrho^2\sigma^2+\sigma_v^2}{\varrho^2\sigma^2+\sigma_v^2+\sigma_e^2}$ *is the steady-state Kalman filter gain,* $\sigma^2 = \frac{1}{2\varrho^2}\left[-(\sigma_v^2+(1-\varrho^2)\sigma_e^2) + \sqrt{(\sigma_v^2+(1-\varrho^2)\sigma_e^2)^2 + 4\varrho^2\sigma_v^2\sigma_e^2}\right]$ *is the steady-state Kalman filter uncertainty, and* $\ln \widehat{e}_t$ *is perceived to be a white noise process.*

Proof. See Appendix B.1 for the application of the Kalman filter. Log linearization around the steady state gives the result. ■

Variables denoted with a "$\widehat{\ }$" express the respective variables in percent deviations from their steady-state value. Equation (3) shows that future house price beliefs depend on the current house price and today's beliefs. As $\varrho \in (0,1)$, the weight on the beliefs increases in the forecast horizon.

[19] Formally, $\mathcal{P} := \mathbb{P}_{-q} \otimes \mathcal{P}_q$, where $\mathbb{P}_{-q}$ is the objective measure over the sequences of the external variables without house prices, $\mathcal{P}_q$ is the measure over the sequences of house prices implied by the described perceived model of house prices, and $\otimes$ is the product measure. Since we are interested in a first-order solution to the model, it does not matter what households perceive to be the dependence structure between house prices and the other external variables.

[20] We assume that the agents' prior variance equals the steady-state Kalman variance.

Current house prices translate one-to-one into house price expectations. Hence, today's house price is extrapolated into the future. Turning to the belief updating Equation (3), we see that beliefs have an autoregressive component and are updated according to past observed house price changes. House price updating is increasing in the Kalman gain, g, and decreasing in the persistence of the beliefs, ϱ.

Asymmetry in house price beliefs. For our analysis, we may also consider time variation in the house price updating process. Specifically, we allow for heterogeneities in the persistence of beliefs (ϱ) and the Kalman gain (g) below and above a certain threshold $\bar{\omega}$.[21] Variations in the persistence of the belief process can be micro-founded by assuming that the agents hold different perceptions of belief persistence (ϱ^h, ϱ^l). Differential Kalman gains can be modeled by allowing the relative variances of the transitory components $((\sigma_e^h)^2, (\sigma_e^l)^2)$ and the persistent components $((\sigma_v^h)^2, (\sigma_v^l)^2)$ to differ above and below the threshold. Augmenting the house price belief model (2) by incorporating differential belief updating above and below the threshold $\bar{\omega}$ and proceeding analogously to the derivation in Lemma (1) yields a threshold version of the belief updating model:

$$\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1} = \mathbb{1}\big(\Delta\widehat{q}_{i,t-1} > \bar{\omega}\big)\Big[q_{i,t} + \varrho^h \widehat{\widehat{m}}_{i,t}\Big] + \mathbb{1}\big(\Delta\widehat{q}_{i,t-1} < \bar{\omega}\big)\Big[q_{i,t} + \varrho^l \widehat{\widehat{m}}_{i,t}\Big] \tag{5}$$

$$\begin{aligned}\widehat{\widehat{m}}_{i,t} = \mathbb{1}\big(\Delta\widehat{q}_{i,t-1} > \bar{\omega}\big)&\Big[\varrho^h(1-g^h)\widehat{\widehat{m}}_{i,t-1} + g^h \Delta\widehat{q}_{i,t-1}\Big] \\ &+ \mathbb{1}\big(\Delta\widehat{q}_{i,t-1} < \bar{\omega}\big)\Big[\varrho^l(1-g^l)\widehat{\widehat{m}}_{i,t-1} + g^l \Delta\widehat{q}_{i,t-1}\Big]\end{aligned} \tag{6}$$

This threshold framework captures distinct dynamics in the belief adjustments, conditional on whether the extrapolation parameter lies above or below the critical value $\bar{\omega}$. Consequently, agents update their beliefs asymmetrically across housing market regimes, reflecting heterogeneity in their responses to house price fluctuations. This extension provides a more nuanced representation of belief formation, particularly under varying market conditions.

Firms and price setting. We assume a continuum of monopolistically competitive firms that produce varieties of intermediate goods and have the same beliefs as households. Firms' beliefs, however, concern only the variables over which households have rational expectations. Therefore, firms are rational. Firm k buys labor $n_{i,t}(k)$ from the representative labor packer and produces variety $y_{i,t}(k)$ with a linear technology where labor is the only production factor. Households from both regions buy the variety produced. The firm sets its retail price $P_{ii,t}(k)$ and maximizes the expected discounted stream of profits, subject to Rotemberg-type adjustment costs. Formally, the

[21] In section (V), we discuss why this might arise and provide empirical evidence.

firm solves

$$\max_{P_{ii,t}(k)} \mathbb{E}_0^{\mathcal{P}} \sum_{t=0}^{\infty} \beta^t \frac{\Lambda_{i,t}}{P_{i,t}} \left[P_{ii,t}(k) y_{ii,t}(k) - (1-\tau^{\ell}) W_{i,t} n_{i,t} k j) - P_{ii,t} \frac{\kappa}{2} \left(\frac{P_{ii,t}(k)}{P_{ii,t-1}(k)} - 1 \right)^2 y_{ii,t} \right]$$

$$\text{with} \quad y_{ii,t}(k) = \left(\frac{P_{ii,t}(k)}{P_{ii,t}} \right)^{-\epsilon} y_{ii,t}$$

with $y_{ii,t}(k) = \xi_{a,t} n_{i,t}(k)$. $\Lambda_{i,t} = u'_{c,i,t}/u'_{c,i,0}$ denotes the stochastic discount factor and τ^{ℓ} is a wage subsidy paid by the government. It is selected such that the distortion in the monopolistic competition is offset in the non-stochastic steady state. The subsidy is financed through a lump-sum tax on the firm. In symmetric equilibrium, all firms choose the same price, $P_{ii,t}(k) = P_{ii,t} \forall k$. It follows that $\Pi_{ij,t}$ is producer price inflation in region i, or in other words, inflation of the goods basket produced in region i and sold to region j.[22] It follows that the terms of trade between region i and j are given by $s_{ij,t} = \frac{P_{ii,t}}{P_{ji,t}}$. The solution of the firm's problem results in a standard currency union version of the Phillips curve.

Monetary authority. The monetary authority sets nominal interest rates according to a standard Taylor rule targeting currency union consumer price inflation:

$$i_t = \frac{1}{\beta} \left(\Pi_t^{cu} \right)^{\phi_\pi} \xi_{r,t} \tag{7}$$

Currency union inflation is the average of region-level consumer price inflation ($\Pi_{i,t}$), weighted by the country size: $\Pi_t^{cu} = \prod_{j=1}^{N} (\Pi_{i,t})^{\gamma_i}$. $\xi_{r,t}$ denotes a monetary policy shock.

Market clearing. To achieve goods market clearing, each goods market for a variety j must clear. Additionally, labor and housing markets must clear within each region. Finally, the balance-of-payments equations must hold. Aggregation and the specific market clearing conditions are relegated to Appendix (B.3).

Equilibrium. We adopt the equilibrium concept of internally rational expectations equilibrium, as defined in Adam and Marcet (2011):

Definition 1 (Internally rational expectations equilibrium)**.** *An IREE consists of three bounded stochastic processes: shocks* $(\xi_t)_{t\geq 0}$*, allocations* $([c_{ij,t}(k)]_{k\in[0,1]}, b_{i,t}, h_{i,t}, x_{i,t}, n_{i,t})_{i\in[1,N],j\in[1,N],t\geq 0}$ *and prices* $(w_{i,t}, q_{i,t}, i_t, [P_{ij,t}(k)]_{k\in[0,1]})_{i\in[1,N],j\in[1,N],t\geq 0}$*, such that in all t*

[22] We assume that $P_{ii,t} = P_{ij,t}, \forall j$.

1. *households choose c, n, h, x, b optimally, given their beliefs $\mathcal{P}$,*

2. *firms set prices optimally, given their beliefs $\mathcal{P}$,*

3. *the monetary authority acts according to the Taylor rule (7),*

4. *markets for varieties of consumption goods, hours, and housing clear, given the prices, and the household budget constraints hold.*

III.B The solution method

We solve our model to the first order around a non-stochastic and efficient steady state. This preserves the analytic tractability at many points in the model and allows us to derive results on household behavior under capital gains extrapolation in a closed form. The model presented in Section III.A admits a unique non-stochastic steady state with zero net inflation and parity of the terms of trade:

Lemma 2 (Non-stochastic steady-state)**.** *Consider the model economy presented in Section III.A. As the variance of the shocks (actual or perceived) fades,* $\mathrm{Var}[\|\xi_t\|] \to 0$, $\xi_t = (\xi_{a,t}, \xi_{r,t}, e_t, v_t)^\top$, *there exists one and only one steady state in which net inflation is zero,* $\pi_{ij,ss} = \pi_{i,ss} = 0, \forall i, j \in [1, N]$, *and in which the terms of trade are at parity,* $s_{i,j,ss} = 1, \forall i, j \in [1, N]$.

Proof. See Appendix B.4 ■

Linearizing models with capital gains extrapolation is not straightforward. In fact, to the best of our knowledge, we are the first to provide a first-order approximation of a model with capital gains extrapolation under the assumption that agents hold rational expectations outside of the asset pricing block.[23] The entire exposition focuses on the typical linearized household problem in region i with all derivations being analogous for the typical household in the remaining regions.[24] For expositional clarity, we omit the specification that housing investment occurs in domestically produced goods. This simplification does not affect the fundamental logic of our solution method. We also drop the regional subscript i.

[23] Winkler (2020) proposes the "conditionally model-consistent expectations" (CMCE) concept as a starting point for linearizing models with capital gains extrapolation. Under CMCE, however, beliefs over all of the external variables are distorted relative to rational expectations. In our approach, linearized decision rules may be obtained under the assumption that belief distortions apply only to the asset prices, allowing us to confine the deviations from rational expectations to exactly those variables where the survey data allows us to discipline the expectations-modeling choices.

[24] For any variable $\mathrm{var}_t \notin \{b_{i,t}, \Sigma_{i,t}\}$, define $\widehat{\mathrm{var}_t} := \frac{\mathrm{var}_t - \mathrm{var}_{ss}}{\mathrm{var}_{ss}} \simeq \ln \mathrm{var}_t - \ln \mathrm{var}_{ss}$ to the first order. For $b_{i,t}, \Sigma_{i,t}$ define $\widehat{\mathrm{var}_t} := \frac{\mathrm{var}_t - \mathrm{var}_{ss}}{y_{ss}}$, which allows for the case that $\mathrm{var}_{ss} = 0$. (That is, we scale the deviations in the bond holdings and profits by the GDP.) Note, furthermore, that $\widehat{1 + r_{t+1}} \simeq \ln(1 + r_{t+1}) - \ln(1 + r_{ss}) \simeq r_{t+1} + \ln \beta$. We abuse the notation slightly and write $\widehat{r}$ instead of $\widehat{1 + r}$.

External and internal variables. Standard first-order solution techniques for models with rational expectations rely on a recursive representation of the equilibrium conditions. For instance, the inter-temporal consumption decision is captured by the forward recursive Euler equation:

$$\widehat{c}_t = \mathbb{E}_t \widehat{c}_{t+1} - \frac{1}{\sigma} \mathbb{E}_t \widehat{r}_{t+1}.$$

Under subjective expectations, an equivalent formulation exists:

$$\widehat{c}_t = \mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+1} - \frac{1}{\sigma} \mathbb{E}_t^{\mathcal{P}} \widehat{r}_{t+1}.$$

At this point, it is important to note that, first, we can characterize the external variables for which households have distorted expectations. In our case, this is only the house price, $\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{t+s}$, for which we derived a subjective expectation model. Second, for all other external variables for which households have rational expectations, we can formulate all of the equilibrium conditions recursively in the usual manner. In this case, the equilibrium-implied distribution measure applies and we can drop $\mathcal{P}$ from the expectations operator. The difficulty arises concerning the expectations for household choices or the internal variables, in our case $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+s}$. These variables depend on the external variables over which households hold subjective and rational expectations. In this case, we need to determine the subjective expectations distribution of $\widehat{c}_{t+1}$.[25] Our solution method allows us to determine these variables in closed form. We provide a detailed derivation in Appendix B.5 and concentrate, here, on conveying the intuition of our solution approach.[26]

Characterizing subjective expectations over internal variables. In solving for the subjectively optimal consumption plan, we exploit two key insights into how households behave to first-order that are valid irrespective of which set of beliefs they hold.[27] First, there is only one inter-temporal trade-off, namely in consumption, c. Given a path for consumption, the first-order conditions for housing, hours worked, and housing investment uniquely pin down a mapping from the external variables to the decisions for these internal variables. This insight allows us to concentrate on finding the optimal path for consumption.

The second insight is that to the first-order, the permanent income hypothesis holds and con-

[25]The reason is that households have distorted expectations over at least one price sequence and, therefore, will make distorted choices; in particular, they plan to make choices in the future that are inconsistent with what these choices will be in equilibrium. If we ignored this; that is, if we exchanged $\mathbb{E}_t^{\mathcal{P}} \widehat{c}_{t+s}$ for $\mathbb{E}_t \widehat{c}_{t+s}$ in the forward iteration above, then the computed equilibrium would be different from the IREE in Definition 1.

[26]For tractability, we set habit formation, $\bar{h}$, to zero and we assume that housing investments are made on aggregate consumption of goods bundle c_t instead of only locally produced goods. Relaxing these assumptions does not affect our methodological approach.

[27]In particular, our solution method can be used to solve RE models.

sumption depends only on the path of real interest rates, an external variable, and the subjectively expected lifetime income. By iterating forward the Euler Equation we obtain

$$\widehat{c}_t = -\frac{1}{\sigma}\sum_{s\geq 0}\mathbb{E}_t\widehat{r}_{t+s+1} + \mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty, \tag{8}$$

where we define $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty = \lim_{s\to\infty}\mathbb{E}_t^{\mathcal{P}}\widehat{c}_{t+s}$. This shows that the only subjective expectation variable to remain is $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$.[28]

At this point, the question arises whether this term is merely a technical artifact originating from our methodological approach or whether it possesses intrinsic economic meaning. For expositional purposes it is informative to consider a transitory shock to the model. Suppose that the shock raises house prices, which leads households to believe that house prices will rise further. Consequently, they believe that they will be wealthier in the future, leading to an increase in their current and planned future consumption. This expectation is reflected in the limit by an increase in $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$, thereby capturing a subjective expectations wealth effect. In essence, due to extrapolation, households believe they will be richer than they actually will be. Under rational expectations, this term also exists. However, households recognize that the shock is purely transitory and will not influence their long-term wealth. Therefore, the wealth effect under rational expectations is always zero.

Characterizing the subjective expectations wealth effect. We now focus on deriving an analytical expression for the subjective expectations wealth effect, $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$. Using the insights from above, the household's first-order conditions are given by the equations in (9). (For expositional brevity, we dropped the exogenous shocks and set time-to-build to zero.)

$$\begin{aligned}
&\text{Rational Expectations} && \text{Subjective Expectations} \\
&\varphi\widehat{n}_t + \sigma\widehat{c}_t = \widehat{w}_t && \varphi\widehat{n}_t + \sigma\widehat{c}_t = \widehat{w}_t \\
&\widehat{q}_t = (1-\eta)\widehat{x}_t && \widehat{q}_t = (1-\eta)\widehat{x}_t \\
&\widehat{c}_t = -\tfrac{1}{\sigma}\textstyle\sum_{s\geq 0}\mathbb{E}_t\widehat{r}_{t+s+1} + \mathbb{E}_t\widehat{c}_\infty && \widehat{c}_t = -\tfrac{1}{\sigma}\textstyle\sum_{s\geq 0}\mathbb{E}_t\widehat{r}_{t+s+1} + \mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty \\
&\widehat{h}_t = \tfrac{\sigma}{\nu(1-\bar{\beta})}(\widehat{c}_t - \bar{\beta}\mathbb{E}_t\widehat{c}_{t+1}) - \tfrac{1}{\nu}\tfrac{\widehat{q}_t - \bar{\beta}\mathbb{E}_t\widehat{q}_{t+1}}{1-\bar{\beta}} && \widehat{h}_t = \tfrac{\sigma}{\nu(1-\bar{\beta})}(\widehat{c}_t - \bar{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{c}_{t+1}) - \tfrac{1}{\nu}\tfrac{\widehat{q}_t - \bar{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_{t+1}}{1-\bar{\beta}},
\end{aligned} \tag{9}$$

where $\overline{\beta} := \beta(1-\delta)$ and under rational expectations $\mathbb{E}_t\widehat{c}_\infty = 0$, as explained above. From this representation, it becomes evident that the difference between subjective and rational expectations

[28] We directly used the fact that households have rational expectations over external variables other than house prices, $\mathbb{E}_t^{\mathcal{P}}\text{var}_{t+1} = \mathbb{E}_t\text{var}_{t+1}$ for any variable $\text{var}_{t+1} \neq q_{t+1}$.

arises from the expectation operator $\mathcal{P}$ applied to next period's consumption and house prices, as well as from the presence of the subjective expectations wealth effect, $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$. In particular, while the labor supply and housing investment decisions remain identical under both expectation formations, the key differences manifest in the Euler equation and the housing demand condition. Subjective expectations with respect to house prices are easy to characterize, as these are pinned down by the subjective expectations model. Further note that subjective expectations for the next period's consumption can be characterized by leading the Euler Equation (8) one period, once we have a representation for the subjective expectations wealth effect. Therefore, to solve the subjective expectations model, it suffices to find a representation for the subjective expectations wealth effect. We find this characterization by combining the first-order conditions with the linearized household budget constraint, *(BC)* in Appendix-equation (B.6). After iterating over this equation, we can find a closed form expression for the subjective expectations wealth effect.

Proposition 1 (Subjective expectations wealth effect)**.** *The first-order around the non-stochastic steady-state subjective expectations wealth effect, $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$, is given by*

$$\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty = \bar{\mathrm{Q}} \cdot \underbrace{\left[\widehat{q}_t + \bar{\mathrm{M}}\widehat{\bar{m}}_t\right]}_{=\Lambda_{t,1}} + \bar{\mathrm{C}}\left[\underbrace{\sum_{n=1}^{\infty}\beta^n \mathbb{E}_t\{\bar{z}_{t+n}\}}_{=\Lambda_{t,2}} + \widehat{b}_{t+1}\right], \tag{10}$$

where $\bar{\mathrm{Q}}, \bar{\mathrm{C}} \in (0,1)$, $\bar{\mathrm{M}} > 0$ and $\bar{z}_{t+n}$ is a function of external variables the household has rational expectations about, as is specifically stated in Appendix B.5.

Proof. See Appendix B.5 ■

The notation "ss" in the subscript of a variable denotes its' steady-state values. Equation (10) shows that the subjective expectations wealth effect consists of three parts. First, there are variables for which the household has subjective expectations, $\Lambda_{t,1}$. In our case this is only the house price, $\widehat{q}_t$, and the posterior beliefs about house prices, $\widehat{\bar{m}}_t$. Second, it depends on the expectations of all of the variables the household has rational expectations about, $\Lambda_{t,2}$, which are collected in z^*_{t+n}. Third, it depends on today's bond choices $\widehat{b}_{t+1}$. Intuitively, households evaluate the wealth effect under subjective expectations by applying the permanent income hypothesis. In doing so they account for their future decisions by using their first-order conditions resulting in a function that depends only on prices and today's saving choices. This formulation elucidates the previously discussed intuition regarding the role of subjective expectations in wealth effects. Specifically, an increase in house prices ($\widehat{q}_t$) or in beliefs about future house prices ($\widehat{\bar{m}}_t$) lead to a rise in $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$, given that $\bar{\mathrm{Q}}, \bar{\mathrm{M}} > 0$. The agents, anticipating higher wealth as a result of these changes, adjust their expectations accordingly and expect an increase in consumption in the long run.

Discussion. Our method has two important advantages over previous approaches to solving asset price learning models. First, we solve the model using a first-order approximation, which makes it fast to solve, easily scalable, and amenable to an analysis of Ramsey-optimal policies. The literature previously relied on non-linear solution techniques (Adam et al., 2017) or hybrid techniques (Adam et al., 2022) to solve these models. Hence, solution procedures are much more involved and limit the complexity of the models that can be numerically solved. Second, our solution method confines subjective expectations to house prices. A previously developed method by Winkler (2020) and Caines and Winkler (2021), which also relies on perturbation, assumes household expectations to conform with the concept of conditionally model-consistent expectations. Under this concept, subjective expectations about one variable lead to spillovers to expectations about other variables. Thus, households will form subjective expectations across all model variables. In our approach, households only hold subjective expectations with respect to one variable, while they remain rational with respect to all other variables. This method of explicitly characterizing choices in terms of lifetime income is general in the sense that it allows for solving household decisions under any time-consistent set of beliefs.

IV. Analytic Insights

We proceed by deriving the primary channel through which house price extrapolation and heterogeneities on the housing supply side can produce the patterns observed in Section (II). Furthermore, we establish that our findings do not hold under a rational expectations framework.

To do so we examine a simplified version of the model introduced above: we consider a one-region, zero liquidity endowment economy with fully sticky prices. We study the path of house prices within an exercise that, while keeping everything tractable, allows us to understand the effects of expansionary monetary policy: the consumption endowment process is selected such that the real rate drops by ε on impact of the shock and returns to 0 afterwards. This is arguably a parsimonious way to model expansionary monetary policy.

We assume that the housing supply is a function of endowment, and moves in response to the shock by $-\iota_i\varepsilon$. Hence, in response to an expansionary monetary policy shock housing supply increases. In the following regional differences in housing supply will be simply modeled as regional variations in ι_i. In Appendix (B.6.3) we also consider the case of demand-side heterogeneities and show that similar results can be attained as under supply-side heterogeneities.

Throughout this section we will consider parameter choices commonly used in the literature. Specifically, we consider parameter choices in line with Section (V), which are stated in Table (5)

and (C.5). Detailed derivations are relegated to Appendix B.6. Under the given assumptions the house price and housing market clearing can be expressed as:

$$\widehat{q}_{i,t} = \mathbb{E}_t^{\mathcal{P}} \widehat{q}_{i,t+1} - \widehat{r}_{t+1} - \nu(1-\bar{\beta})\widehat{h}_{i,t} + \nu(1-\bar{\beta})^2 \sum_{s=0}^{\infty} \bar{\beta}^s \mathbb{E}_t^{\mathcal{P}} \widehat{h}_{i,t+s+1} \tag{11}$$

$$\widehat{h}_{i,t} - (1-\delta)\widehat{h}_{i,t-1} = -\iota_i \varepsilon_t \tag{12}$$

House price dynamics under rational expectations. As we focus on housing supply side heterogeneities we simplify the housing demand side and assume that $\nu = 1$. The house price is pinned down jointly by the equations (11) and (12). The differences between rational and subjective expectations arise due to the expectation formation on future house prices, $\mathbb{E}_t^{\mathcal{P}} \widehat{q}_{t+1}$, and future housing choices, $\mathbb{E}_t^{\mathcal{P}} \widehat{h}_{t+s+1}$. Under rational expectations, we can simply use the housing market clearing condition (12) to pin down house prices. Doing so yields the following proposition.

Proposition 2 (House prices under Rational Expectations)**.** *In the outlined one-region, zero liquidity endowment economy with exogenous housing supply, and considering a shock monetary policy shock such that $\varepsilon_1 < 0$ and $\varepsilon_t = 0$ for $t > 1$. House prices are given by:*

$$\widehat{q}_{i,t} = \mathbb{E}\widehat{q}_{i,t+1} - \widehat{r}_{t+1} - (1-\bar{\beta})\widehat{h}_{i,t} + (1-\bar{\beta})^2(1-\delta)^t \frac{1}{1-\bar{\beta}(1-\delta)}(-\iota_i \varepsilon_1)$$

And for the first two periods we have:

$$\widehat{q}_{i,1} = -\varepsilon_1 + \omega_h \iota_i \varepsilon_1$$
$$\widehat{q}_{i,2} = (1-\delta)\omega_h \iota_i \varepsilon_1$$

with $\omega_h = \frac{1-\bar{\beta}}{1-\bar{\beta}(1-\delta)} \in (0,1)$.

Proof. See Appendix B.6.1 ■

Proposition (2) makes clear that an expansionary monetary policy shock raises house prices. The shock fades after the first period. If housing supply increases simultaneously ($\iota_i > 0$), the house price increase is less pronounced in the first period. Due to the increase in housing supply, house prices will be below the steady-state level after the first period, converging back to the steady-state going forward. The coefficient ω_h, which scales the effect through housing supply, captures the notion that greater housing supply exerts downward pressure on both current and future prices, owing to the durability of housing.[29]

[29] In the housing demand equation, $\mathbb{E}\widehat{q}_{i,t+1}$ appears. Iterating forward on this equation yields the closed-form

House price dynamics under subjective expectations. Moving on to the subjective expectation case, house price expectations, $\mathbb{E}_t^{\mathcal{P}}\widehat{q}_{i,t+1}$, are pinned down by the subjective expectations model, equations (3) and (4). Future expected housing choices, $\mathbb{E}_t^{\mathcal{P}}\widehat{h}_{i,t+s+1}$, can be characterized using the solution method described in Section (III.B). We arrive at the following proposition:

Proposition 3 (House prices under Subjective Expectations)**.** *In the outlined one-region, zero liquidity endowment economy with exogenous housing supply, and considering a shock monetary policy shock such that $\varepsilon_1 < 0$ and $\varepsilon_t = 0$ for $t > 1$. House prices are given by:*

$$\widehat{q}_{i,t} = -\widehat{h}_{i,t} - \omega_r \widehat{r}_{t+1} + \omega_m \widehat{\bar{m}}_{i,t}$$

And for the first two periods we have:

$$\widehat{q}_{i,1} = -\omega_r \varepsilon_1 + \iota_i \varepsilon_1$$
$$\widehat{q}_{i,2} = (1-\delta)\iota_i \varepsilon_1 + \omega_m g \widehat{q}_{i,1}$$

with $\omega_r = 1 + \frac{y_{ss}(1-\beta)}{y_{ss}+h_{ss}q_{ss}\delta} > 1$, $\omega_m > 1$, and $\frac{\partial \omega_m}{\partial \rho} < 0$.

Proof. See Appendix B.6.2 ■

Proposition (3) elucidates several key dynamics. First, an expansionary monetary policy shock raises house prices, and given that $\omega_r > 1$, this increase is more pronounced than under RE. The amplified impact response of house prices under SE follows directly from the dual role of prices: they signal not only current scarcity but also the expected future path of asset values. Under RE, agents anticipate mean-reversion in house prices; under SE, agents apply their perceived model and project elevated prices to persist. Formally, the heightened pass-through of the monetary shock to house prices, captured by $\omega_r > 1$ under SE relative to unity under RE, arises because expectations affect the house price not only directly through the future house price in the housing demand equation, but also indirectly through agents' choices.[30] Second, a positive housing supply response attenuates the monetary-policy-induced rise in house prices, and this mitigating effect is stronger than under RE, as reflected by $\omega_h < 1$. As noted above, under RE agents implicitly internalize the fact that higher housing supply depresses future prices, thereby restraining current prices as well; this forward-looking channel is absent under SE. Third, belief extrapolation introduces a backward-looking component that exerts upward pressure on house prices. An increase in the Kalman gain

expression for ω_h: since additional supply today depresses future prices, this feeds back into current prices. Proposition (2) derives this channel explicitly.

[30] Solving for $\sum_{s=0}^{\infty}\bar{\beta}^s \mathbb{E}_t^{\mathcal{P}}\widehat{h}_{i,t+s+1}$ introduces a distinct interest rate pass-through, as future expected housing choices depend on all exogenous variables, including the path of interest rates.

(g) amplifies this channel, as does a decrease in the persistence of belief formation (ϱ). While both mechanisms generate amplification in theory, the quantitative exercises below demonstrate that the Kalman gain is the quantitatively dominant channel.

Regional heterogeneities. Let us now turn towards understanding how these dynamics affect regional dispersions. Consider an expansionary monetary policy shock that is symmetric across regions and results in a heterogeneous increase in housing supply as ι_i differs across regions. Using the results from above, we can derive the regional standard deviation two periods after the impact of the shock:

Proposition 4 (Regional standard deviation)**.** *The regional standard deviation of house price growth in period two under rational expectations is given by:*

$$\sigma_{\Delta q,2} = \delta\omega_h \sqrt{\frac{1}{N}\sum_i^N [(\bar{\iota} - \iota_i)\epsilon_1]^2}, \quad \bar{\iota} = \frac{1}{N}\sum_i^N \iota_i$$

The regional standard deviation of house price growth in period two under subjective expectations is given by:

$$\sigma_{\Delta q,2} = (\omega_m g - \delta)\sqrt{\frac{1}{N}\sum_i^N [(\iota_i - \bar{\iota})\epsilon_1]^2}, \quad \bar{\iota} = \frac{1}{N}\sum_i^N \iota_i$$

where $\omega_m g >> \delta$ generally holds. And a higher Kalman gain, g, or lower belief persistence, ϱ, amplifies regional differences:

$$\frac{\partial\sigma_{\Delta q,2}}{\partial g} = \omega_m \sqrt{\frac{1}{N}\sum_i^N [(\iota_i - \bar{\iota})\epsilon_1]^2} > 0, \quad \frac{\partial\sigma_{\Delta q,2}}{\partial \varrho} = g\frac{\partial\omega_m}{\partial\rho}\sqrt{\frac{1}{N}\sum_i^N [(\iota_i - \bar{\iota})\epsilon_1]^2} < 0$$

Proof. The regional standard deviations are obtained by first computing house price growth from Propositions (2) and (3), respectively, and subsequently calculating the corresponding standard deviations across regions. ■

Proposition (4) summarizes the paper's main argument. Under both rational and subjective expectations, the presence of regional heterogeneity, $\sum_i^N (\iota_i - \bar{\iota}) \neq 0$, generates dispersion in house price growth. With rational expectations, this dispersion remains limited, as $\omega_h \in (0, 1)$ and δ is small and positive. Under subjective expectations, however, dispersion can become substantial since $(\omega_m g - \delta) > 0$ and $\omega_m g$ may be large.[31] Moreover, it becomes apparent that an increase in

[31]Numerical simulations indicate that $\omega_m g$ can reach values up to 73, implying strong amplification. For realistic

the Kalman gain g, or a decrease in the perceived persistence ϱ, leads to an increase in house price growth dispersion.

This mechanism explains why the dispersion in regional house price growth is greater during busts than in booms. As we will demonstrate below, the formation of house price beliefs differs between booms and busts. In particular, the Kalman gain (g) is higher during busts. As a result, the amplification effect of house price extrapolation is stronger in bust periods, leading to increased regional dispersion in house price growth. Importantly, this mechanism is absent in the rational expectations model, rendering it incapable of accounting for the asymmetry in regional dispersion in house price growth rates between booms and busts. Therefore, the rational expectations framework fails to capture the asymmetric regional house price growth dynamics observed in the data.

V. House price beliefs and regional heterogeneities in the data

In this section, we focus on our primary empirical moment: house price belief updating during house price booms and busts. We directly estimate the belief persistence (ϱ) and the Kalman gain (g) from our subjective expectations model laid out in Section (III.A): Notably, we find that in busts the Kalman gain is larger than in booms. Further, provide potential explanations why these patterns arise and study it's direct implications for house prices. We use these findings to calibrate our quantitative model. We then calibrate the regional heterogeneities on the housing supply side using 20 regions, each region representing one metropolitan area from the Case-Shiller U.S. National Home Price Index. Finally, we calibrate the slope of the Phillips-Curve such that the model matches the peak response of a 25 bp monetary policy shock, unconditional on a house price boom or bust. The remaining parameters a chosen in line with the literature.

V.A House price beliefs and boom-bust dynamics

Violation of rational expectations. In Section (III), we construct a model that breaks the rational expectations hypothesis with respect to house prices and introduces a Bayesian belief updating model to describe the house price belief formation. This modeling approach is grounded in a substantial body of empirical literature that examines the dynamics of belief formation in the context of housing markets.[32] To assess whether house price expectations conform to full-information

Kalman gains of $g \in [0.01, 0.025]$, we use $g = 0.02$ in our baseline example.

[32]See: Case et al. (2012); Armona et al. (2019); Kuchler and Zafar (2019); Ma (2020)

rational expectations, we follow the test proposed by Coibion and Gorodnichenko (2015) and extended to house price expectations by Adam et al. (2022), conditioning additionally on booms and busts. Specifically, we estimate the following regression using house price expectations from the Survey of Consumers and house price data from the Case-Shiller index:

$$q_{t+12} - \mathbb{E}_t^{\mathcal{P}} q_{t+12} = a^{boom} + a^{bust} + b^{boom}\mathbb{1}(\Delta q_{t-1} > 0)(\mathbb{E}_t^{\mathcal{P}} q_{t+12} - \mathbb{E}_{t-1}^{\mathcal{P}} q_{t+12}) + b^{bust}\mathbb{1}(\Delta q_{t-1} < 0)(\mathbb{E}_t^{\mathcal{P}} q_{t+12} - \mathbb{E}_{t-1}^{\mathcal{P}} q_{t+12}) + u_t$$

The regression projects forecast errors on revisions in booms and busts. Under full-information rational expectations, the coefficients b^{boom} and b^{bust} should be zero.[33] Table (2) presents the

Table 2: Test for rational expectations

b^{boom}	b^{bust}	R^2	N
1.76^{*}	3.26^{***}	0.45	150
(1.00)	(1.17)		

Notes: Estimation of the Coibion and Gorodnichenko (2015) regression for house prices, conditioning on boom and bust regimes. Reported standard errors are Newey-West. Significant levels: $***p < 0.01, **p < 0.05, *p < 0.1$.

results. Two findings stand out. First, the coefficients are statistically significant in both regimes, providing clear evidence against full-information rational expectations. Second, the positive point estimates indicate that expectation revisions are insufficiently large relative to the information contained in forecast errors—that is, households update their house price beliefs too sluggishly. These findings are in line with Adam et al. (2022).

In appendix (C.3), we further demonstrate that our modeling approach remains valid when conditioning on monetary policy shocks. Specifically, we examine the response of forecast errors to these shocks. Under the assumption of rational expectations, this response should theoretically be zero; however, our analysis reveals a substantial and statistically significant response. We observe a sluggish adjustment of beliefs following the shock, with evidence of initial over-pessimism in the first month, which subsequently transitions to over-optimism in the longer term. These findings are consistent with the literature and suggest that a Bayesian belief updating model is well-suited to capturing the observed dynamics in house price expectations. The findings therefore confirm our modeling choice from Section (III).

[33]The Survey of Consumers only reports expectations at one fixed horizon. We follow Adam et al. (2022) and use the previous month's expectations to proxy for $\mathbb{E}_{t-1}^{\mathcal{P}} q_{t+12}$. Under full-information rational expectations b^{boom} and b^{bust} remain zero under this proxy.

House price belief formation. Our findings in Section (IV) suggest that stronger belief updating in busts are essential to explain diverging regional house price growth rates in busts. Therefore, we estimate the belief updating process described in Section (III) and allow for potential differences in the belief updating process. Specifically, we test two specifications: a linear one and a threshold specification allowing for regime switching. The linear process reads as follows:

$$\mathbb{E}_t^{\mathcal{P}} \Delta q_{t+1} = c + \varrho(1-g) \times \mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t + \varrho g \times \Delta q_{t-1}. \tag{13}$$

Equation (13) can be derived by combining equations (3) and (4). c is a constant that can be linked to the long-run house price growth expectations. The belief updating model is in monthly frequencies, and the sample is for the period 2007 to 2020. We obtain the month-on-month percentage change in house price expectations, $\mathbb{E}_t^{\mathcal{P}} \Delta q_{t+1}$, by dividing the year-on-year percentage changes from the Michigan survey by 12. House price growth data is taken from the Case-Shiller National House Price Index. To estimate this model, we replicate the belief updating process employed by the agents in our model, which follows a recursive updating procedure. Specifically, to estimate this process, we feed in the realized past house price growth data, Δq_{t-1}. For a given tuple (ϱ, g, c) and by recursively updating $\mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t$, we obtain a sequence of house price growth expectations. We then use a solver choosing (ϱ, g, c) such that the mean square error (MSE) between the data and the fitted values is minimized.[34] For the estimation procedure, we also impose that $\varrho, g \in (0, 1)$.

Equation (14) allows for heterogeneity in ϱ and g below and above a certain threshold, based on equations (5) and (6). The estimation procedure is equivalent to the linear model, only now we choose $(\varrho^h, \varrho^l, g^h, g^l, c^h, c^l, \omega)$ to minimize the MSE between the fitted values and the data. Notice that the threshold is also estimated and that the regime depends on past observed house price growth, Δq_{t-1}.

$$\begin{aligned} \mathbb{E}_t^{\mathcal{P}} \Delta q_{t+1} = \mathbb{1}(\Delta q_{t-1} > \omega) \left(c^h + \varrho^h (1-g^h) \times \mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t + \varrho^h g^h \times \Delta q_{t-1} \right) + \\ \mathbb{1}(\Delta q_{t-1} < \omega) \left(c^l + \varrho^l (1-g^l) \times \mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t + \varrho^l g^l \times \Delta q_{t-1} \right) \end{aligned} \tag{14}$$

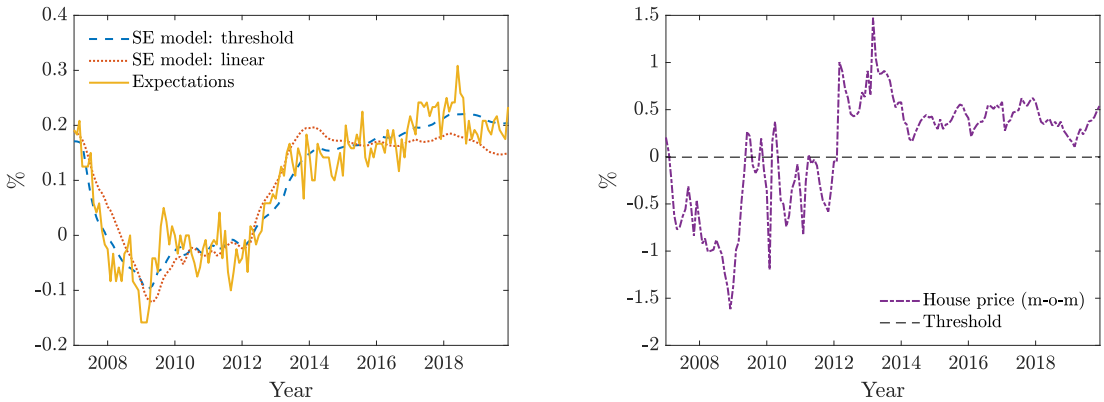
The estimation results are presented in Figure (2). The solid yellow line in panel (a) plots the expectations data from the Michigan survey, the blue dashed line represents the threshold model estimates, and the dotted red line shows the linear model. The dashed-dotted purple line in panel (b) depicts the month-on-month house price growth, while the dashed black line indicates the threshold

[34] The estimation procedure follows Adam et al. (2017), who apply this to stock prices. It demands a starting value for $\mathbb{E}_{t-1}^{\mathcal{P}} \Delta q_t$, which can affect the outcomes. To minimize the effect of the starting values on the estimated parameters, we feed in monthly house price growth data starting with 1987, thereby obtaining the initial values for 2007M1. Changing the starting value to 1987 has insignificant effects on our results.

for the threshold model. Turning to the linear model first, we find that it does a reasonable job of explaining the expectation formation process. However, it lags behind during the bust episode and underestimates the expectations for house price growth during the recovery following the global financial crisis. In comparison, the threshold model improves on these dimensions. It captures the bust without an obvious lag and performs better in recovery episodes. The estimated threshold for this model is close to zero, suggesting that the updating behavior differs during periods when house prices rise (booms) compared to when they fall (busts).

Figure 2: House price belief model, U.S.

(a) Expectations (b) House price growth



Notes: SE model, threshold (blue dashed): fitted values of Equation (14); subjective expectations model, linear (red dotted): fitted values of Equation (13); expectations (yellow solid): mean expectations data from the Michigan survey; house price (m-o-m) (purple dashed-dotted): month-on-month percentage change in the house price. The black dashed line in panel (b) depicts the threshold in the threshold model.

The estimated parameters from the models are shown in Table (3). We find a ϱ of 0.95 and a Kalman gain of 0.018 for the linear model. For the threshold model, we find a ϱ^h of 0.99 and a Kalman gain of g^h, 0.013 in booms. In busts, ϱ^l decreases to 0.84, while g^l increases to 0.022. All parameters are roughly in line with the literature. The differences in the parameters across regimes transparently show why the threshold model can fit the data better than the linear model: In busts when house prices and beliefs fall drastically in a short period, a higher Kalman gain and a lower persistence parameter enable faster pass-through from the observed house prices to the beliefs. In booms, when house prices and beliefs recover steadily over a longer period, the expectations are better matched by a slow-moving process with high persistence. This is achieved with a higher degree of persistence ϱ^h and a lower degree of updating g^h. The inclusion of the threshold component decreases the MSE by roughly 45%.

We also perform a robustness exercise to study whether the parameter differences across regimes affect model performance. We do so by subsequently imposing the estimated parameters from above

and below the threshold on the linear model. We find a significant increase in the MSE, implying that the differences in the estimated parameters of the threshold model matter for the model's performance.

Table 3: House price belief model: parameters

$Specification$	ϱ^{lin}	ϱ^{h}	ϱ^{l}	g^{lin}	g^{h}	g^{l}	$Threshold$	MSE^{i}/MSE^{lin}
$Baseline:$								
$linear$	0.95			0.018				1.00
$threshold$		0.99	0.84		0.013	0.022	−0.004	0.55
$Robustness:$								
$high\ regime\ beliefs$	0.99			0.013				5.9
$low\ regime\ beliefs$	0.84			0.022				5.5

Notes: Linear: estimated parameters from Equation (13); threshold: estimated parameters from (14); robustness exercises are explained in the main text.

As a final robustness check, we follow Case et al. (2012), who document the pass-through from realized house prices to expectations by regressing expectations on lagged house price growth, and extend their specification by conditioning on booms and busts:

$$\mathbb{E}_t^{\mathcal{P}} \Delta q_{t+1} = a^{boom} + a^{bust} + b^{boom} \mathbb{1}(\Delta q_{t-1} > 0) \Delta q_{t-1} + b^{bust} \mathbb{1}(\Delta q_{t-1} < 0) \Delta q_{t-1} + u_t$$

Table (4) presents the results. The pass-through is statistically indistinguishable from zero in booms, while positive and significant in busts.[35] Crucially, we reject the null hypothesis of equal pass-through across regimes $(H_0 : boom = bust)$. These results are consistent with the findings documented above. Further, the large R^2 is in line with Case et al. (2012), confirming that lagged house price growth accounts for a substantial share of variation in expectations.

Table 4: Pass-through from lagged prices to expectations

b^{boom}	b^{bust}	R^2	N	$p-val.\ H_0 : b^{boom} = b^{bust}$
−0.05	0.06^{**}	0.61	150	0.03
(0.04)	(0.03)			

Notes: Estimation of the Case et al. (2012) regression for house prices, conditioning on boom and bust regimes. Reported standard errors are Newey-West. Significant levels: $***p < 0.01, **p < 0.05, *p < 0.1$. Column 5 shows the p-value for the null hypothesis that both coefficients are equal.

[35] Although the coefficient for the boom is insignificant, its sign is counterintuitive. We show in the Appendix that this anomaly is attributable to small sample limitations. For details see the discussion below and Appendix (C.6).

Sources of asymmetric updating. The results above raise the question of why households update more strongly during house price busts. A potential reason why updating differs in booms and busts could be that the actual law of motion differs across regimes, which in turn affects the perceived law of motion, hence ϱ, σ_e, and σ_v. In our first exercise, we test this hypothesis by regressing house price growth on its lag, conditioning on booms and busts. Consistent with our belief formation results, we find that house price growth is more persistent in booms than in busts, suggesting a lower ϱ in busts. Given the definition of the Kalman gain g, a decline in perceived persistence ϱ—holding σ_v and σ_e constant—increases g when σ_v is not too large. Thus, less persistent dynamics of house prices in busts can induce stronger updating.

As a second exercise, we can directly rerun the threshold exercise (equation (14)) using actual house prices instead of expectations. We find that indeed the persistence (ϱ) is smaller in a bust, while the updating (g) is larger. Both of these exercises reveal that the expectation formation process inherits the properties from the actual law of motion. The results can be found in Appendix (C.4).

Consequences of asymmetric updating. Our findings indicate that house price expectations updating is more pronounced in busts. This, in turn, can affect house prices themselves: a larger reaction in expectations translates into economic decision-making and will eventually be reflected in prices. We test whether this is the case by studying the effect of a monetary policy shock on house prices during booms and busts. We find that house prices respond more strongly during times of a house price bust (see Appendix (C.5). These findings directly validate our results from Proposition (3). We will rely on this empirical moment in the quantitative exercise.

Additional evidence. An alternative explanation for asymmetric belief updating is heightened attention during busts. The Bayesian learning framework admits an equivalent representation within a Rational Inattention framework (Pfäuti, 2025), where the Kalman gain can be interpreted as a measure of attention (Appendix (C.4)). Supporting this interpretation, Appendix (C.4) shows that mentions of "house price" in The New York Times increase substantially during busts, suggesting that public attention to housing market developments is significantly higher in busts than in booms.

Glaeser and Nathanson (2017), focusing on 1980-2011 using annual data, point out that their model would fit the data better if the extrapolation were larger in a boom.[36] Crucially, our data set lacks the run-up to the financial crisis, raising the question of whether this impacts our findings. Unfortunately, there is no house price expectations data available at an adequate frequency. We

[36]They construct a partial equilibrium housing model allowing for naive extrapolative expectations. Note that they do not directly estimate updating of expectations.

therefore rely on a proxy proposed by Jacobson (2022), combining backcasts until 1993 with the actual expectations data from 2007 onward. Re-estimating equation (14) leaves the results qualitatively unchanged: the threshold remains near zero, with lower persistence and a higher Kalman gain in busts. When explicitly distinguishing between booms, busts, and normal times, we find: $g^{bust} > g^{boom} > g^{normal}$ and $\varrho^{boom} > \varrho^{normal} > \varrho^{bust}$. Three findings are noteworthy. First, consistent with our baseline extrapolation remains strongest in busts. Second, in line with Glaeser and Nathanson (2017), extrapolation is elevated in booms relative to normal times. Third, while belief updating is amplified in both booms and busts, persistence is markedly higher in booms than busts—reflecting that boom-phase house price dynamics are gradual and persistent, whereas bust-phase dynamics are sharp but short-lived. Finally, our findings are robust when re-estimating the regressions from Case et al. (2012) using the extended data series, as shown in Table (4). All results are presented in Appendix (C.6).

V.B Cross-regional heterogeneity and housing supply

Our findings in Section (IV) indicated that structural regional disparities on the housing supply side are crucial in explaining why house price growth exhibits greater dispersion during economic downturns than in periods of expansion. We rely on a large empirical literature that has documented the link between regional house prices and housing supply side heterogeneities.[37] More specifically, recent research by Aastveit and Anundsen (2022) has linked regional variations in housing supply elasticities across US metropolitan areas to differential responses in house prices to monetary policy shocks. This directly relates to our mechanism.[38]

Calibrating supply side heterogeneities. To calibrate supply-side heterogeneity, we draw on time-to-build data from the U.S. Census, which measures the duration between building authorization and construction completion.[39] However, these data are only available at the Census Division level. To obtain corresponding measures at the metropolitan level, we employ the housing supply elasticities from Guren et al. (2021), aggregate them to the Census Division level, and regress them onto the Census time-to-build measures. As shown in Appendix (C.7), these variables exhibit a strong positive correlation. We then use the estimated slope coefficient from this regression to infer

[37] See, for example, Glaeser et al. (2008); Saiz (2010); Mian et al. (2013); Mian and Sufi (2014); Guren et al. (2021).

[38] Glaeser and Gyourko (2005) argue that housing supply is fixed on the downward margin, though their analysis operates in the long-run rather than business cycle frequencies. Fernandes and Rigato (2023) model time-to-build as an endogenous decision, showing that firms decelerate projects during downturns. This implies that high time-to-build regions exhibit prolonged supply responses as construction remains ongoing, while low time-to-build regions reach the extensive margin earlier, halting construction. This aligns with our narrative.

[39] We use annual averages from 1999 to 2017.

time-to-build measures at the city level. Table (C.4) shows the results. Since the final model is calibrated to quarterly frequency, we round the monthly time-to-build data to the nearest quarter.

V.C Remaining model parameters

Table (5) lists the parameter values, all of which are symmetric across regions. First, focusing on the household sector, the labor disutility shifter, the inverse Frisch elasticity, and the intertemporal elasticity of substitution are set to standard values in line with the literature. The discount factor is set to achieve a 2.0% steady-state interest rate. We further assume the regions are symmetric in size. We set housing depreciation to 2% per quarter. Moving to the production sector, we set the home bias, the elasticity of substitution across regions and goods in accordance with Bletzinger and von Thadden (2021). We will choose the persistence parameter of house price beliefs and the Kalman gain as presented in Table (3). The Taylor rule weight on inflation is 1.27, in line with Iacoviello (2005). In terms of price adjustment, we choose the price adjustment costs such that the peak response of house prices to a monetary policy shock, unconditional on booms or busts, matches the response in the data. The slope of the Phillips Curve is given by $\frac{\epsilon-1}{\kappa} = 0.006$. This is in line with the recent literature.[40]

Table (C.5) in the Appendix shows the allocation and prices in the steady state. The steady-state value for the housing preference shifter is chosen such that we attain a symmetric steady-state in the allocation variables. This modeling choice ensures that all cross-regional differences result from the structural heterogeneity on the housing supply side. The steady-state value for the house price is the only variable that differs across countries.[41] Symmetric steady-state values for bond levels, which are zero for both countries, also imply that there is no net-borrower or net-saver country in the steady-state. Changes in monetary policy will therefore not lead to Fisherian debt revaluation effects. We parameterize ξ_x to match the housing sector in the US economy and target a steady-state housing investment-to-consumption ratio, $\frac{x_{ss}}{c_{ss}}$, of 6.5% as in Adam et al. (2022).

[40] Adam and Billi (2006) set the slope of the Phillips Curve to 0.057. However, it has been argued that the slope of the Phillips Curve has decreased, see i.e Del Negro et al. (2020) and Hazell et al. (2022). Our parameter choice is in line with their estimates.

[41] Numerically, these differences are small. The region with the smallest time-to-build has a steady-state house price of 0.395, while the one with the largest time-to-build has a steady-state house price of 0.405. Accordingly, the relative differences in ξ_h are relatively small, ranging from 0.104 to 0.106.

Table 5: Model parameters (symmetric parameters)

Parameter		Value	Description	Source/ Target
Households	χ	1.000	labor disutility shifter	standard
	φ	1.250	inverse Frisch elasticity	standard
	σ	2.000	inverse of intertemporal EOS	standard
	ν	1.000	housing utility elasticity	Iacoviello (2005)
	δ	0.020	housing depreciation	2% quarterly depreciation
	β	0.995	discount factor	standard for quarterly frequency
	γ	0.500	relative region size	symmetric regions
Goods aggregation & production	λ	0.800	home bias	
	ς	1.000	EOS across regions	Bletzinger and von Thadden (2021)
	ϵ	6.000	EOS across varieties	
	κ	858.40	price adjustment costs	slope of 0.006 for the Phillips curve
	η	0.800	elasticity of housing production	Adam et al. (2022)
Policy	ϕ_π	1.270	Taylor coefficient	Iacoviello (2005)

Notes: All parameters depicted above are equal across countries. One period in the model is one quarter.

VI. Quantitative Results

In this section, we will present our findings. First, we show that the model is able to replicate aggregate house price responses to a monetary policy shock, conditional and unconditional on booms and busts. Second, the model captures the documented boom-bust-asymmetry in regional house price growth dispersion.

VI.A Aggregate house prices in booms and busts

We will start by studying the model performance on an aggregate level. All aggregate variables are a convex combination of the regional variables weighted by their size, i.e. $\widehat{q}_t^{agg} = \sum_{i=1}^{N} \gamma_i \widehat{q}_{i,t}$. We will study the aggregate house price response in the model and the data, unconditionally and conditionally on being in a boom or a bust. As discussed above, we target the peak response unconditionally of being in a boom or bust. We then study the performance of the rational expectations model and the subjective expectations model conditional on being in a boom or bust.

For the unconditional response in the model we simply consider the linear model using the

parameterization described above. The unconditional empirical response is given by estimating Equation (C.3), dropping the conditionality on the boom-bust regimes. Equation (C.3) gives the conditional empirical response. An appropriate model counterpart produces the response to a monetary policy shock conditional on being in either a boom or bust regime. To construct this conditionality in the model, we use a linear model but adjust the model parameters such that the model captures either boom or bust dynamics in house price updating. We use the parameters estimated in Table (3). Specifically, the boom calibration sets $\varrho^h = 0.99$ and $g^h = 0.013$, while the bust calibration sets $\varrho^l = 0.84$ and $g^l = 0.022$. Importantly, the respective parameters have no effect on the steady-state. Therefore, the shock will hit both models in the same steady-state but the dynamics will be different due to differential updating behavior. We consider a 25 basis points expansionary monetary policy shock on impact across all models and their empirical counterparts.[42] As discussed previously, these shocks are empirically too small to affect the transition from one regime to the other, thereby validating our quantitative exercise.

Table (6) shows the peak responses in the model and the data, unconditional and conditional on booms and busts. In the data, we see a strong response to the shock, both unconditional and conditional on boom and bust regimes. This reflects the fact that asset prices, including house prices, are generally very volatile. The rational expectations model is unable to match the magnitude of the response observed in the data. It misses the peak response in the data by roughly a factor of 3. This is not surprising, as rational expectations models generally have issues generating sufficient asset price volatility (Adam et al., 2017). Further, the response is equivalent across boom and bust periods as there is no source of asymmetry in the rational expectations model. In contrast, the subjective expectations models conditioning on booms and busts match their empirical counterparts quite well. The model peak response lies within one standard deviation of the empirical estimates for both the boom and the bust case. The results reflect our findings from Proposition (2) and (3). Under subjective expectations, the response to a monetary policy shock is scaled up compared to the rational expectations counterpart. Additionally, extrapolation further dynamically amplifies house price responses. Both of these channels seem to be very important to match the magnitude of the house price response in the data.

Figure (3) plots the IRFs of the models associated with table (6). In terms of dynamics, we typically observe a very sluggish response in the data (see for instance Appendix (C.5)).[43] The rational expectations model produces no hump-shaped dynamics, with house prices peaking on impact and returning immediately to steady state. In the subjective expectations model, we do

[42] To determine the size of 25 bp monetary policy shock in the data, we estimate a local projection exercise regressing the FFR on the monetary policy shock. We choose the minimum of the IRF in the first quarter as a reference point to determine the empirical shock size. We use this approach unconditional and conditional on booms/busts.

[43] In the case of the local projections shown in Appendix (C.5), the responses are extremely sluggish.

Table 6: Aggregate house price response to monetary policy shock at peak

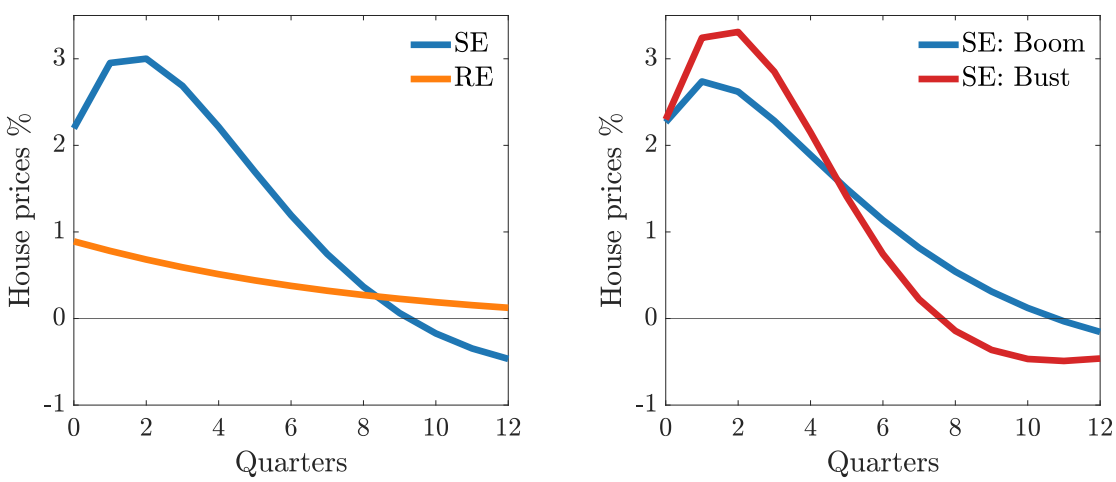
	Unconditional	Boom	Bust
Data	3.00	2.21	4.28
	(1.43, 4.57)	(0.46, 3.97)	(1.40, 4.28)
Subjective Expectations	3.00	2.74	3.31
Rational Expectations	0.89	0.89	0.89

Notes: The unconditional model uses the calibration of the linear from Table (3), Boom and Bust the respective calibrations from the same table. Confidence intervals in the data are one standard deviation from the local projections exercise.

observe some sluggishness in the house price dynamics. House prices respond on impact and increase for the first four quarters, after converging back to the steady state. The extrapolative belief structure drives this dynamic: following the initial shock, agents anticipate further price increases and invest in housing until their expectations are not met, at which point beliefs are revised and the model returns to steady state. Quantitatively, however, this persistence falls short of what is observed in the data. This is a well-known limitation of DSGE models that typically requires additional backward-looking components, such as habit formation to fully match empirical persistence (Smets and Wouters, 2007).

Figure 3: House price response to monetary policy shock

(a) Unconditional (b) Conditional



Notes: Responses to expansionary monetary policy shock (25 bp on impact).

VI.B Cross-regional heterogeneity in booms and busts

Having demonstrated that our model can replicate aggregate house price responses during both boom and bust periods, we now turn our attention to analyzing cross-regional heterogeneities. We evaluate the model's performance in response to a monetary policy shock conditional on booms and busts.

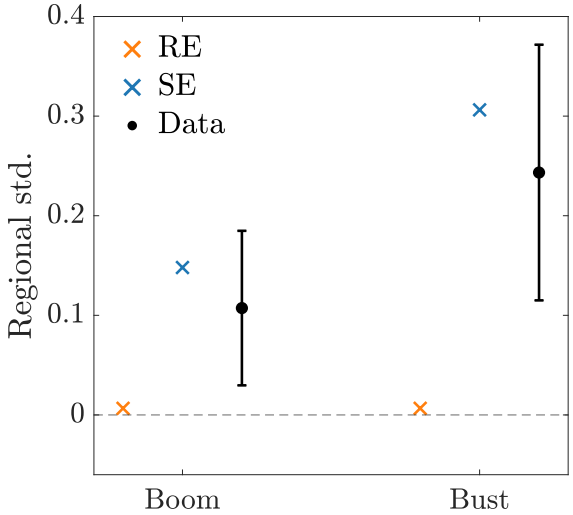
Regional house price growth dispersion. We begin by comparing the cross-regional house price growth standard deviation in the data to that in the model. Empirically, we use the local projections presented in Figure (1). As for house prices, we focus on the impulse responses in booms and busts at their respective peaks. We compare the empirically estimated peak responses with their counterparts in both a rational and a subjective expectations model, conditional on being in a boom or a bust. Figure (4) presents the results.

Focusing on the rational expectations model, we find that the model is unable to generate any sizable regional differences: The peak response is almost zero. As before, the rational expectations model is also unable to generate any differences between booms and busts. Moving to the subjective expectations model, we find that the model slightly overstates the peak response in a boom, relative to the data. However, accounting for estimation uncertainty, we are well within the one standard deviation confidence interval. Similarly, in a bust the model lies above the estimated point estimate of the local projections, but again well within the confidence intervals. Generally, our model does well in replicating the observed cross-regional variations in booms and busts, in particular in comparison to the rational expectations model.

These results are in line with Proposition (4): Housing supply-side heterogeneities create differences in house price growth, which in turn is amplified by extrapolation. Larger extrapolation in busts means stronger amplification and, therefore, larger cross-regional differences. Under rational expectations, the amplification mechanism is absent; therefore, the model is only able to create minor cross-regional differences.

Figure (5) illustrates the responses of house price levels across cities. Panel (a) presents the standard deviation of house prices across booms and busts for each metropolitan area. Consistent with our theoretical findings, the responsiveness of house prices in cities with more inelastic housing supply is stronger, leading to larger variations between booms and busts. Panel (b) compares the model-implied house price peak responses with their empirical counterparts, which are estimated using local projections and are unconditional on booms and busts. The figure reveals a strong positive correlation between the model-generated and empirical peak responses. Notably, this close correspondence arises even though time-to-build constitutes the sole source of heterogeneity in the model, indicating that it captures a key dimension of regional variation in housing markets.

Figure 4: Peak response of the standard deviation in regional house price growth to a monetary policy shock, model versus data

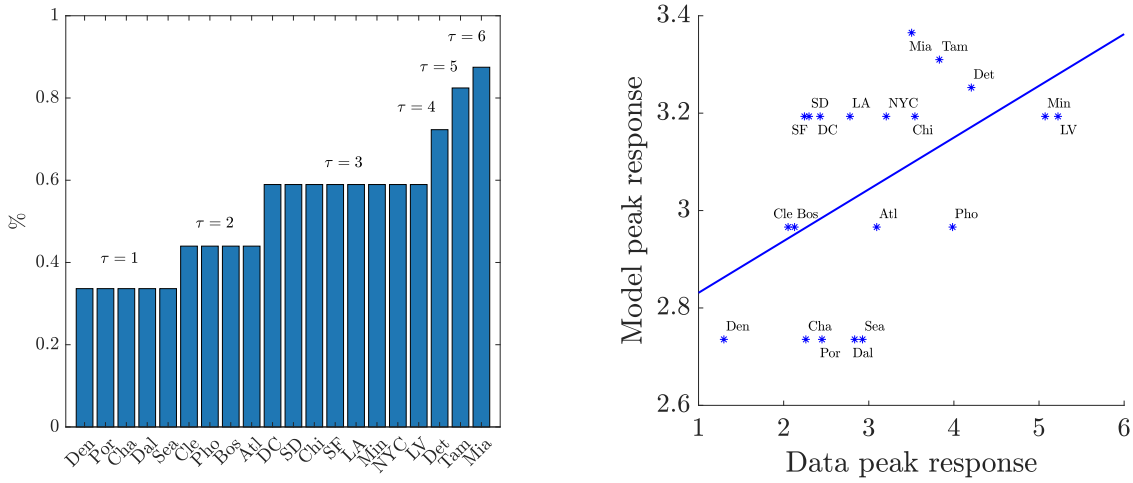


Notes: Response to a monetary policy shock (25 bp on impact); confidence intervals: 68% (Newey-West).

These results are in line with the empirical evidence reported by Aastveit and Anundsen (2022).

Figure 5: City level responses

(a) Standard deviation across Boom & Bust (b) Unconditional peak responses: Model & data

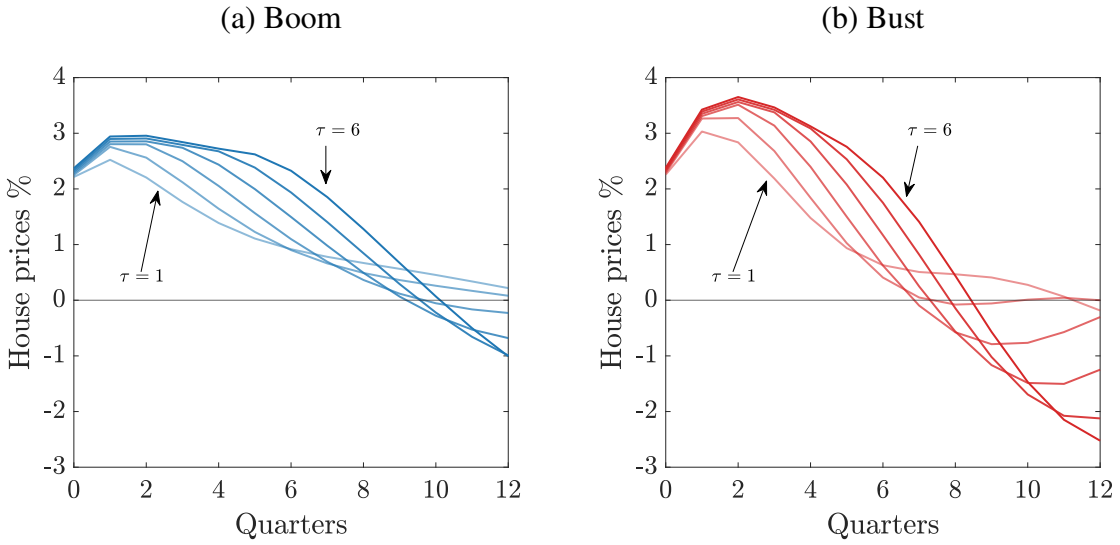


Notes: Panel(a): City level standard deviations of house prices across booms and busts to an expansionary monetary policy shock (25 bp on impact). The city name abbreviations are defined in Table (C.4). Panel (b): Peak response to the same shock in the data and model. The data response is generated by a local projections exercise. Specifically, we replace the aggregate house price in appendix (C.5) with the city counterpart unconditional on being in a boom or bust.

Figure (6) presents the corresponding impulse responses of house prices to a monetary policy shock, conditional on booms and busts. Because of the quarterly calibration, several cities share

identical time-to-build durations, leading to overlapping IRFs. The dynamics confirm that regions with longer time-to-build exhibit stronger responses, and that regional dispersion is greater in busts—consistent with the analytical results in Section (IV). For comparison, Appendix (D.1) shows that under rational expectations, the model generates no substantial regional dispersion.

Figure 6: Regional house price responses to a monetary policy shock in booms and busts



Notes: Responses to an expansionary monetary policy shock (25 bp on impact).

House prices and economic activity. Having established the boom-bust asymmetry in regional house price responses, we now examine the implications for aggregate economic activity. These dynamics are of particular interest due to their implications for welfare and, consequently, their relevance for policy considerations. House prices directly affect output through housing investment: from the investment condition (9), setting time-to-build to zero for brevity, we have:

$$\widehat{q}_{i,t} = (1 - \eta)\widehat{x}_{i,t}$$

Higher house prices translate into increased housing investment which, through market clearing, raises aggregate output. Table (7) reports the peak responses of aggregate output and its cross-regional standard deviation. Both mirror the patterns observed in house price dynamics, reflecting the spillover from housing markets to output. As these regional disparities propagate through the model, they affect consumption, labor markets, and inflation, with direct implications for welfare and monetary policy, which we discuss in the next section.

Table 7: Peak output response to monetary policy shock

	Boom	Bust
Aggregate Response	1.47	1.71
Cross − regional Variation	0.13	0.21

Notes: Responses to an expansionary monetary policy shock (25 bp on impact). The Boom and Bust models use the calibrations from Table (3). The cross-regional variation is computed using the regional standard deviation.

VII. Policy analysis

Turning to policy implications, we study how targeting house prices in the Taylor rule changes aggregate variation and cross-regional variation. The linearized Taylor rule we consider is given by:

$$i_t = \phi^{\pi} \widehat{\pi}_t^{agg} + \phi^{q} \widehat{q}_t^{agg} \tag{15}$$

For our baseline case, we consider a rule with $\phi_\pi = 1.27$ and $\phi_q = 0$, as we did for all exercises above. Under house price targeting, we increase the weight on house prices such that $\phi_q > 0$. In the following, we will first focus on how targeting house prices changes the responses of key model variables. We then turn to the welfare implications of this policy. Going forward, we study the response of house prices to an expansionary productivity shock.

Consequences of house price targeting. We assess the effects of house price targeting by comparing aggregate and cross-regional volatility of inflation, output, and house prices in response to a productivity shock, computed over the first 24 quarters after the shock.

Under subjective expectations, house price targeting reduces aggregate volatility in both output and house prices, with the effect being particularly pronounced during busts. The reduction in output volatility follows from lower fluctuations in housing investment, which dampen aggregate output. This comes at the cost of higher inflation volatility, as the greater emphasis on house prices in the policy rule reduces the relative weight placed on inflation stabilization. Under rational expectations, qualitatively similar dynamics arise, but the changes in volatility are smaller and symmetric across booms and busts, reflecting the absence of any source of asymmetry in that framework.

Turning to regional dispersion, house price targeting significantly reduces cross-regional differences in output and house prices under subjective expectations, again with stronger effects in busts. The intuition is straightforward: extrapolation amplifies house price movements that spill over to the real economy; consequently, stabilizing house prices dampens the amplifying force and thereby

also the differential regional responses they generate. The regional dispersion in inflation remains low throughout, owing to the flat Phillips curve. Under rational expectations, the model generates no meaningful regional variation to begin with, so targeting has little effect on dispersion. The results are reported in Tables (D.1) and (D.2) in the Appendix.

Welfare implications. We derive a welfare-based loss function as a second-order approximation of household utility around the efficient steady state, weighted by region size, and express welfare losses in terms of the equivalent permanent consumption decline as a fraction of steady-state consumption, following Galí (2015). This loss function can be expressed in terms of aggregate and cross-regional volatility (see Appendix (D.2) for details):

$$\frac{\mathbb{W}_{-1}}{c_{ss}} = -\frac{c_{ss}{}^{\sigma}}{2}\mathbb{E}_{-1}\sum_{t=0}^{\infty}\beta^t\left[\Omega_t^{agg} + \Omega_t^{var} + \Gamma^s\sum_{j\neq i}^{N}\widehat{s}_{ijt}^2 + O(3) + \text{t.i.p.}\right] \tag{16}$$

Ω_t^{agg} collects the aggregate variation on a currency area level and Ω_t^{var} is the cross-regional variation:

$$\Omega_t^{agg} = \Gamma^{\pi}(\widehat{\pi}_t^{agg})^2 + \Gamma^c(\widehat{c}_t^{agg})^2 + \Gamma^h(\widehat{h}_t^{agg})^2 + \Gamma^n(\widehat{n}_t^{agg})^2 + \Gamma^x(\widehat{x}_t^{agg})^2 - \Gamma^{\xi_a}\widehat{\xi}_{a,t}\widehat{n}_t^{agg}$$
$$\Omega_t^{var} = \Gamma^{\pi}\widehat{\sigma}_{\pi,t}^2 + \Gamma^c\widehat{\sigma}_{c,t}^2 + \Gamma^h\widehat{\sigma}_{h,t}^2 + \Gamma^n\widehat{\sigma}_{n,t}^2 + \Gamma^x\widehat{\sigma}_{x,t}^2 + \Gamma^b\widehat{\sigma}_{b,t}^2$$

Before we turn to the results, we briefly review the sources of inefficiency in this model. Three important inefficiencies arise. First, under subjective expectations, agents' choices are distorted due to their misspecified beliefs. In particular, households wishing to achieve capital gains in the future may be willing to shift resources from consumption units to housing demand and investment, eventually also distorting labor supply choices. In the welfare loss functions, this channel contributes to the quadratic deviations of $(\widehat{c}_t^{agg})^2$, $(\widehat{h}_t^{agg})^2$, $(\widehat{n}_t^{agg})^2$, and $(\widehat{x}_t^{agg})^2$ from their efficient benchmarks. While such allocation gaps may also arise from other frictions—such as nominal rigidities—even under rational expectations, misspecified beliefs represent an additional source of these inefficiencies.[44] Second, welfare losses arise from regional heterogeneities through the convexity of the loss function: reducing dispersion by shifting volatility from more to less volatile regions enhances aggregate welfare. This is captured by the term Ω_t^{var}. Moreover, welfare losses directly occur through deviations in relative prices, $(\widehat{s}_{ij,t})$. Finally, the model features the standard inefficiency associated with inflation, which arises from price adjustment costs.

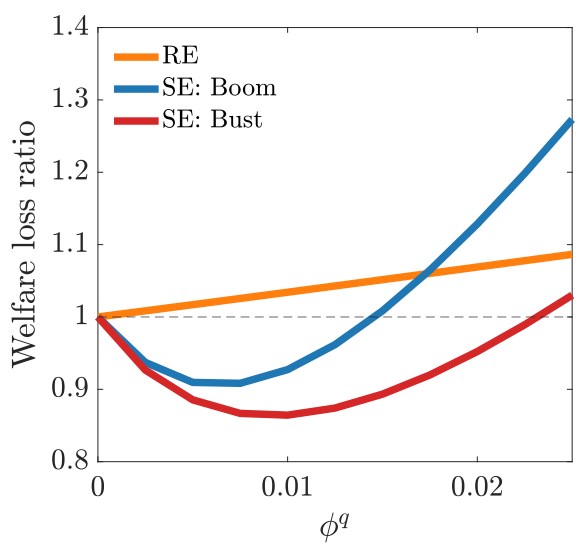
As discussed above, house price targeting reduces aggregate variance in house prices and output

[44]See Adam and Woodford (2021) for more details.

and compresses regional dispersion across all variables, but at the cost of higher inflation volatility. Under subjective expectations, this creates a trade-off: the policy corrects belief-distorted choices and mitigates regional heterogeneities, but raises inflation. Under rational expectations, both of the former two inefficiencies are either absent or negligible, leaving the policymaker with little incentive to lean against house prices.

We evaluate welfare losses under both the baseline and house price targeting regimes by computing the welfare loss ratio $\frac{\mathbb{W}(\phi^q=z)}{\mathbb{W}(\phi^q=0)}$, where z captures the degree of house price targeting. Values below unity indicate welfare improvements relative to inflation targeting. Figure (7) plots the results. Under subjective expectations, house price targeting enhances welfare, as both belief distortions and regional heterogeneities are present. The gains are particularly pronounced in busts, when stronger expectation updating amplifies both inefficiencies, and leaning against house prices dampens actual and expected price dynamics, thereby mitigating them. In contrast, under rational expectations, house price targeting does not yield welfare improvements, as the primary source of inefficiency stems solely from inflation. This finding aligns with the existing literature (Adam and Woodford, 2021; Bianchi et al., 2024b).

Figure 7: Welfare loss ratio across different degrees of house price targeting



Notes: The welfare loss ratio is defined as $\frac{\mathbb{W}(\phi^q=z)}{\mathbb{W}(\phi^q=0)}$. A welfare loss ratio smaller than unity indicates a policy that improves welfare relative to inflation targeting.

VIII. Conclusion

In this paper, we examine regional heterogeneity in housing cycles and their implications for monetary policy. We document that regional dispersion in house price growth is larger in busts

compared to booms. We then develop a two-region New Keynesian model and show that including stronger belief updating and regional housing supply side heterogeneities can jointly explain the increase of regional house price growth dispersion in busts. We empirically provide evidence that supports our modeling choices, and specifically document a higher degree of belief updating in house price busts. Our findings suggest that placing a greater emphasis on house prices by the monetary authority can reduce volatility in output and house prices, as well as regional dispersions. Under subjective beliefs, this policy proves to be welfare-improving.

References

Aastveit, K. A., B. Albuquerque, and A. K. Anundsen (2023): "Changing supply elasticities and regional housing booms," **Journal of Money, Credit and Banking**, 55, 1749–1783.

Aastveit, K. A. and A. K. Anundsen (2022): "Asymmetric Effects of Monetary Policy in Regional Housing Markets," **American Economic Journal: Macroeconomics**, 14, 499–529.

Adam, K. and R. M. Billi (2006): "Optimal monetary policy under commitment with a zero bound on nominal interest rates," **Journal of Money, credit and Banking**, 1877–1905.

Adam, K., P. Kuang, and A. Marcet (2012): "House Price Booms and the Current Account," **NBER Macroeconomics Annual**, 26, 77–122.

Adam, K. and A. Marcet (2011): "Internal Rationality, Imperfect Market Knowledge and Asset Prices," **Journal of Economic Theory**, 146, 1224–1252.

Adam, K., A. Marcet, and J. Beutel (2017): "Stock price booms and expected capital gains," **American Economic Review**, 107, 2352–2408.

Adam, K., O. Pfäuti, and T. Reinelt (2022): "Subjective Housing Price Expectations, Falling Natural Rates and the Optimal Inflation Target," .

Adam, K. and M. Woodford (2021): "Robustly optimal monetary policy in a new Keynesian model with housing," **Journal of Economic Theory**, 198, 105352.

Albuquerque, B., E. Cerutti, Y. Kido, and R. Varghese (2025): "Not all housing cycles are created equal: Macroeconomic consequences of housing booms," **Journal of International Money and Finance**, 103496.

Almgren, M., J.-E. Gallegos, J. Kramer, and R. Lima (2022): "Monetary policy and liquidity constraints: Evidence from the euro area," **American Economic Journal: Macroeconomics**, 14, 309–340.

Angeletos, G.-M., F. Collard, and H. Dellas (2020): "Business-cycle anatomy," **American Economic Review**, 110, 3030–3070.

Angeletos, G.-M., Z. Huo, and K. A. Sastry (2021): "Imperfect macroeconomic expectations: Evidence and theory," **NBER Macroeconomics Annual**, 35, 1–86.

Armona, L., A. Fuster, and B. Zafar (2019): "Home price expectations and behaviour: Evidence from a randomized information experiment," **The Review of Economic Studies**, 86, 1371–1410.

Benigno, P. (2004): "Optimal Monetary Policy in a Currency Area," **Journal of International Economics**, 63, 293–320.

Beraja, M., E. Hurst, and J. Ospina (2019): "The aggregate implications of regional business cycles," **Econometrica**, 87, 1789–1833.

Bianchi, F., C. L. Ilut, and H. Saijo (2024a): "Smooth Diagnostic Expectations," Tech. rep., National Bureau of Economic Research.

Bianchi, J., A. McKay, and N. Mehrotra (2024b): "How Should Monetary Policy Respond to Housing Inflation?" Tech. rep., Federal Reserve Bank of Minneapolis.

Bletzinger, T. and L. von Thadden (2021): "Designing QE in a fiscally sound monetary union," **European Economic Review**, 132, 103611.

Bordalo, P., N. Gennaioli, S. Y. Kwon, and A. Shleifer (2021): "Diagnostic bubbles," **Journal of Financial Economics**, 141, 1060–1077.

Bordalo, P., N. Gennaioli, and A. Shleifer (2018): "Diagnostic expectations and credit cycles," **The Journal of Finance**, 73, 199–227.

Burnside, C., M. Eichenbaum, and S. Rebelo (2016): "Understanding booms and busts in housing markets," **Journal of Political Economy**, 124, 1088–1147.

Caballero, R. J. and A. Simsek (2019): "Prudential monetary policy," Tech. rep., National Bureau of Economic Research.

——— (2020): "A risk-centric model of demand recessions and speculation," **The Quarterly Journal of Economics**, 135, 1493–1566.

Caines, C. and F. Winkler (2021): "Asset Price Beliefs and Optimal Monetary Policy," **Journal of Monetary Economics**, 123, 53–67.

Calza, A., T. Monacelli, and L. Stracca (2013): "Housing finance and monetary policy," **Journal of the European Economic Association**, 11, 101–122.

Case, K. E., R. J. Shiller, and A. Thompson (2012): "What have they been thinking? Home buyer behavior in hot and cold markets," **NBER WP 18400**.

Chen, V. W. (2019): "Fiscal multipliers and regional reallocation," .

Coibion, O. and Y. Gorodnichenko (2015): "Information rigidity and the expectations formation process: A simple framework and new facts," **American Economic Review**, 105, 2644–2678.

Corsetti, G., J. B. Duarte, and S. Mann (2022): "One money, many markets," **Journal of the European Economic Association**, 20, 513–548.

Degasperi, R. and G. Ricco (2021): "Information and policy shocks in monetary surprises," Tech. rep., Working Paper.

Del Negro, M., M. Lenza, G. E. Primiceri, and A. Tambalotti (2020): "What's up with the Phillips Curve?" Tech. rep., National Bureau of Economic Research.

Dettling, L., J. Hsu, and E. Llanes (2018): "FED Notes: A Wealthless Recovery? Asset Ownership and the Uneven Recovery from the Great Recession," Tech. rep., The Board of Governors of the Federal Reserve System.

Farhi, E. and I. Werning (2020): "Taming a Minsky cycle," **Unpublished manuscript, Harvard University**.

Fernandes, A. and R. Rigato (2023): "K wasn't built in a day: Investment with endogenous time to build," **Job Market Paper**.

Fontanier, P. (2022): "Optimal policy for behavioral financial crises," **Available at SSRN 4282972**.

Galí, J. (2015): **Monetary policy, inflation, and the business cycle: an introduction to the new Keynesian framework and its applications**, Princeton University Press.

Galí, J. and T. Monacelli (2008): "Optimal monetary and fiscal policy in a currency union," **Journal of International Economics**, 76, 116–132.

Gitti, G. (2024): "Nonlinearities in the Regional Phillips Curve with Labor Market Tightness," .

Glaeser, E. L. and J. Gyourko (2005): "Urban decline and durable housing," **Journal of political economy**, 113, 345–375.

Glaeser, E. L., J. Gyourko, and A. Saiz (2008): "Housing supply and housing bubbles," **Journal of urban Economics**, 64, 198–217.

Glaeser, E. L. and C. G. Nathanson (2017): "An extrapolative model of house price dynamics," **Journal of Financial Economics**, 126, 147–170.

Gourio, F., A. K. Kashyap, and J. W. Sim (2018): "The trade offs in leaning against the wind," **IMF Economic Review**, 66, 70–115.

Guren, A. M. (2018): "House price momentum and strategic complementarity," **Journal of Political Economy**, 126, 1172–1218.

Guren, A. M., A. McKay, E. Nakamura, and J. Steinsson (2021): "Housing wealth effects: The long view," **The Review of Economic Studies**, 88, 669–707.

Hazell, J., J. Herreno, E. Nakamura, and J. Steinsson (2022): "The slope of the Phillips Curve: evidence from US states," **The Quarterly Journal of Economics**, 137, 1299–1344.

Iacoviello, M. (2005): "House prices, borrowing constraints, and monetary policy in the business cycle," **American economic review**, 95, 739–764.

Jacobson, M. (2022): "Beliefs, aggregate risk, and the us housing boom," .

Kaplan, G., K. Mitman, and G. L. Violante (2020): "The housing boom and bust: Model meets evidence," **Journal of Political Economy**, 128, 3285–3345.

Kekre, R. (2022): "Optimal Currency Areas with Labor Market Frictions," **American Economic Journal: Macroeconomics**, 14, 44–95.

Krishnamurthy, A. and W. Li (2020): "Dissecting mechanisms of financial crises: Intermediation and sentiment," Tech. rep., National Bureau of Economic Research Cambridge.

Kuchler, T. and B. Zafar (2019): "Personal experiences and expectations about aggregate outcomes," **The Journal of Finance**, 74, 2491–2542.

L'Huillier, J.-P., S. R. Singh, and D. Yoo (2023): "Incorporating diagnostic expectations into the New Keynesian framework," **Review of Economic Studies**, rdad101.

Ma, C. (2020): "Momentum and Reversion to Fundamentals: Are They Captured by Subjective Expectations of House Prices?" **Journal of Housing Economics**, 49, 101687.

Maxted, P. (2024): "A macro-finance model with sentiment," **Review of Economic Studies**, 91, 438–475.

Mian, A., K. Rao, and A. Sufi (2013): "Household balance sheets, consumption, and the economic slump," **The Quarterly Journal of Economics**, 128, 1687–1726.

Mian, A. and A. Sufi (2014): "What explains the 2007–2009 drop in employment?" **Econometrica**, 82, 2197–2223.

——— (2015): **House of Debt: How They (and You) Caused the Great Recession, and How We Can Prevent it from Happening Again**, University of Chicago Press.

Pfäuti, O. (2025): "Inflation—who cares? Monetary policy in times of low attention," **Journal of Money, Credit and Banking**, 57, 1211–1239.

Pica, S. (2021): "Housing Markets and the Heterogeneous Effects of Monetary Policy Across the Euro Area," **Available at SSRN**.

Saiz, A. (2010): "The geographic determinants of housing supply," **The Quarterly Journal of Economics**, 125, 1253–1296.

Schmitt, N. and F. Westerhoff (2019): "Short-run momentum, long-run mean reversion and excess volatility: an elementary housing model," **Economics Letters**, 176, 43–46.

Schmitt-Grohé, S. and M. Uribe (2003): "Closing small open economy models," **Journal of International Economics**, 61, 163–185.

Slacalek, J., O. Tristani, and G. L. Violante (2020): "Household balance sheet channels of monetary policy: A back of the envelope calculation for the euro area," **Journal of Economic Dynamics and Control**, 115, 103879.

Smets, F. and R. Wouters (2007): "Shocks and frictions in US business cycles: A Bayesian DSGE approach," **American economic review**, 97, 586–606.

Stein, J. C. (2012): "Monetary policy as financial stability regulation," **The Quarterly Journal of Economics**, 127, 57–95.

Svensson, L. E. (2017): "Cost-benefit analysis of leaning against the wind," **Journal of Monetary Economics**, 90, 193–213.

Weidmann, J. (2018): "The relationship between monetary and macroprudential policies-black and white or shades of grey?: dinner speech at the annual meeting of the cebra, frankfurt am main," .

Winkler, F. (2020): "The role of learning for asset prices and business cycles," **Journal of Monetary Economics**, 114, 42–58.

Online Appendix

A. Micro-founding the debt-elastic interest rate

Without loss of generality, we focus in this section on a two-region case, a home country H and a foreign country F, with all variables for the foreign region denoted by an asterisk. In the model, households in country H receive on their bond holdings the effective nominal interest rate $1 + i_{t-1} - \psi b_t$, with b_t being the real value of the aggregate bond holding in country H; households in country F receive the effective nominal rate $1 + i_{t-1} - \psi b_t^*$. Moreover, the intermediation of bond positions entails a real cost $\gamma(1+\pi_t)^{-1}\frac{\psi}{2}(b_t)^2 + (1-\gamma)(1+\pi_t)^{-1}\frac{\psi}{2}(b_t^*)^2$, of which $(1+\pi_t)^{-1}\frac{\psi}{2}(b_t)^2$ is paid by each consumer in H and $(1+\pi_t)^{-1}\frac{\psi}{2}(b_t)^2$ is paid by each consumer in F. In this appendix, we detail how these debt-elastic interest rates and associated intermediation costs can be parsimoniously micro-founded. We achieve this by introducing two competitive bond clearing houses, one in each country, that represent households' only access to financial markets and where they incur a real cost that is quadratic in the size of their balance sheet. The specific market arrangement is as follows: Households hold a consol and may hold liquid bonds.

Consol. Each household in H is endowed with $\bar{b} \in \mathbb{R}$ units of a non-marketable consol[45] that pays as a coupon $(\beta^{-1}-1)(\gamma P_{t-1} + (1-\gamma)P_{t-1}^*)P_t^{-1}$ units of H's consumption basket each period, per unit of consol. This implies that the nominal coupon rate, applied to the nominal coupon value $P_{t-1}\bar{b}$, is $(\beta^{-1}-1)(\gamma P_{t-1} + (1-\gamma)P_{t-1}^*)/P_{t-1}$; the real coupon rate applied to the real value $\bar{b}$, in turn, is $(\beta^{-1}-1)\left(\gamma + (1-\gamma)\frac{P_{t-1}^*}{P_{t-1}}\right)(1+\pi_t)^{-1}$. The situation in country F is symmetric: Each household is endowed with $\bar{b}^*$ units of a consol that pays $(\beta^{-1}-1)(\gamma P_{t-1} + (1-\gamma)P_{t-1}^*)(P_t^*)^{-1}$ units of F's consumption basket each period, per unit of consol. $\bar{b}$, $\bar{b}^*$ are model parameters selected such that (i) $\gamma\bar{b} + (1-\gamma)\bar{b}^* = 0$ and (ii) all markets clear in the non-stochastic steady state with zero net inflation and terms-of-trade parity *without* households holding any liquid bonds. The latter fact ensures that there are no costs of financial intermediation in the steady state, shutting down this particular friction. The specific choice of the coupon payment scheme ensures two facts: (1) condition (i) implies that the nominal payments between H and F associated with the two consols exactly cancel out–whatever H/F receives as coupon payments on its consol endowment is paid for by F/H as a coupon service on its (endowed) short position of the consol; and (2) the real coupon rates paid by/ to the consol endowment only depend on the real exchange rate and the inflation rates and not on the price levels. Households cannot trade their consol holdings.

[45] A consol is a type of bond that has infinite maturity and perpetually pays a constant or varying coupon rate.

Bonds. Households do have the possibility, however, to vary their position in the liquid bond. This liquid bond is a nominal, one-period, zero-coupon bond and the positions of the representative H-, respectively, the representative F-household, are denominated as b_t, b_t^*. If a household wants to hold a net balance of liquid bonds different from zero, they must go to one of the clearing houses in their country: In the H-country, there is a continuum of mass γ (respectively mass $1-\gamma$ in F) of competitive clearing houses buying and selling bonds from and to the government and the respective country's citizens. Households themselves cannot directly buy/sell government bonds without having an account at the clearing house. The clearing house can costlessly buy/sell bonds but it incurs an operating cost that is quadratic in the size of its balance sheet, making this a model of costly financial intermediation. Thus, the interest rate that each citizen receives on their bond holdings is determined by the nominal rate paid on government bonds and the aggregate holding of liquid bonds. Each clearing house is owned equally by all citizens of the respective country so that it pays its profits to those citizens.[46] Consider an arbitrary clearing house in H (with symmetric arrangements in F). Denoting as $B_{c,t+1}$ the nominal value of the clearing house's net liabilities against H's citizens and as $B_{g,t+1}$ the nominal value of the clearing house's position in the government bond, the profit maximization program is

$$\max_{B_{c,t+1},B_{g,t+1}\in\mathbb{R}} -(1+i_t^b)B_{c,t+1} + (1+i_t)B_{g,t+1} - \frac{\psi}{2}P_t^{-1}(B_{c,t+1})^2, \text{ s.t. } B_{c,t+1} = B_{g,t+1},$$

where i_t^b is the nominal rate clearing of the market for household bond positions and i_t is the nominal rate the monetary authority sets on government bonds. $\frac{\psi}{2}P_t^{-1}(B_{c,t+1})^2 = \frac{\psi}{2}P_t(b_{c,t+1})^2$ is the nominal cost of intermediating–crucially, the real cost of intermediation does not directly depend on the price level. The first order conditions for this program are

$$\begin{aligned} 1 + i_t^b + \psi P_t^{-1} B_{c,t+1} &= \mu_t, \\ 1 + i_t &= \mu_t, \\ B_{c,t+1} &= B_{g,t+1}, \end{aligned}$$

[46] In equilibrium, each clearing house makes a non-negative profit, and along the transition path back to the steady state after some shock, each clearing house makes a strictly positive profit. This fact is, in principle, incompatible with the notion of competitiveness (there is an incentive to open up more clearing houses or, equivalently, it is strictly profitable to split each clearing house). Therefore, it is better to interpret the clearing house's program as reflecting the capacity constraints: The program presented here can be thought of as the inner problem of a profit maximization program with an additional factor (say, managerial effort), that which (i) makes the intermediation service production function exhibit constant returns to scale (instead of decreasing RTS), (ii) is provided by households, and (iii) is in perfectly inelastic supply. When the clearing house is modeled this way, it behaves exactly as modeled here: In the current way of modelling, the clearing house always makes zero profits and the household's remuneration for providing the additional factor is the profit .

where μ_t is the Lagrange multiplier on the balance-sheet constraint $B_{c,t+1} = B_{g,t+1}$. Market clearing in the household's bond positions in H requires

$$\gamma B_{c,t+1} = \gamma P_t b_{t+1},$$

and market clearing in the government's bond positions requires

$$\gamma B_{g,t+1} + (1-\gamma) B^*_{g,t+1} = 0,$$

so that by using the balance-sheet constraints $B_{c,t+1} = B_{g,t+1}, B^*_{c,t+1} = B^*_{g,t+1}$ and the clearing conditions for household bond positions in H and F, we recover the market clearing condition for government bonds in the main model:

$$\gamma P_t b_{t+1} + (1-\gamma) P^*_t b^*_{t+1} = 0.$$

In sum, the aggregate conditions implied by this market arrangement are

$$\begin{aligned} 1 + i^b_t + \psi b_{t+1} &= 1 + i_t, \\ 1 + i^{b,*}_t + \psi b^*_{t+1} &= 1 + i_t, \\ \gamma P_t b_{t+1} + (1-\gamma) P^*_t b^*_{t+1} &= 0. \end{aligned}$$

The nominal profits of the typical clearing house in H in equilibrium are

$$\begin{aligned} \text{Profit}_{t+1} &= (i_t - i^b_t) B_{c,t+1} - \frac{\psi}{2} P_t^{-1} (B_{c,t+1})^2 \quad \text{with optimal } B_{c,t+1} = \frac{i_t - i^b_t}{\psi} P_t \\ &= (i_t - i^b_t)^2 \psi^{-1} P_t - (i_t - i^b_t)^2 \psi^{-1} P_t \cdot \frac{1}{2} \\ &= P_t \frac{\psi}{2} (b_{t+1})^2, \quad \text{using market clearing in the household bond positions.} \end{aligned}$$

Of the $1 + i_t\%$ nominal interest collected on its (paid for) position of government bonds, each clearing house withholds $\psi b_{t+1}\%$ of the interest from its customers (respectively, charges $-\psi b_{t+1}\%$ of additional interest if $b_{t+1} < 0$). Half of these $\psi b_{t+1}\%$ are used for covering the operating cost (by buying this amount of H's final basket and selling it in exchange for the numéraire), and the other half is paid as profit to the owners of the clearing house (which, in equilibrium, are its customers).

B. Proofs and Derivations

B.1 Proof of Lemma (1)

Without loss of generality, we drop the region-specific indexation in the following. The agents apply the optimal Bayesian filter, that is, the Kalman filter, to arrive at the observable system:[47]

$$\begin{aligned}\ln \frac{q_{t+1}}{q_t} &= \varrho \ln \overline{m}_t + \ln \widehat{e}_{t+1} \\ \ln \overline{m}_t &= \varrho \ln \overline{m}_{t-1} - \frac{\sigma_v^2}{2} + g \cdot \left(\ln \widehat{e}_t + \frac{\sigma_e^2 + \sigma_v^2}{2} \right),\end{aligned}$$

where $\ln \overline{m}_t := \mathbb{E}_t^{\mathcal{P}}(\ln m_t)$ is the posterior mean, $g = \frac{\varrho^2\sigma^2+\sigma_v^2}{\varrho^2\sigma^2+\sigma_v^2+\sigma_e^2}$ is the steady-state Kalman filter gain, $\sigma^2 = \frac{1}{2\varrho^2}[-(\sigma_v^2 + (1-\varrho^2)\sigma_e^2) + \sqrt{(\sigma_v^2 + (1-\varrho^2)\sigma_e^2)^2 + 4\varrho^2\sigma_v^2\sigma_e^2}]$ is the steady-state Kalman filter uncertainty, and $\ln \widehat{e}_t$ is perceived to be a white noise process.

To avoid simultaneity in the house price, we modify the belief setup following Adam et al. (2017).[48] We obtain the same observable system but with lagged information being used in the posterior mean updating equation:

$$\ln \overline{m}_t = \varrho(1-g)\left(\ln \overline{m}_{t-1} - \frac{\sigma_v^2}{2} \right) + g \left(\ln \frac{q_{t-1}}{q_{t-2}} + \frac{\sigma_e^2}{2} \right). \tag{B.1}$$

Under this formulation, the posterior mean is predetermined. We may now derive the posterior mean on the $s > 0$ periods-ahead of price:

$$\mathbb{E}_t^{\mathcal{P}} q_{t+s} = q_t \cdot \exp\left(\ln \overline{m}_t \cdot \varrho \tfrac{1-\varrho^s}{1-\varrho} + \frac{1}{2}\sigma^2 \left(\varrho \tfrac{1-\varrho^s}{1-\varrho} \right)^2 \right) \cdot \exp(V), \quad V \propto \sigma_v^2 \tag{B.2}$$

For the derivation of equation(B.2) see Appendix (B.2). This completes the proof.

[47] We assume the the agents' prior variance equals the steady-state Kalman variance.

[48] q_t appears twice: in the forecast equation, and in the Kalman-updating Equation through $\ln \widehat{e}_t$. Since q_t depends on $\overline{m}_t$, but the latter also depends on the former, it is not assured that at any point an equilibrium asset price exists and whether it is unique. See Adam et al. (2017) for the details. The idea of the modification is to alter the agents' perceived information setup in that they observe each period one component of the lagged transitory price growth.

B.2 Derivation of Equation (B.2)

Without loss of generality, we drop the region-specific indexation in the following. Equation (B.2) is the result of the following calculations:

$$
\begin{aligned}
\mathbb{E}_t^{\mathcal{P}} q_{t+s} &= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \frac{q_{t+s}}{q_t} = q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp\left(\ln q_{t+s} - \ln q_t \right) = q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp\left(\sum_{n=1}^{s} \Delta \ln q_{t+n} \right) \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp\left(\sum_{n=1}^{s} \ln m_{t+n} \right) \cdot \underbrace{\mathbb{E}_t^{\mathcal{P}} \left[\prod_{n=1}^{s} e_{t+n} \right]}_{= \prod_n \mathbb{E}_t^{\mathcal{P}} e_{t+n} = 1} \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp\left(\sum_{n=1}^{s} \left[\sum_{j=0}^{n-1} \varrho^j \ln v_{t+n-j} + \varrho^n \ln m_t \right] \right) \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp\left(\ln m_t \cdot \sum_{n=1}^{s} \varrho^n \right) \cdot \mathbb{E}_t^{\mathcal{P}} \exp\left(\underbrace{\sum_{n=1}^{s} \sum_{j=0}^{n-1} \varrho^j \ln v_{t+n-j}}_{\sim \mathcal{N}} \right) \\
&= q_t \cdot \mathbb{E}_t^{\mathcal{P}} \exp\left(\varrho \frac{1-\varrho^s}{1-\varrho} \ln m_t \right) \cdot \exp(V), \quad V \propto \sigma_v^2 \\
\iff \mathbb{E}_t^{\mathcal{P}} q_{t+s} &= q_t \cdot \exp\left(\ln \overline{m}_t \cdot \varrho \tfrac{1-\varrho^s}{1-\varrho} + \frac{1}{2} \sigma^2 \left(\varrho \tfrac{1-\varrho^s}{1-\varrho} \right)^2 \right) \cdot \exp(V), \quad V \propto \sigma_v^2
\end{aligned}
$$

B.3 Aggregation and market clearing

To achieve goods market clearing, each goods market for a variety k must clear. $\Psi_{i,t} := (1 + \pi_t)^{-1} \frac{\psi}{2} (b_{i,t})^2$ is the real cost of intermediating the position of an i-citizen in the union-wide bond. This cost, just like consumption and housing investment, gets passed along down to the varieties $\Psi_{i,t} := \left(\frac{P_{ii,t}}{P_{i,t}}\right)^{-\frac{1}{\varsigma}} \lambda \Psi_t$. Goods market clearing across all goods markets requires

$$
\int y_{it}(k) dk = \gamma_i \Big(\int [c_{iit}(k) + x_{iit}(k) + \Psi_{iit}(k)] dk + \sum_{j \neq i}^{N} \gamma_j \int [c_{jit}(k) + \Psi_{jit}(k)] dk \Big) + \int \Phi_t(k) dk
$$

where $\Phi_t(k) = \left(\frac{P_{ii,t}(j)}{P_{i,t}} \right)^{-\epsilon} \Phi_t$ and $\Phi_t = \frac{\kappa}{2} \left(\Pi_{ii,t} - 1 \right)^2 y_{ii,t}$ account for the price adjustment costs from the firm side. Since housing investment requires domestically produced goods, we have that $x_i = x_{ii,t}$. Aggregation and successive substitution eventually yield the domestic and foreign

aggregate goods market clearing conditions:

$$\gamma_i\Big(1-\frac{\kappa}{2}(\Pi_{ii,t}-1)^2\Big)y_{i,t}=\gamma_i(\frac{P_{ii,t}}{P_{i,t}})^{(-1/\varsigma)}\lambda_{ii}(c_{i,t}+\psi_{i,t})+\gamma_i x_{i,t}+\sum_{j\neq i}\gamma_j(\frac{P_{ii,t}}{P_{j,t}})^{(-1/\varsigma)}\lambda_{ji}(c_{j,t}+\psi_{j,t})$$

Further, the bond market clearing condition is given by

$$\sum_{j}^{N}\gamma_j P_{j,t}b_{j,t+1}=0.$$

Market clearing in the housing sectors for any region i is given by

$$H(x_{i,t-\tau})=(h_{i,t}-(1-\delta)h_{i,t-1})$$

Finally, the household budget constraints are satisfied.

B.4 Proof of Lemma 2

Without loss of generality, we focus in this section on a two-region case, a home country H and a foreign country F, with all variables for the foreign region denoted by an asterisk. For simplicity we also assume that time-to-build is zero in both regions. We prove that there exists a unique non-stochastic steady state with zero net inflation and parity in the terms of trade $P_{H,t}/P_{F,t}$ (which implies parity in the real exchange rate P_t/P_t^*, see the definition of $\mathfrak{p}$).[49] We prove this first for arbitrary parameters $\eta,\eta^*,\xi_{h,ss},\xi_{x,ss},\xi^*_{h,ss},\xi^*_{x,ss}$ and then show that it is possible to select parameters such that the steady-state allocation is symmetric. The non-stochastic steady state with zero net inflation and real exchange rate parity ("SS" for short) is obtained by setting $\mathrm{Var}[\|\xi_t\|]=0$, where the shock vector contains both actual shocks and shocks that are only perceived (and never observed) by the household within their perceived house price model: $\xi_t=(\xi_{a,t},\xi_{c,t},\xi_{h,t},\xi_{x,t},\xi_{i,t},e_t,v_t)^\top$. Thus, the non-stochastic steady state represents the time-invariant equilibrium that obtains if the agents do not expect any shock to ever materialize and indeed no shock ever does materialize, and we have $\xi_t=(1,1,\xi_{h,ss},\xi_{x,ss},1,1,1)^\top$ almost surely where $\xi_{h,ss},\xi_{x,ss}$ are the model parameters. We are interested in, and prove the existence and uniqueness of a non-stochastic steady state in which the net rates of inflation are zero and the terms of trade are at parity. The latter assumption implies $\mathfrak{p}_H(1)=\mathfrak{p}^*_H(1)=\mathfrak{p}_F(1)=\mathfrak{p}^*_F(1)=\mathfrak{p}(1)=1$.

[49] With H being region i and F being region j. Define $P_{H,t}=P_{ii,t},P_{F,t}=P_{ji,t},P_t=P_{i,t},P^*_t=P_{j,t}$. The relative prices are denoted by $\mathfrak{p}_{H,t}=\frac{P_{H,t}}{P_t},\mathfrak{p}_{F,t}=\frac{P_{F,t}}{P_t},\mathfrak{p}^*_{H,t}=\frac{P_{H,t}}{P^*_t},\mathfrak{p}^*_{F,t}=\frac{P_{F,t}}{P^*_t},\mathfrak{p}_t=\frac{P_t}{P^*_t}$.

Beliefs are irrelevant in the non-stochastic steady state. The first relevant insight is that in the non-stochastic steady state, defined as above, the presence and precise parameterization of subjective beliefs over house prices is irrelevant. To see this, recall that the subjective house price expectation dynamics are fully characterized by

$$\begin{aligned}
\forall s > 0,\ \mathbb{E}_t^{\mathcal{P}} q_{t+s} &= q_t \cdot \exp\left(\ln \overline{m}_t \cdot \varrho \tfrac{1-\varrho^s}{1-\varrho} + \frac{1}{2}\sigma^2 \left(\varrho \tfrac{1-\varrho^s}{1-\varrho}\right)^2 \right) \cdot \exp(V), \quad V \propto \sigma_v^2 \\
\ln \overline{m}_t &= (1-g)\left(\ln \overline{m}_{t-1} - \frac{\sigma_v^2}{2} \right) + g \left(\ln \frac{q_{t-1}}{q_{t-2}} + \frac{\sigma_e^2}{2} \right), \\
&+ \text{equations determining the equilibrium evolution of the price level, } q_t.
\end{aligned} \tag{B.3}$$

Now recall that the non-stochastic steady state represents the unique equilibrium of the economy when the variances of actual and perceived external shocks tend to zero and the initial conditions are selected such that they give rise to a constant path of equilibrium variables. Formally, the SS arises by replacing $\mathrm{Var}[\|\xi_t\|]$ with $\wp^2 \cdot \mathrm{Var}[\|\xi_t\|]$ in the model, taking the limit $\wp \to 0$ and solving for the fixed point of the equilibrium equations. Applying this logic, $\sigma_e, \sigma_v, \sigma \to 0$, to the equations (B.3) delivers

$$\begin{aligned}
q_{ss} &= q_{ss} \cdot \exp\left(\ln \overline{m}_{ss} \cdot \varrho \tfrac{1-\varrho^s}{1-\varrho} + 0 \right) \cdot \exp(0) \\
\ln \overline{m}_{ss} &= (1-g) \ln \overline{m}_{ss} + g \cdot 0,
\end{aligned}$$

from which we see that there exists a unique fixed point, which is $\overline{m}_{ss} = 1$, regardless of the level of q_{ss}. This means that the presence of subjective beliefs has no consequences for the non-stochastic steady state provided the agents have the prior that house prices do not change, $\overline{m}_{ss} = 1$, and that the absence of perceived and actual shocks means that house prices indeed never change, which in turn means that the prior of constant house prices is never changed.

In summary, if a non-stochastic steady state with zero net inflation and terms of trade parity exists and is unique, then it solves the following system of equations.

Household

$$\begin{aligned}
&(h) \quad q_{ss} = \xi_{h,ss} h_{ss}^{-\nu} c_{ss}^{\sigma} + \beta(1-\delta) q_{ss}, \\
&(n) \quad \chi n_{ss}^{\varphi} c_{ss}^{\sigma} = w_{ss}, \\
&(b) \quad 1 = \beta(1 + r_{ss}), \\
&(x) \quad 1 = q_{ss} \cdot \xi_{x,ss} x_{ss}^{\eta-1},
\end{aligned} \tag{B.4}$$

Household*

$(h^*)\quad q_{ss} = \xi^*_{h,ss}(h^*_{ss})^{-\nu}(c^*_{ss})^{\sigma} + \beta(1-\delta)q^*_{ss},$

$(n^*)\quad \chi(n^*_{ss})^{\varphi}(c^*_{ss})^{\sigma} = w^*_{ss},$

$(b^*)\quad 1 = \beta(1 + r^*_{ss}),$

$(x^*)\quad 1 = q^*_{ss} \cdot \xi^*_{x,ss}(x^*_{ss})^{\eta-1},$

Interest rates

$(r)\quad 1 + r_{ss} = 1 + i_{ss} - \psi b_{ss},$

$(r^*)\quad 1 + r^*_{ss} = 1 + i_{ss} - \psi b^*_{ss},$

Firm

$(PC)\quad w_{ss} = 1 \quad$ using that $\tau^{\ell} = 1/\epsilon,$

Firm*

$(PC^*)\quad w^*_{ss} = 1 \quad$ using that $\tau^{\ell} = 1/\epsilon,$

Bond market clearing

$(B)\quad \gamma b_{ss} + (1-\gamma)b^*_{ss} = 0,$

Goods market clearing

$(GMC)\quad n_{ss}\gamma = \gamma\lambda \mathrm{y}_{ss} + (1-\gamma)(1-\lambda^*)\mathrm{y}^*_{ss},$

$(GMC^*)\quad n^*_{ss}(1-\gamma) = \gamma(1-\lambda)\mathrm{y}_{ss} + (1-\gamma)\lambda^*\mathrm{y}^*_{ss},$

$(BOP)\quad \gamma(1-\lambda)\mathrm{y}_{ss} - (1-\gamma)(1-\lambda^*)\mathrm{y}^*_{ss} - \gamma(i_{ss}b_{ss} + (\beta^{-1}-1)\bar{b}) = 0,$

Housing market clearing

$(HMC)\quad \eta^{-1}\xi_{x,ss}(x_{ss})^{\eta} = \delta h_{ss},$

$(HMC^*)\quad (\eta^*)^{-1}\xi^*_{x,ss}(x^*_{ss})^{\eta^*} = \delta h^*_{ss},$

where $\bar{b}, \bar{b}^*$ are the model parameters (see Appendix A for an interpretation) chosen so as to (i) ensure equilibrium existence, and (ii) ensure that (§1) $\gamma\bar{b} + (1-\gamma)\bar{b}^* = 0$. We solve for the SS in 4 steps:

1. First, we explicitly solve a number of equations, thus, substituting out a number of variables:

   (a) (b) and (b^*) together with (r) and (r^*) and (B) imply $b_{ss} = b^*_{ss} = 0$ and $i_{ss} = \beta^{-1} - 1$, hence it follows that $\mathrm{y}_{ss} = c_{ss} + x_{ss}$ and analogously *. (§1) then implies $\bar{b} = -\bar{b}^* \cdot (1-\gamma)/\gamma$, where $\bar{b}^*$ is not pinned down yet. We solve for this in the very last step.

   (b) (PC) and (PC^*) imply $w_{ss} = w^*_{ss} = 1$;

(c) together with (n) and (n^*), this implies $n_{ss} = (\chi^{-1})^{1/\varphi} \cdot c_{ss}^{-\sigma/\varphi} =: \phi(c_{ss})$ with $\phi' < 0$ and analogously for n_{ss}^*, with the *same* function ϕ;

(d) (HMC) and (HMC^*) imply $h_{ss} = (\delta)^{-1}\xi_{x,ss}\eta^{-1}x_{ss}^{\eta}$ and analogously for h_{ss}^*;

(e) (h) and (h^*) imply (with $\bar{\beta} := \beta(1-\delta)$)

$$q_{ss} = \left(1-\bar{\beta}\right)^{-1} \xi_{h,ss} \left[\delta^{-1}\xi_{x,ss}\eta^{-1}x_{ss}^{\eta}\right]^{-\nu} c_{ss}^{\sigma}$$
$$q_{ss}^* = \left(1-\bar{\beta}\right)^{-1} \xi_{h,ss}^* \left[\delta^{-1}\xi_{x,ss}^*(\eta^*)^{-1}\left(x_{ss}^*\right)^{\eta^*}\right]^{-\nu} (c_{ss}^*)^{\sigma}.$$

(f) (BOP) now reads $\gamma(1-\lambda)\mathrm{y}_{ss} = (1-\gamma)(1-\lambda^*)\mathrm{y}_{ss}^* - (1-\gamma)(\beta^{-1}-1)\bar{b}^*$, and the symmetric (BOP^*), which is redundant by Walras's law, reads $(1-\gamma)(1-\lambda^*)y_{ss}^* = \gamma(1-\lambda)y_{ss} + (1-\gamma)(\beta^{-1}-1)\bar{b}^*$; using this in $(GMC), (GMC^*)$ produces

$$(GMC) \quad \phi(c_{ss}) = \mathrm{y}_{ss} + (1-\gamma)/\gamma \cdot (\beta^{-1}-1)\bar{b}^*,$$
$$(GMC^*) \quad \phi(c_{ss}^*) = \mathrm{y}_{ss}^* - (\beta^{-1}-1)\bar{b}^*.$$

2. The remaining equations are (x), (x^*) and (GMC), (GMC^*), (BOP) with unknowns $x_{ss}, x_{ss}^*, c_{ss}, c_{ss}^*, \bar{b}^*$. In this step, we show there are strictly increasing functions that yield x_{ss}, x_{ss}^*, given c_{ss}, c_{ss}^*, respectively. We start by plugging q_{ss} into (x):

$$\left(1-\bar{\beta}\right)^{-1} \xi_{h,ss} \left[\delta^{-1}\xi_{x,ss}\eta^{-1}x_{ss}^{\eta}\right]^{-\nu} c_{ss}^{\sigma}\xi_{x,ss}x_{ss}^{\eta-1} = 1$$

(the equation for * is symmetric.) Now since $\eta \in (0,1)$ and $\nu > 0$, the expression on the left-hand side is a strictly decreasing function of x_{ss} for any c_{ss}. Moreover, for $x_{ss} \to 0$, the $LHS \to +\infty$ and for $x_{ss} \to \infty$, the $LHS \to 0$, hence Bolzano's intermediate value theorem (and continuity) ensures that for each c_{ss} there exists a unique x_{ss}. We call this implicitly defined mapping $x_{ss} = \psi(c_{ss})$. As the implicit function theorem shows, $\eta, \nu, \sigma > 0$ imply $\psi' > 0$. Analogous arguments hold for *.

3. We now insert our previous findings into the only remaining equations:

$$(GMC) \quad \phi(c_{ss}) - c_{ss} - \psi(c_{ss}) - (1-\gamma)/\gamma \cdot (\beta^{-1}-1)\bar{b}^* =: \zeta(c_{ss}, \bar{b}^*) = 0,$$
$$(GMC^*) \quad \phi(c_{ss}^*) - c_{ss}^* - \psi^*(c_{ss}^*) + (\beta^{-1}-1)\bar{b}^* =: \zeta^*(c_{ss}^*, \bar{b}^*) = 0,$$

Observe, now, that $c_{ss} \mapsto \zeta$ is continuous and strictly decreasing with $\lim_{c\to 0}\zeta = +\infty$ (by $\lim_{c\to 0}\phi = +\infty$ and $\lim_{c\to 0}\psi < +\infty$) and $\lim_{c\to +\infty}\zeta = -\infty$ (by $\lim_{c\to +\infty} c, \psi = +\infty$ and $\lim_{c\to\infty}\phi = 0$). Therefore, Bolzano's intermediate value theorem ensures that there exists a

unique c_{ss} for each $\bar{b}^*$. The exactly analogous argument ensures the existence and uniqueness of c^*_{ss}. We call these mappings $\varpi : \bar{b}^* \mapsto c_{ss}, \varpi^* : \bar{b}^* \mapsto c^*_{ss}$. The implicit function theorem now yields

$$\partial\varpi/\partial\bar{b}^* < 0 \text{ and } \partial\varpi^*/\partial\bar{b}^* > 0.$$

4. Finally, only one equation remains, (BOP), with only one variable, $\bar{b}^*$:

$$\begin{aligned}\gamma(1-\lambda)[\varpi(\bar{b}^*)+\psi(\varpi(\bar{b}^*))] - (1-\gamma)(1-\lambda^*)[\varpi^*(\bar{b}^*)+\psi^*(\varpi^*(\bar{b}^*))] \\ + (1-\gamma)(\beta^{-1}-1)\bar{b}^* =: \mathcal{H}(\bar{b}^*) = 0\end{aligned}$$

with $\bar{b}^* \mapsto \mathcal{H}$ continuous. It also holds that $\lim_{\bar{b}^*\to-\infty}\mathcal{H}(\bar{b}^*) = -\infty$.[50] On the other hand, as $\bar{b}^* \to +\infty$, $\mathcal{H} \to +\infty$.[51] Thus, Bolzano's intermediate value theorem ensures the **existence** of a $\bar{b}^* \in \mathbb{R}$, which satisfies the BOP-equation and, thus, the existence of a non-stochastic steady state.

Uniqueness of the steady-state can be shown by establishing the strict positive monotonicity of $\mathcal{H}$: (we suppress our arguments, for brevity)

$$\begin{aligned}
\frac{\partial\mathcal{H}}{\partial\bar{b}^*} &= \gamma(1-\lambda)(1+\psi')\varpi' - (1-\gamma)(1-\lambda^*)(1+\psi^{*\prime})\varpi^{*\prime} \\
&\quad + (1-\gamma)(\beta^{-1}-1) \\
\text{step 2, cf. notes below:} \quad &> -(1-\gamma)(1-\lambda)(\beta^{-1}-1) - (1-\gamma)(1-\lambda^*)(\beta^{-1}-1) \\
&\quad + (1-\gamma)(\beta^{-1}-1) \\
&= (1-\gamma)(\beta^{-1}-1)[1-1+\lambda-1+\lambda^*] \\
&\overset{\text{sign}}{=} \lambda+\lambda^*-1 \\
\text{step 3, cf. notes below:} \quad &= \lambda + 1 - \frac{\gamma}{1-\gamma}(1-\lambda) - 1 \\
&\overset{\text{sign}}{=} (1-\gamma)\lambda - \gamma(1-\lambda) \\
&= \lambda - \gamma
\end{aligned}$$

[50]Proof: (1) $\gamma(1-\lambda)[\varpi^*(\bar{b}^*)+\psi^*(\varpi^*(\bar{b}^*))] \geq 0$ by non-negativity of consumption & housing investment; (2) $\gamma(1-\lambda)[\varpi+\psi\circ\varpi]+(1-\gamma)(\beta^{-1}-1)\bar{b}^* = \gamma(1-\lambda)[\phi-\frac{1-\gamma}{\gamma}(\beta^{-1}-1)\bar{b}^*]+(1-\gamma)(\beta^{-1}-1)\bar{b}^* = \gamma(1-\lambda)\phi+\lambda(1-\gamma)(\beta^{-1}-1)\bar{b}^*$, where the second equality is a consequence of (GMC); (3) $\lim_{\bar{b}^*\to-\infty}\varpi = +\infty$ (assuming the contrary will produce a contradiction with (GMC)); (4) $\lim_{c\to\infty}\phi = 0$; (5) steps 1–4 now imply $\mathcal{H}(\bar{b}^*) \leq \gamma(1-\lambda)\phi+\lambda(1-\gamma)(\beta^{-1}-1)\bar{b}^* \to -\infty$ as $\bar{b}^* \to -\infty$. □

[51]Proof: (1) $\gamma(1-\lambda)[\varpi(\bar{b}^*)+\psi(\varpi(\bar{b}^*))] \geq 0$ by non-negativity of consumption & housing investment; (2) $-(1-\lambda^*)[\varpi^*(\bar{b}^*)+\psi^*(\varpi^*(\bar{b}^*))]+(\beta^{-1}-1)\bar{b}^* = -(1-\lambda^*)[\phi(c^*_{ss})+(\beta^{-1}-1)\bar{b}^*]+(\beta^{-1}-1)\bar{b}^* = -(1-\lambda^*)\phi(c^*_{ss})+\lambda^*(\beta^{-1}-1)\bar{b}^*$ where the substitution is made using the definition of ϖ^*; (3) $\lim_{\bar{b}^*\to+\infty}\varpi^* = +\infty$ (assuming the contrary will produce a contradiction with (GMC^*)); (4) Fact 3 and $\lim_{c\to+\infty}\phi(c) = 0$ imply $\mathcal{H}(\bar{b}^*) \geq -(1-\lambda^*)\phi(c^*_{ss}) + \lambda^*(\beta^{-1}-1)\bar{b}^* \to +\infty$ as $\bar{b}^* \to +\infty$. □

$$\geq 0 \text{ by assumption.}$$

Step 3 follows by symmetric home bias, that is, $\gamma(1-\lambda) = (1-\gamma)(1-\lambda^*) \iff \lambda^* = 1 - \frac{\gamma}{1-\gamma}(1-\lambda)$; Step 2 requires slightly more work: First, we use the implicit function theorem on $(GMC)\&(GMC^*)$, respectively, to obtain

$$\varpi' = -\frac{\partial\zeta/\partial\bar{b}^*}{\partial\zeta/\partial c_{ss}} = \frac{\frac{1-\gamma}{\gamma}(\beta^{-1}-1)}{\phi' - (1+\psi')} < 0,$$
$$\varpi^{*\prime} = -\frac{\partial\zeta^*/\partial\bar{b}^*}{\partial\zeta^*/\partial c^*_{ss}} = \frac{-(\beta^{-1}-1)}{\phi' - (1+\psi^{*\prime})} > 0;$$

Second, we recognize that since $\phi' < 0$ then

$$\frac{1+\psi'}{1+\psi'-\phi'} < 1 \iff \frac{1+\psi'}{-(1+\psi')+\phi'} > -1$$

and symmetrically for *. This shows that $\frac{\partial\mathcal{H}}{\partial\bar{b}^*} > 0$ and the SS is unique.

Finally, notice that since existence and uniqueness follow for arbitrary parameters $\eta \in (0,1), \eta^* \in (0,1), \xi_{h,ss}, \xi_{x,ss}, \xi^*_{h,ss}, \xi^*_{x,ss} > 0$, it is possible to set $\eta^* > \eta$ and then choose $\xi_{x,ss}, \xi^*_{x,ss}$ to ensure that both (HMC) and $(HMC)^*$ hold if $h_{ss} = h^*_{ss}$ and $x_{ss} = x^*_{ss}$. Given this choice of $\xi_{x,ss}, \xi^*_{x,ss}$, and the symmetry in the housing stock and housing investment, it is then possible to select $\xi_{h,ss}, \xi^*_{h,ss}$ such that equations $(x) and (x^*)$ hold. It then follows that $c_{ss} = c^*_{ss}, n_{ss} = n^*_{ss}, \bar{b} = \bar{b}^* = 0$, and the allocation is symmetric. This completes the proof.

B.5 Derivation of the household's subjectively optimal plans

In this appendix, we provide a formal derivation of the linearized subjectively optimal household decision rules presented in equations (9) and (10). Throughout the derivation, we concentrate on the representative household in i, with the understanding that the situation in all other regions is symmetric. We therefore omit the regional subscript i for the sake of conciseness. For simplicity, we omit domestically produced housing investment goods. This does not affect the layout of the proof below. We further allow for potential external innovations to ξ_c, ξ_h, ξ_x.

Consider the household program presented in Section III.A, which is restated here for convenience.[52] We first clarify the shape of the underlying probability space to set the appropriate frame for the following derivations: denote $\Omega \ni \omega_t := (\xi_t, r_t, w_t, \Sigma_t, \pi_t, (P_{ij,t})_{i,j\in[1,N]}, q_t)^\top$ as

[52] We ignore the terms $T_{t+s} and \mathrm{b}_{t+s}$ since they will be zero in the equilibrium we analyze. All derivations go through if these terms are non-zero.

the vector of the external decision-relevant variables that the household takes as given; denote $\Omega^t \ni \omega^t := (\omega_{t-s})_{s\geq 0}$ as the one-sided infinite history of past external variables; and denote $\Omega^\infty \ni \omega := (\omega_t)_{t\in\mathbb{Z}}$ as the typical elements from the set of possible realizations of the full sequences of the external variables;[53] and denote $\mathcal{B}^\infty$ as the Borel-sigma-algebra over Ω^∞. Each household is now endowed with a probability measure $\mathcal{P}$ over $(\Omega^\infty, \mathcal{B}^\infty)$ which encodes their subjective beliefs over the realizations of the external variables ω. Rational expectations, denoted $\mathcal{P} = \mathbb{P}$, are a special case of this setup, where $\mathbb{P}$ is the (unique) measure generated by the distribution of $\xi := (\xi_t)_{t\in\mathbb{Z}}$ and the equilibrium conditions that allow us to compute $\omega\backslash\xi$ as a deterministic function of ξ.[54] As explained in Section III.A, we assume $\mathcal{P}$ to be of a particular form: that is, we assume that $\mathcal{P} = \mathcal{P}_q \otimes \mathbb{P}_{-q}$, where $\otimes$ is the product measure (statistical independence), $\mathcal{P}_q$ is the distribution over $(q_t)_{t\in\mathbb{Z}}$ generated by the unobserved components model (2), and $\mathbb{P}_{-q}$ is the rational expectations measure without house prices–that is, the measure over $\omega\backslash q = (\xi_t, r_t, w_t, \Sigma_t, \pi_t, (P_{ij,t})_{i,j\in[1,N]})_{t\in\mathbb{Z}}$ that which is consistent with the equilibrium-implied joint probability distribution of $\omega\backslash q$. Consequently, for any two measurable functions $f : \mathbb{R} \to \mathbb{R}, g : \mathbb{R}^{\dim\Omega-1} \to \mathbb{R}$, we have

$$\mathbb{E}^{\mathcal{P}}[f(q_t) \cdot g(\omega_t\backslash q_t)] = \int f\mathcal{P}_q(\mathrm{d}q) \cdot \int g\mathbb{P}_{-q}(\mathrm{d}\omega\backslash q).$$

Finally, $\mathbb{E}_t^{\mathcal{P}}$ is the expectation implied by $\mathcal{P}$ conditional on the sigma-algebra generated by ω^t.

Now, at each calendar date $t \in \mathbb{Z}$, the household takes as given ω^t and chooses today's consumption, labor, housing, housing investment, and bond levels, $(c_t, n_t, h_t, x_t, b_{t+1})$, as well as contingent plans for the future, $\{(c_{t+s}, n_{t+s}, h_{t+s}, x_{t+s}, b_{t+s+1}) : \Omega^{t+s} \to \mathbb{R}_+^4 \times \mathbb{R}\}_{s\in\mathbb{N}_+}$, to maximize

$$\begin{aligned}
&\mathbb{E}_t^{\mathfrak{p}} \sum_{s=0}^{\infty} \beta^s \left(\frac{\xi_{c,t+s} c_{t+s}^{1-\sigma}}{1-\sigma} + \frac{\xi_{h,t+s} h_{t+s}^{1-\nu}}{1-\nu} - \chi \frac{n_{t+s}^{1+\varphi}}{1+\varphi} \right) \text{ subject to} \\
&c_{t+s} + q_{t+s}(h_{t+s} - (1-\delta)h_{t+s-1}) + b_{t+s+1} + x_{t+s} = w_{t+s} n_{t+s} \\
&\quad + (1 + r_{t+s}) b_{t+s} + q_{t+s} \cdot \tfrac{\xi_{x,t+s}}{\eta} x_{t+s-\tau}^{\eta} + \Sigma_{t+s}, \quad \forall s \geq 0,\ \mathcal{P}\text{-almost surely,}
\end{aligned} \tag{B.5}$$

as well as being subject to a standard no-Ponzi-game condition on b. The first-order conditions for

[53] We restrict Ω to be the set of absolutely summable sequences.

[54] For any three vectors x, y, z where $x = (y, z)$, we define $x\backslash y := z$.

program (B.5) (suppressing the transversality conditions) are

$$
\begin{aligned}
\forall s \geq 0, \quad & (h) \quad \xi_{c,t+s} c_{t+s}^{-\sigma} q_{t+s} = \xi_{h,t+s} h_{t+s}^{-\nu} + \beta(1-\delta)\mathbb{E}_{t+s}^{\mathcal{P}}\{\xi_{c,t+s+1} c_{t+s+1}^{-\sigma} q_{t+s+1}\}, \\
& (n) \quad \chi n_{t+s}^{\varphi} \xi_{c,t+s}^{-1} c_{t+s}^{\sigma} = w_{t+s}, \\
& (b) \quad \xi_{c,t+s} c_{t+s}^{-\sigma} = \beta \mathbb{E}_{t+s}^{\mathcal{P}} \left\{(1+r_{t+s+1})\xi_{c,t+s+1} c_{t+s+1}^{-\sigma}\right\}, \\
& (x) \quad \beta^{\tau} \mathbb{E}_{t+s}^{\mathcal{P}} \left\{q_{t+s+\tau}\xi_{c,t+s+\tau} c_{t+s+\tau}^{-\sigma} \cdot \xi_{x,t+s+\tau} x_{t+s}^{\eta-1}\right\} = \xi_{c,t+s} c_{t+s}^{-\sigma}, \\
& (BC) \quad c_{t+s} + q_{t+s}(h_{t+s} - (1-\delta)h_{t+s-1}) + b_{t+s+1} + x_{t+s} = w_{t+s} n_{t+s} \\
& \qquad + (1+r_{t+s}) b_{t+s} + q_{t+s} \cdot \tfrac{\xi_{x,t+s}}{\eta} x_{t+s-\tau}^{\eta} + \Sigma_{t+s}.
\end{aligned}
\tag{B.6}
$$

In close analogy to the standard procedure in a model with fully rational expectations, $\mathcal{P} = \mathbb{P}$, we now derive a linear approximation to (B.6) that–together with the other linearized equilibrium conditions–allows us to solve the model to the first order in the amplitude of the shocks. The specific challenge here, with $\mathcal{P} = \mathcal{P}_q \otimes \mathbb{P}_{-q}$, will be to compute all of the expectations explicitly that depend on house prices, q. This includes the house prices, themselves, $(q_{t+s})_{s>0}$, as well as the expectations over future contingent choices, $(c_{t+s}, n_{t+s}, h_{t+s}, x_{t+s}, b_{t+s+1})_{s>0}$.

To the first order around the steady state[55] from Lemma 2, in which we have $1 + r_{ss} = \beta^{-1}, \chi n_{ss}^{\varphi} c_{ss}^{\sigma} = w_{ss}(=1), q_{ss} = \xi_{h,ss} h_{ss}^{-\nu} c_{ss}^{\sigma}/(1-\bar{\beta}), \beta^{\tau} q_{ss} \xi_{x,ss} x_{ss}^{\eta-1} = 1$, it holds that, $\forall s \geq 0$:

$$
\begin{aligned}
& (h) \quad \widehat{h}_{t+s} = \tfrac{1}{\nu}\xi_{h,t+s} + \tfrac{\sigma}{\nu} \frac{\widehat{c}_{t+s} - \frac{1}{\sigma}\widehat{\xi}_{c,t+s} - \bar{\beta}\mathbb{E}_{t+s}^{\mathcal{P}}\left\{\widehat{c}_{t+s+1} - \frac{1}{\sigma}\widehat{\xi}_{c,t+s+1}\right\}}{1-\bar{\beta}} - \tfrac{1}{\nu}\frac{\widehat{q}_{t+s} - \bar{\beta}\mathbb{E}_{t+s}^{\mathcal{P}}\{\widehat{q}_{t+s+1}\}}{1-\bar{\beta}}, \\
& (n) \quad \varphi \widehat{n}_{t+s} + \sigma \widehat{c}_{t+s} - \widehat{\xi}_{c,t+s} = \widehat{w}_{t+s}, \\
& (b) \quad \widehat{c}_{t+s} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s} = -\tfrac{1}{\sigma}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+1}\} + \mathbb{E}_{t+s}^{\mathcal{P}}\left\{\widehat{c}_{t+s+1} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s+1}\right\}, \\
& (x) \quad \mathbb{E}_{t+s}^{\mathcal{P}}\widehat{q}_{t+s+\tau} - \mathbb{E}_{t+s}^{\mathcal{P}}\left\{\sigma\widehat{c}_{t+s+\tau} - \widehat{\xi}_{c,t+s+\tau}\right\} \\
& \qquad + \left(\sigma\widehat{c}_{t+s} - \widehat{\xi}_{c,t+s}\right) + \mathbb{E}_{t+s}\widehat{\xi}_{x,t+s+\tau} = (1-\eta)\widehat{x}_{t+s}, \\
& (BC) \quad \widehat{b}_{t+s+1} = \tfrac{1}{\beta}\widehat{b}_{t+s} + \tfrac{w_{ss}n_{ss}}{y_{ss}}(\widehat{w}_{t+s} + \widehat{n}_{t+s}) + \widehat{\Sigma}_{t+s} - \tfrac{c_{ss}}{y_{ss}}\widehat{c}_{t+s} \\
& \qquad - \tfrac{q_{ss}h_{ss}}{y_{ss}}[\widehat{h}_{t+s} - (1-\delta)\widehat{h}_{t+s-1}] + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\left(\widehat{x}_{t+s-\tau} - \beta^{\tau}\widehat{x}_{t+s} + \widehat{\xi}_{x,t+s}\right).
\end{aligned}
\tag{B.7}
$$

To characterize the household's expectations over their own choice variables, we start with the optimality condition for liquid bonds. Iterations over future instances of the condition reveal

$$
(b) \quad \widehat{c}_{t+s} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s} = -\tfrac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n}\} + \lim_{n\to\infty}\mathbb{E}_{t+s}^{\mathcal{P}}\widehat{c}_{t+s+n}.
$$

To keep the notation concise, we define, for any random process, $(k_t)_t$: $\mathbb{E}_{t+s}^{\mathcal{P}} k_{\infty} := \lim_{n\to\infty}\mathbb{E}_{t+s}^{\mathcal{P}} k_{t+s+n}$.

[55] Technically, we are scaling $\mathrm{Var}^{\mathcal{P}}[\|\omega\|] \to 0$. This is consistent, however, with the definition of $\mathcal{P}$ and $\mathrm{Var}[\|\xi\|] \to 0$.

Here, we implicitly assume that the law of iterated expectations holds for $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_{t+s}$–at the end of this section, we verify that the $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$ we recover does indeed satisfy a law of iterated expectations. Careful inspection of equations (B.7) reveals that $\mathbb{E}_{t+s}^{\mathcal{P}}\widehat{c}_\infty$ is the only subjective expectation of a choice variable left unknown. This is because the law of iterated expectations for $\mathcal{P}$ allows substituting (b) into

$$
\begin{aligned}
&(h) \quad \widehat{h}_{t+s} = \tfrac{1}{\nu}\xi_{h,t+s} - \tfrac{\sigma}{\nu}\frac{\frac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n}\} - \bar{\beta}\frac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n+1}\}}{1-\bar{\beta}} - \tfrac{1}{\nu}\frac{\widehat{q}_{t+s} - \bar{\beta}\mathbb{E}_{t+s}^{\mathcal{P}}\{\widehat{q}_{t+s+1}\}}{1-\bar{\beta}} + \tfrac{\sigma}{\nu}\mathbb{E}_{t+s}^{\mathcal{P}}\widehat{c}_\infty,\\
&(n) \quad \varphi\widehat{n}_{t+s} + \sigma\widehat{c}_{t+s} - \widehat{\xi}_{c,t+s} = \widehat{w}_{t+s},\\
&(b) \quad \widehat{c}_{t+s} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s} = -\tfrac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n}\} + \mathbb{E}_{t+s}^{\mathcal{P}}\widehat{c}_\infty,\\
&(x) \quad \mathbb{E}_{t+s}^{\mathcal{P}}\widehat{q}_{t+s+\tau} - \sum_{n=1}^{\tau}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n}\} + \mathbb{E}_{t+s}\widehat{\xi}_{x,t+s+\tau} = (1-\eta)\widehat{x}_{t+s},\\
&(BC) \quad \widehat{b}_{t+s+1} = \tfrac{1}{\beta}\widehat{b}_{t+s} + \tfrac{w_{ss}n_{ss}}{y_{ss}}\left(\widehat{w}_{t+s} + \widehat{n}_{t+s}\right) + \widehat{\Sigma}_{t+s} - \tfrac{c_{ss}}{y_{ss}}\widehat{c}_{t+s}\\
&\qquad - \tfrac{q_{ss}h_{ss}}{y_{ss}}\left[\widehat{h}_{t+s} - (1-\delta)\widehat{h}_{t+s-1}\right] + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\left(\widehat{x}_{t+s-\tau} - \beta^\tau\widehat{x}_{t+s} + \widehat{\xi}_{x,t+s}\right).
\end{aligned}
\tag{B.8}
$$

Now to find $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$ (and characterize the choices at calendar date t), we first use the budget constraint for some $s > \tau$. We start by plugging in the following optimality conditions:

$$
\begin{aligned}
(BC) \quad \widehat{b}_{t+s+1} &= \tfrac{1}{\beta}\widehat{b}_{t+s} + \tfrac{w_{ss}n_{ss}}{y_{ss}}(1+1/\varphi)\widehat{w}_{t+s} + \widehat{\Sigma}_{t+s} - \tfrac{c_{ss}+n_{ss}w_{ss}\sigma/\varphi}{y_{ss}}\left(\widehat{c}^*_{t+s} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s}\right) - \tfrac{c_{ss}}{y_{ss}\sigma}\widehat{\xi}_{c,t+s}\\
&\quad - \tfrac{q_{ss}h_{ss}}{y_{ss}}\left[\widehat{h}^*_{t+s} - (1-\delta)\widehat{h}^*_{t+s-1}\right] + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\left(\widehat{x}^*_{t+s-\tau} - \beta^\tau\widehat{x}^*_{t+s} + \widehat{\xi}_{x,t+s}\right)\\
&\quad - \tfrac{c_{ss}+n_{ss}w_{ss}\sigma/\varphi+\delta q_{ss}h_{ss}\sigma/\nu}{y_{ss}}\mathbb{E}_{t+s}^{\mathcal{P}}\widehat{c}_\infty\\
&\quad + \tfrac{q_{ss}h_{ss}}{y_{ss}}\tfrac{1}{\nu}\tfrac{1}{1-\bar{\beta}}\left(\widehat{q}_{t+s} - \bar{\beta}\mathbb{E}_{t+s}^{\mathcal{P}}\{\widehat{q}_{t+s+1}\} - (1-\delta)\left(\widehat{q}_{t+s-1} - \bar{\beta}\mathbb{E}_{t+s-1}^{\mathcal{P}}\{\widehat{q}_{t+s}\}\right)\right)\\
&\quad + \tfrac{x_{ss}}{y_{ss}}\tfrac{1}{\beta^\tau(1-\eta)}\left(\mathbb{E}_{t+s-\tau}^{\mathcal{P}}\{\widehat{q}_{t+s}\} - \beta^\tau\mathbb{E}_{t+s}^{\mathcal{P}}\{\widehat{q}_{t+s+\tau}\}\right),
\end{aligned}
$$

where we defined the auxiliary variables

$$
\begin{aligned}
&\widehat{h}^*_{t+s} := \tfrac{1}{\nu}\xi_{h,t+s} - \tfrac{\sigma}{\nu}\frac{\frac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n}\} - \bar{\beta}\frac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n+1}\}}{1-\bar{\beta}},\\
&\widehat{c}^*_{t+s} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s} := -\tfrac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n}\},\\
&(1-\eta)\widehat{x}^*_{t+s} := -\sum_{n=1}^{\tau}\mathbb{E}_{t+s}\{\widehat{r}_{t+s+n}\} + \mathbb{E}_{t+s}\widehat{\xi}_{x,t+s+\tau}.
\end{aligned}
$$

Notice that the auxiliary variables are composed only of terms over which the household has rational expectations; thus, the terms that fade to zero as $s \to \infty$.

Next, we apply the operator $\mathbb{E}_t^{\mathcal{P}}$ to both sides of (BC) and consider the limit as $s \to \infty$. This delivers

$$
\begin{aligned}
(BC) \quad & \mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty = \tfrac{1}{\beta}\mathbb{E}_t^{\mathcal{P}} \widehat{b}_\infty - \tfrac{c_{ss}+n_{ss}w_{ss}\sigma/\varphi+\delta q_{ss}h_{ss}\sigma/\nu}{y_{ss}}\mathbb{E}_{t+s}^{\mathcal{P}}\widehat{c}_\infty \\
& \quad + \tfrac{\delta q_{ss}h_{ss}/\nu + x_{ss}(\beta^{-\tau}-1)/(1-\eta)}{y_{ss}}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty \\
\iff \quad & \mathbb{E}_t^{\mathcal{P}} \widehat{c}_\infty = \tfrac{y_{ss}}{c_{ss}+n_{ss}w_{ss}\sigma/\varphi+\delta q_{ss}h_{ss}\sigma/\nu}\left[\tfrac{1-\beta}{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty + \tfrac{\delta q_{ss}h_{ss}/\nu + x_{ss}(\beta^{-\tau}-1)/(1-\eta)}{y_{ss}}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty\right].
\end{aligned}
$$

Now since $\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty$ can be computed from the subjectively perceived house price model, the only unknown left is $\mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty$. To find this, we plug $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$ into (BC). At some $s > 0$, we have

$$
\begin{aligned}
(BC) \quad \mathbb{E}_t^{\mathcal{P}}\widehat{b}_{t+s+1} = {} & \tfrac{1}{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{b}_{t+s} + \tfrac{w_{ss}n_{ss}}{y_{ss}}(1+1/\varphi)\mathbb{E}_t\widehat{w}_{t+s} + \mathbb{E}_t\widehat{\Sigma}_{t+s} - \tfrac{c_{ss}+n_{ss}w_{ss}\sigma/\varphi}{y_{ss}}\mathbb{E}_t\left(\widehat{c}^*_{t+s} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s}\right) - \tfrac{c_{ss}}{y_{ss}\sigma}\mathbb{E}_t\widehat{\xi}_{c,t+s} \\
& - \tfrac{q_{ss}h_{ss}}{y_{ss}}\mathbb{E}_t\left[\widehat{h}^*_{t+s} - (1-\delta)\widehat{h}^*_{t+s-1}\right] + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\mathbb{E}_t^{\mathcal{P}}\left(\widehat{x}_{t+s-\tau} - \beta^\tau\widehat{x}_{t+s} + \widehat{\xi}_{x,t+s}\right) \\
& - \tfrac{1-\beta}{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty - \tfrac{\delta q_{ss}h_{ss}/\nu + x_{ss}(\beta^{-\tau}-1)/(1-\eta)}{y_{ss}}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty \\
& + \tfrac{q_{ss}h_{ss}}{y_{ss}}\tfrac{1}{\nu}\tfrac{1}{1-\bar{\beta}}\mathbb{E}_t^{\mathcal{P}}\left(\widehat{q}_{t+s} - \bar{\beta}\mathbb{E}_{t+s}^{\mathcal{P}}\{\widehat{q}_{t+s+1}\} - (1-\delta)\left(\widehat{q}_{t+s-1} - \bar{\beta}\mathbb{E}_{t+s-1}^{\mathcal{P}}\{\widehat{q}_{t+s}\}\right)\right).
\end{aligned}
$$

After defining the auxiliary variables

$$
\begin{aligned}
z^*_{t+s} := {} & \tfrac{w_{ss}n_{ss}}{y_{ss}}(1+1/\varphi)\widehat{w}_{t+s} + \widehat{\Sigma}_{t+s} - \tfrac{c_{ss}+n_{ss}w_{ss}\sigma/\varphi}{y_{ss}}\left(\widehat{c}^*_{t+s} - \tfrac{1}{\sigma}\widehat{\xi}_{c,t+s}\right) - \tfrac{c_{ss}}{y_{ss}\sigma}\widehat{\xi}_{c,t+s} \\
& - \tfrac{q_{ss}h_{ss}}{y_{ss}}\left[\widehat{h}^*_{t+s} - (1-\delta)\widehat{h}^*_{t+s-1}\right] + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\widehat{\xi}_{x,t+s}, \qquad \text{(B.9)} \\
Q_{t+s} := {} & \tfrac{q_{ss}h_{ss}}{y_{ss}}\tfrac{1}{\nu}\tfrac{1}{1-\bar{\beta}}\left(\widehat{q}_{t+s} - \bar{\beta}\mathbb{E}_{t+s}^{\mathcal{P}}\{\widehat{q}_{t+s+1}\} - (1-\delta)\left(\widehat{q}_{t+s-1} - \bar{\beta}\mathbb{E}_{t+s-1}^{\mathcal{P}}\{\widehat{q}_{t+s}\}\right)\right),
\end{aligned}
$$

we can rewrite (BC) into

$$
\begin{aligned}
(BC) \quad \mathbb{E}_t^{\mathcal{P}}\widehat{b}_{t+s+1} = {} & \tfrac{1}{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{b}_{t+s} + \mathbb{E}_t z^*_{t+s} + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\mathbb{E}_t^{\mathcal{P}}\left(\widehat{x}_{t+s-\tau} - \beta^\tau\widehat{x}_{t+s}\right) + \mathbb{E}_t^{\mathcal{P}}Q_{t+s} \\
& - \tfrac{1-\beta}{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty - \tfrac{\delta q_{ss}h_{ss}/\nu + x_{ss}(\beta^{-\tau}-1)/(1-\eta)}{y_{ss}}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty.
\end{aligned}
$$

Performing backward substitution until $s = 1$ delivers

$$
\begin{aligned}
(BC) \quad & \mathbb{E}_t^{\mathcal{P}}\widehat{b}_{t+s+1} = \beta^{-s}\widehat{b}_{t+1} - \beta^{-s}(1-\beta^s)\tfrac{\beta}{1-\beta}\left[\tfrac{1-\beta}{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty + \tfrac{\delta q_{ss}h_{ss}/\nu + x_{ss}(\beta^{-\tau}-1)/(1-\eta)}{y_{ss}}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty\right] \\
& \quad + \beta^{-s}\sum_{n=1}^{s}\beta^n\left(\mathbb{E}_t z^*_{t+n} + \mathbb{E}_t^{\mathcal{P}}Q_{t+n}\right) + \beta^{-s}\tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\sum_{n=1}^{s}\beta^n\mathbb{E}_t^{\mathcal{P}}\left(\widehat{x}_{t+n-\tau} - \beta^\tau\widehat{x}_{t+n}\right) \\
\iff \quad & \mathbb{E}_t^{\mathcal{P}}\widehat{b}_{t+s+1} - \mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty = \beta^{-s}\cdot\Bigg[\widehat{b}_{t+1} + \sum_{n=1}^{s}\beta^n\left(\mathbb{E}_t z^*_{t+n} + \mathbb{E}_t^{\mathcal{P}}Q_{t+n}\right) + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\sum_{n=1}^{s}\beta^n\mathbb{E}_t^{\mathcal{P}}\left(\widehat{x}_{t+n-\tau} - \beta^\tau\widehat{x}_{t+n}\right) \\
& \qquad - \mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty - (1-\beta^s)\tfrac{\beta}{1-\beta}\tfrac{\delta q_{ss}h_{ss}/\nu + x_{ss}(\beta^{-\tau}-1)/(1-\eta)}{y_{ss}}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty\Bigg].
\end{aligned}
$$

Now since we know that $\lim_{s\to\infty}\mathbb{E}_t^{\mathcal{P}}\widehat{b}_{t+s}$ exists, it must be that the left-hand side of this equation tends to zero as $s\to\infty$. But since $\beta^{-s}\to+\infty$ as $s\to\infty$, this implies that

$$\mathbb{E}_t^{\mathcal{P}}\widehat{b}_\infty = \widehat{b}_{t+1}+\sum_{n=1}^{\infty}\beta^n\left(\mathbb{E}_t z^*_{t+n}+\mathbb{E}_t^{\mathcal{P}}Q_{t+n}\right)+\tfrac{x_{ss}}{y_{ss}}\underbrace{\beta^{-\tau}\sum_{n=1}^{\infty}\beta^n\mathbb{E}_t^{\mathcal{P}}\left(\widehat{x}_{t+n-\tau}-\beta^{\tau}\widehat{x}_{t+n}\right)}_{=\beta^{-\tau}\sum_{n=1}^{\tau}\beta^n\widehat{x}_{t+n-\tau}\text{ by rearranging a convergent sum}}-\tfrac{\beta}{1-\beta}\tfrac{\delta q_{ss}h_{ss}/\nu+x_{ss}(\beta^{-\tau}-1)/(1-\eta)}{y_{ss}}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_\infty,$$

so that we can characterize $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$ entirely in terms of the variables that are either predetermined, current choice variables (i.e., known under $\mathbb{E}_t^{\mathcal{P}}$) or variables of which we can compute expectations in closed form (be they rational or not):

$$\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty = \tfrac{y_{ss}}{c_{ss}+n_{ss}w_{ss}\sigma/\varphi+\delta q_{ss}h_{ss}\sigma/\nu}\tfrac{1-\beta}{\beta}\left[\widehat{b}_{t+1}+\sum_{n=1}^{\infty}\beta^n\left(\mathbb{E}_t z^*_{t+n}+\mathbb{E}_t^{\mathcal{P}}Q_{t+n}\right)+\tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\sum_{n=1}^{\tau}\beta^n\widehat{x}_{t+n-\tau}\right].$$

In the last step, we explicitly characterize the subjective expectation $\mathbb{E}_t^{\mathcal{P}}\sum_{n=1}^{\infty}\beta^n Q_{t+n}$ in terms of the processes governing subjective house price expectations, namely $\widehat{q}_t,\widehat{\bar{m}}_t$.

Recall that for any $s>0$, we have $\mathbb{E}_t^{\mathcal{P}}\widehat{q}_{t+s}=\widehat{q}_t+(1-\varrho^s)\frac{\varrho}{1-\varrho}\widehat{\bar{m}}_t$, so that we have

$$\begin{aligned}
\mathbb{E}_t^{\mathcal{P}}Q_{t+s} &= \tfrac{q_{ss}h_{ss}}{y_{ss}}\tfrac{1}{\nu}\tfrac{1}{1-\bar{\beta}}\left(\mathbb{E}_t^{\mathcal{P}}\widehat{q}_{t+s}-\bar{\beta}\mathbb{E}_t^{\mathcal{P}}\{\widehat{q}_{t+s+1}\}-(1-\delta)\left(\mathbb{E}_t^{\mathcal{P}}\widehat{q}_{t+s-1}-\bar{\beta}\mathbb{E}_t^{\mathcal{P}}\{\widehat{q}_{t+s}\}\right)\right)\\
&= \tfrac{q_{ss}h_{ss}}{y_{ss}}\tfrac{1}{\nu}\tfrac{1}{1-\bar{\beta}}\Big(\delta(1-\bar{\beta})\widehat{q}_t+(1-\varrho^s)\tfrac{\varrho}{1-\varrho}\widehat{\bar{m}}_t-\bar{\beta}(1-\varrho^{s+1})\tfrac{\varrho}{1-\varrho}\widehat{\bar{m}}_t\\
&\quad-(1-\delta)(1-\varrho^{s-1})\tfrac{\varrho}{1-\varrho}\widehat{\bar{m}}_t+(1-\delta)\bar{\beta}(1-\varrho^s)\tfrac{\varrho}{1-\varrho}\widehat{\bar{m}}_t\Big)\\
&= \tfrac{q_{ss}h_{ss}}{y_{ss}}\tfrac{1}{\nu}\tfrac{1}{1-\bar{\beta}}\left(\delta(1-\bar{\beta})\widehat{q}_t+\widehat{\bar{m}}_t\tfrac{\varrho}{1-\varrho}\left[\delta(1-\bar{\beta})+(1-\bar{\beta}\varrho)(1-\varrho-\delta)\varrho^{s-1}\right]\right),
\end{aligned}$$

which in turn implies

$$\sum_{s\geq1}\beta^s\mathbb{E}_t^{\mathcal{P}}Q_{t+s}=\beta\tfrac{q_{ss}h_{ss}}{y_{ss}}\tfrac{1}{\nu}\left(\tfrac{\delta}{1-\beta}\widehat{q}_t+\widehat{\bar{m}}_t\tfrac{\varrho}{1-\varrho}\left[\tfrac{\delta}{1-\beta}+\tfrac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\tfrac{1-\varrho-\delta}{1-\beta\varrho}\right]\right). \tag{B.10}$$

Notice that, as claimed in the main text, that

$$\tfrac{\delta}{1-\beta}+\tfrac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\tfrac{1-\varrho-\delta}{1-\beta\varrho}>0,\ \ \forall\beta,\delta,\varrho\in(0,1),$$ [56]

so that the coefficient of the posterior expected house price growth rate onto the series is positive.

We have now characterized the household's decisions at an arbitrary calendar date t up to first

[56] *Proof.* This follows from the claim being equivalent to the statement $f(\varrho):=\frac{\delta}{1-\beta}(1-\beta\varrho)+\frac{1-\bar{\beta}\varrho}{1-\bar{\beta}}(1-\varrho-\delta)>0,\ \forall\beta,\delta,\varrho\in(0,1)$, from f being an upward-open parabola with $f(1)=0$, and $f'(1)<0$. □

order around the deterministic steady state:

$$
\begin{aligned}
&(h) \quad \widehat{h}_t = \tfrac{1}{\nu}\xi_{h,t} - \tfrac{\sigma}{\nu}\frac{\frac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_t\{\widehat{r}_{t+n}\} - \bar{\beta}\frac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_t\{\widehat{r}_{t+n+1}\}}{1-\bar{\beta}} - \tfrac{1}{\nu}\widehat{q}_t + \tfrac{1}{\nu}\tfrac{\bar{\beta}}{1-\bar{\beta}}\varrho\widehat{\bar{m}}_t + \tfrac{\sigma}{\nu}\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty, \\
&(n) \quad \varphi\widehat{n}_t + \sigma\widehat{c}_t - \widehat{\xi}_{c,t} = \widehat{w}_t, \\
&(b) \quad \widehat{c}_t - \tfrac{1}{\sigma}\widehat{\xi}_{c,t} = -\tfrac{1}{\sigma}\sum_{n\geq 1}\mathbb{E}_t\{\widehat{r}_{t+n}\} + \mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty, \\
&(x) \quad \widehat{q}_t + (1-\varrho^\tau)\frac{\varrho}{1-\varrho}\widehat{\bar{m}}_t - \sum_{n=1}^{\tau}\mathbb{E}_t\{\widehat{r}_{t+n}\} + \mathbb{E}_t\widehat{\xi}_{x,t+\tau} = (1-\eta)\widehat{x}_t, \\
&(BC) \quad \widehat{b}_{t+1} = \tfrac{1}{\beta}\widehat{b}_t + \tfrac{w_{ss}n_{ss}}{y_{ss}}(\widehat{w}_t + \widehat{n}_t) + \widehat{\Sigma}_t - \tfrac{c_{ss}}{y_{ss}}\widehat{c}_t \\
&\qquad - \tfrac{q_{ss}h_{ss}}{y_{ss}}\left[\widehat{h}_t - (1-\delta)\widehat{h}_{t-1}\right] + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\left(\widehat{x}_{t-\tau} - \beta^\tau\widehat{x}_t + \widehat{\xi}_{x,t}\right),
\end{aligned}
\tag{B.11}
$$

where, after a few rearrangements,

$$
\begin{aligned}
\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty = {} & \frac{\delta q_{ss}h_{ss}/\nu}{c_{ss} + \sigma/\varphi\cdot n_{ss}w_{ss} + \delta q_{ss}h_{ss}\sigma/\nu}\cdot\left[\widehat{q}_t + \widehat{\bar{m}}_t\cdot\frac{\varrho}{1-\varrho}\underbrace{\left(1 + \frac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\frac{1-\varrho-\delta}{1-\beta\varrho}\frac{1-\beta}{\delta}\right)}_{>0\ \forall\beta,\delta,\varrho\in(0,1)}\right] \\
& + \frac{y_{ss}}{c_{ss} + \sigma/\varphi\cdot n_{ss}w_{ss} + \delta q_{ss}h_{ss}\sigma/\nu}\frac{1-\beta}{\beta}\cdot\left[\sum_{n=1}^{\infty}\beta^n\mathbb{E}_t\{z_{t+n}^*\} + \tfrac{x_{ss}}{y_{ss}}\sum_{n=0}^{\tau-1}\beta^{-n}\widehat{x}_{t-n} + \widehat{b}_{t+1}\right].
\end{aligned}
$$

Define $\bar{\mathrm{Q}} = \frac{\delta q_{ss}h_{ss}/\nu}{c_{ss}+\sigma/\varphi\cdot n_{ss}w_{ss}+\delta q_{ss}h_{ss}\sigma/\nu}$, $\bar{\mathrm{M}} = \frac{\varrho}{1-\varrho}\left(1 + \frac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\frac{1-\varrho-\delta}{1-\beta\varrho}\frac{1-\beta}{\delta}\right)$, and $\bar{\mathrm{C}} = \frac{y_{ss}}{c_{ss}+\sigma/\varphi\cdot n_{ss}w_{ss}+\delta q_{ss}h_{ss}\sigma/\nu}\frac{1-\beta}{\beta}$. Further, $\sum_{n=1}^{\infty}\beta^n\bar{z}_{t+n} = \sum_{n=1}^{\infty}\beta^n\mathbb{E}_t\{z_{t+n}^*\} + \frac{x_{ss}}{y_{ss}}\sum_{n=0}^{\tau-1}\beta^{-n}\widehat{x}_{t-n}$. Lastly, we verify that the law of iterated expectations holds for the explicit formula given for $\mathbb{E}_t^{\mathcal{P}}\widehat{c}_\infty$, above. We define

$$
A_t := \widehat{b}_{t+1} + \sum_{n=1}^{\infty}\beta^n\left(\mathbb{E}_t z_{t+n}^* + \mathbb{E}_t^{\mathcal{P}}Q_{t+n}\right) + \tfrac{x_{ss}}{y_{ss}}\beta^{-\tau}\sum_{n=1}^{\tau}\beta^n\widehat{x}_{t+n-\tau}
$$

and note that the law of iterated expectations holds if and only if $\mathbb{E}_t^{\mathcal{P}}A_{t+1} = A_t$. Using equation (BC) to substitute $\widehat{b}_{t+2}$ in $\mathbb{E}_t^{\mathcal{P}}A_{t+1}$ we arrive at

$$
\begin{aligned}
\mathbb{E}_t^{\mathcal{P}}A_{t+1} = A_t/\beta - {} & \tfrac{c_{ss}+\sigma/\varphi\cdot n_{ss}w_{ss}}{c_{ss}+\sigma/\varphi\cdot n_{ss}w_{ss}+\delta q_{ss}h_{ss}\sigma/\nu}\tfrac{1-\beta}{\beta}\mathbb{E}_t^{\mathcal{P}}A_{t+1} - \tfrac{q_{ss}h_{ss}\sigma/\nu}{c_{ss}+\sigma/\varphi\cdot n_{ss}w_{ss}+\delta q_{ss}h_{ss}\sigma/\nu}\tfrac{1-\beta}{\beta}\mathbb{E}_t^{\mathcal{P}}A_{t+1} \\
& + \tfrac{(1-\delta)q_{ss}h_{ss}\sigma/\nu}{c_{ss}+\sigma/\varphi\cdot n_{ss}w_{ss}+\delta q_{ss}h_{ss}\sigma/\nu}\tfrac{1-\beta}{\beta}A_t,
\end{aligned}
$$

which can be rearranged into $\mathbb{E}_t^{\mathcal{P}}A_{t+1} = A_t$.

B.6 Derivations for Section IV

In this appendix we derive analytical results on the behavior of house prices in a one-region, zero-liquidity endowment economy with instantaneous housing production.

The equations describing the economy are

$$
\begin{aligned}
&(h) \quad \widehat{q}_{i,t} - \bar{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{q}_{i,t+1} = \sigma(\widehat{c}_{i,t} - \bar{\beta}\mathbb{E}_t^{\mathcal{P}}\widehat{c}_{i,t+1}) - \nu(1-\bar{\beta})\widehat{h}_{i,t},\\
&(b) \quad \widehat{c}_{i,t} = -\sigma^{-1}\mathbb{E}_t\widehat{r}_{t+1} + \mathbb{E}_t^{\mathcal{P}}\widehat{c}_{i,t+1},\\
&(BC) \quad \widehat{b}_{i,t+1} = \beta^{-1}\widehat{b}_{i,t} + \widehat{y}_t - \tfrac{c}{y}\widehat{c}_{i,t} - \tfrac{qh}{y}(\widehat{h}_{i,t} - (1-\delta)\widehat{h}_{i,t-1}) + \tfrac{qh\delta}{y}\widehat{q}_{i,t},\\
&(B) \quad \widehat{b}_{i,t+1} = 0,\\
&(HMC) \quad \widehat{h}_{i,t} - (1-\delta)\widehat{h}_{i,t-1} = -\iota_i\varepsilon_t,
\end{aligned}
\tag{B.12}
$$

with the endowment process $\widehat{y}_t$ being the only exogenous disturbance to the economy. Iterating on (h) and plugging the results into the Euler equation gives our key equation:

$$
\widehat{q}_{i,t} = \mathbb{E}_t^{\mathcal{P}}\widehat{q}_{i,t+1} - \widehat{r}_{t+1} - \nu(1-\bar{\beta})\widehat{h}_{i,t} + \nu(1-\bar{\beta})^2\sum_{s=0}^{\infty}\bar{\beta}^s\mathbb{E}_t^{\mathcal{P}}\widehat{h}_{i,t+s+1}.
$$

B.6.1 *Proof of Proposition (2)*

Under rational expectations, and using the fact that the shocks only hit in period one, we can use housing market clearing to rewrite the house price equation as

$$
\widehat{q}_{i,t} = \mathbb{E}\widehat{q}_{i,t+1} - \widehat{r}_{t+1} - (1-\bar{\beta})\widehat{h}_{i,t} + (1-\bar{\beta})^2(1-\delta)^t\frac{1}{1-\bar{\beta}(1-\delta)}(-\iota_i\varepsilon_1)
$$

Iteration on this equation and using the fact that the economy returns to the steady-state in expectations gives

$$
\begin{aligned}
\widehat{q}_{i,1} &= -\varepsilon_1 - \omega_h(-\iota_i\varepsilon_1)\\
\widehat{q}_{i,2} &= -(1-\delta)\omega_h(-\iota_i\varepsilon_1),
\end{aligned}
$$

with $\omega_h = \frac{1-\bar{\beta}}{1-\bar{\beta}(1-\delta)} \in (0,1)$. This completes the proof.

B.6.2 *Proof of Proposition (3)*

Under subjective beliefs, we need to specifically characterize $\mathbb{E}_t^{\mathcal{P}}\widehat{h}_{i,t+s+1}$. We can make use of the method described in section (III.B). Importantly, we can make use of the fact that the shocks on the endowment and, hence, the interest rate only hit in the first period. As households have rational expectations about these processes, they possess knowledge of this fact. Applying our solution

method and substituting the subjective beliefs housing model, we get

$$\widehat{q}_{i,t} = \mathbb{E}_t^{\mathcal{P}}\widehat{q}_{i,t+1} - (1-\bar{\beta})\widehat{h}_{i,t} - \widehat{r}_{t+1} + (1-\bar{\beta})\Bigg[-\widehat{q}_{i,t} - \frac{y_{ss}}{y_{ss}+\delta q_{ss}h_{ss}}\frac{1-\beta}{1-\bar{\beta}}\widehat{r}_{t+1} + \Big(\beta\frac{q_{ss}h_{ss}}{y_{ss}}\frac{\varrho}{1-\varrho}\big(\frac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\frac{1-\varrho-\delta}{1-\beta\varrho} + \frac{1}{1-\beta\varrho}\big) + \frac{1-\varrho}{1-\varrho\bar{\beta}} - \frac{1}{1-\varrho}\Big)\widehat{\bar{m}}_{i,t}\Bigg]$$

Rearranging yields

$$\widehat{q}_{i,t} = -\widehat{h}_{i,t} - \omega_r\widehat{r}_{t+1} + \omega_m\widehat{\bar{m}}_{i,t},$$

where $\omega_r = 1 + \frac{y(1-\beta)}{y_{ss}+h_{ss}q_{ss}\delta}$ and $\omega_m = (1-\bar{\beta})^{-1}\Big[\varrho + (1-\bar{\beta})(\beta\frac{q_{ss}h_{ss}}{y_{ss}}\frac{\varrho}{1-\varrho}(\frac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\frac{1-\varrho-\delta}{1-\beta\varrho} + \frac{1}{1-\beta\varrho}) + \frac{1-\varrho}{1-\varrho\bar{\beta}} - \frac{1}{1-\varrho})\Big]$. Further, we have that

$$\frac{\partial\omega_m}{\partial\varrho} = (1-\bar{\beta})^{-1}\Bigg[1 + (1-\bar{\beta})\beta\frac{q_{ss}h_{ss}}{y_{ss}}\Big(\frac{\varrho}{1-\varrho}\big(\frac{(1-\bar{\beta})(\beta\bar{\beta}\rho(\varrho+\delta)-1)}{((1-\bar{\beta})(1-\beta\varrho))^2} + \frac{\beta}{(1-\beta\varrho)^2}\big) \frac{1-2\varrho}{(1-\varrho)^2}\big(\frac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\frac{1-\varrho-\delta}{1-\beta\varrho} + \frac{1}{1-\beta\varrho}\big) - \frac{1-\bar{\beta}}{(1-\bar{\beta}\varrho)^2} + \frac{1}{(1-\varrho)^2}\Big),\Bigg]$$

which is smaller than zero for standard parameter choices. The first two periods of house prices are given by

$$\widehat{q}_{i,1} = -\omega_r\varepsilon_1 - (-\iota_i\varepsilon_1)$$
$$\widehat{q}_{i,2} = -(1-\delta)(-\iota_i\varepsilon_t) + \omega_m g\widehat{q}_1$$

This completes the proof.

B.6.3 *Demand-side heterogeneity*

Turning to regional differences on the housing demand side, we focus on the differences in the housing demand elasticity (ν). Following the same logic as in propositions (2) and (3), we derive closed-form characterizations of the house price with respect to a monetary policy shock.

Under rational expectations, the house price is given by

$$\widehat{q}_{i,t} = \mathbb{E}\widehat{q}_{i,t+1} - \widehat{r}_{t+1} - \nu_i(1-\bar{\beta})\widehat{h}_{i,t} + \nu_i(1-\bar{\beta})^2(1-\delta)^t\frac{1}{1-\bar{\beta}(1-\delta)}(-\iota\varepsilon_1).$$

Using the same steps as described in the Proof of Proposition (B.6.1) we get

$$\widehat{q}_{i,1} = -\varepsilon_1 - \omega_h \nu_i(-\iota\varepsilon_t)$$
$$\widehat{q}_{i,2} = -(1-\delta)\omega_h \nu_i(-\iota\varepsilon_t)$$

Given that $\iota > 0$, a higher elasticity of housing demand, hence a lower (ν_i), leads to a stronger response in house prices to a monetary policy shock.

Moving to the subjective beliefs model, we can use the same approach as in the proof of Proposition (B.6.2) and arrive at

$$\widehat{q}_{i,t} = -\nu_i \widehat{h}_{i,t} - \tilde{\omega}_{r,i}\widehat{r}_{t+1} + \tilde{\omega}_{i,m}\widehat{\bar{m}}_{i,t},$$

where $\tilde{\omega}_{i,r} = 1 + \frac{y_{ss}(1-\beta)}{y_{ss}+h_{ss}q_{ss}\delta\nu_i^{-1}}$ and $\tilde{\omega}_{i,m} = (1-\bar{\beta})^{-1}\Big[\varrho + (1-\bar{\beta})(\beta\frac{q_{ss}h_{ss}\nu_i^{-1}}{y_{ss}}\frac{\varrho}{1-\varrho}(\frac{1-\bar{\beta}\varrho}{1-\bar{\beta}}\frac{1-\varrho-\delta}{1-\beta\varrho} + \frac{1}{1-\beta\varrho}) + \frac{1-\varrho}{1-\varrho\bar{\beta}} - \frac{1}{1-\varrho})\Big]$. Assuming that $\iota = 0$, the house price response to the shock in the first two periods is given by

$$\widehat{q}_{i,1} = -\tilde{\omega}_{i,r}\varepsilon_1$$
$$\widehat{q}_{i,2} = \tilde{\omega}_{i,m} g \widehat{q}_{i,1}$$

Under subjective beliefs, we arrive at a similar result without conditioning on a positive supply response ($\iota \geq 0$). As under the supply-side differences, the subjective beliefs model is more responsive to a monetary policy shock. Furthermore, regional heterogeneities enter the model through $\tilde{\omega}_{i,r}$ and $\tilde{\omega}_{i,m}$, and are scaled by the factor g. This yields a structure analogous to that in Propositions (2) and (3), whereby regional heterogeneities give rise to regional variations, which are subsequently amplified by extrapolation.

C. Empirical part and model calibration

C.1 Data sources

For the monthly house price data on the federal level, we use the SP CoreLogic Case-Shiller U.S. National Home Price Index, seasonally adjusted from the FRED database (CSUSHPISA). The FFR (FEDFUNDS), industrial production (INDPRO), and CPI (CPIAUCSL) on a monthly

frequency are also taken from the FRED. The house price expectations data are the mean of the expected change in home values during the next year, taken from Michigan University's Survey of Consumers. For the quarterly state-level data, we use all transactions house price indices ("State code"STHPI) from the FRED and seasonally adjust them. The remaining state-level data on building permits ("State code"BPPRIVSA), employment in construction ("Stade code"CONS), unemployment ("Stade code"UR), and employment in retail are all taken from the FRED and are already seasonally adjusted. We omit the code for employment in retail, as these vary across states. GDP (GDP) and the GDP deflator (GDPDEF) are also obtained from the FRED. As already mentioned in the main text, the house price sensitivity indicator is taken from Guren et al. (2021), and the monetary policy shock from Degasperi and Ricco (2021). The time to build measures was taken from the Census.

C.2 Robustness: Cross-regional heterogeneity in booms and busts

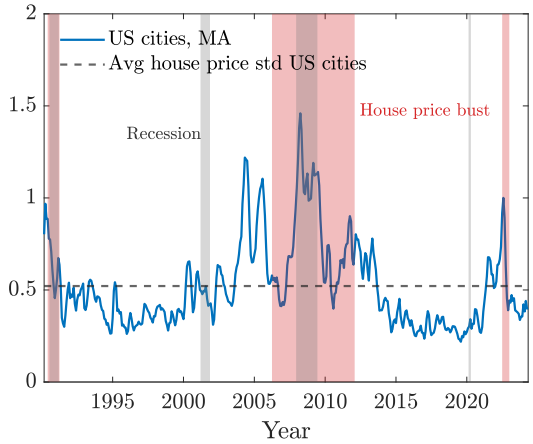
For the euro area, the sample covers the period 2000-2019, also on a quarterly basis. The countries included in the euro area sample are Austria, Germany, Spain, Finland, France, Ireland, Italy, the Netherlands, and Portugal.

Table C.1: Cross-regional house price growth standard deviation (σ_c) in booms and busts, euro area, U.S: cities with alternative boom/bust measure

	E.A. countries		*U.S. cities* :	
	standard measure		Albuquerque et al. (2025) measure	
	mean	*median*	*mean*	*median*
$\frac{Bust}{Boom}$	**1.30**	**1.27**	**1.17**	**1.16**
σ_c : *Boom*	1.57	1.59	0.49	0.41
σ_c : *Bust*	2.04	2.02	0.57	0.48
$p - Val.\ Bust > Boom$	0.002	0.004	0.000	0.006
Number regions	8		20	
Obs. boom	55		225	
Obs. bust	24		183	
Sample	$2000Q1 - 2019Q3$		$1990M1 - 2023M12$	

Notes: The table reports the means and medians, across time, of the estimates of cross-sectional standard deviations of house price growth rates for the euro area countries and U.S. cities using the boom/bust measure by Albuquerque et al. (2025). The test for *Bust* > *Boom* is based on a one-sided t-test.

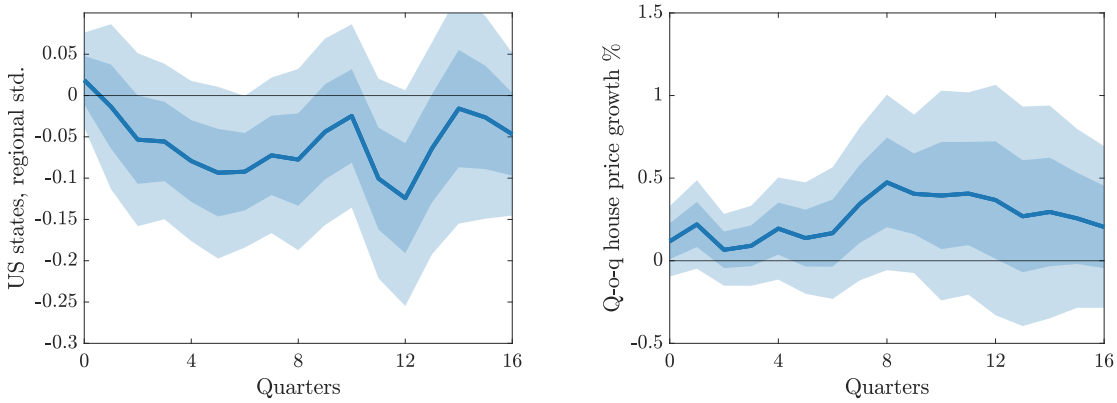
Figure C.1: Regional house price variations in booms and busts



Notes: Three- month moving average of cross-city std.

Figure C.2: State-level std. house price growth response to main business cycle shock

(a) *Std. of House price growth* (b) *House price growth*



Notes: Responses to a main business cycle shock (1 std) ; confidence intervals: 68% and 95% (Newey-West); three-month moving average of std. across states of quarter-on-quarter house price growth (left); q-o-q house price growth at the federal level (right.)

C.3 Forecast error response to monetary policy shock

We are first interested in whether house price expectations are formed according to rational expectations. To answer this question, we study the response of forecast errors to a monetary policy shock. Forecast errors are interesting for two reasons. First, they show whether house price expectations follow rational expectations. Under rational expectations, forecast errors should not respond to the

shock. This becomes obvious in our local projections setup. We define forecast errors as as follows:

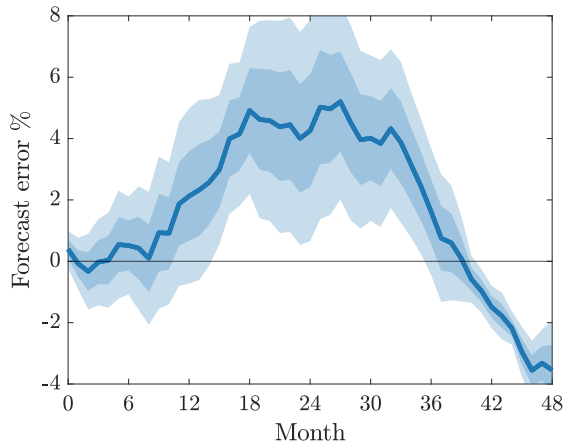
$$fe_t = \Delta q^y_{t+12} - \mathbb{E}^{\mathcal{P}}_t \Delta q^y_{t+12}$$

, where Δq^y_{t+12} is the year-on-year percentage change in house prices and $\mathbb{E}^{\mathcal{P}}_t \Delta q^y_{t+12}$ is the expected year-on-year percentage change in 12 month formed today. The local projections we estimate are given by Equation (C.1). At the time the shock hits, in period t, the forecast errors seen on the left-hand side already contain the forecast formed today for 12 months ahead. In other words, the shock is contained in the agent's information set and should be taken into account. Under rational expectations, agents will perfectly use this information and forecast errors should not respond. The second insight we may gain from this exercise is on how house price expectations are formed. Given the response of the forecast errors is non-zero, the dynamics reveal how households' expectations evolve over time. This will enable us to zoom in on a specific belief updating process that explains our observations. Due to data availability, we focus on the U.S. in this exercise. The monetary policy shock is taken from Degasperi and Ricco (2021). We use the expectations data from the Michigan Survey on Consumer Sentiment. The frequency is monthly and the sample runs from 2007 to 2019. As controls, we include six lags of the forecast error and the monetary policy shock. We estimate the following equation:

$$fe_{t+h} = \alpha^h + \beta^h \epsilon^{MP}_t + x_t + u_{t+h} \qquad h = 0, 1, ..., H \tag{C.1}$$

Figure (C.3) shows the results. The impulse response reveals a non-zero and highly significant response in forecast errors. We can therefore reject rational expectations. On the dynamic behavior, we find that for the first 30 months the forecast error is positive, indicating over-pessimism on the side of the agents. After 30 months, the forecast error turns negative, indicating over-optimism. For both cases, over-pessimism and over-optimism, the IRF is significant at the 95% confidence level. Further, the hump-shaped dynamics in the forecast errors indicate sluggish updating in the expectations formation process. These findings are in line with Adam et al. (2022). Angeletos et al. (2021) document similar dynamics in an SVAR setup for unemployment and inflation in response to an unemployment and inflation shock.

Figure C.3: Forecast error response to a monetary policy shock, U.S.



Notes: Response to an expansionary monetary shock (25 bp); confidence Intervals: 68% and 95% Newey-West).

C.4 Potential sources of heterogeneity in the expectation formation

Heterogeneity in the actual law of motion. In our first exercise, we estimate an $AR(1)$ in house prices growth conditioning on booms and busts:

$$\Delta q_t = c + \beta_1 \mathbb{1}(\Delta q_{t-1} > 0)\Delta q_{t-1} + \beta_2 \mathbb{1}(\Delta q_{t-1} < 0)\Delta q_{t-1} + u_t$$

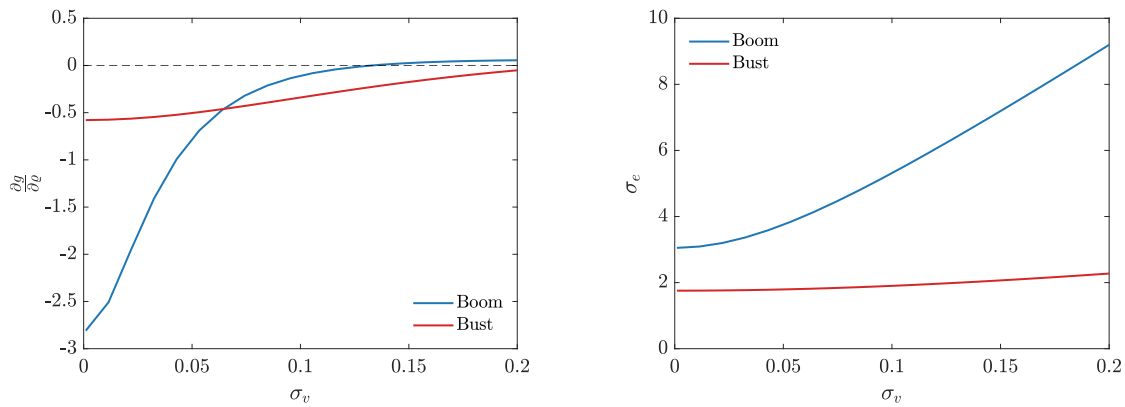
We find that house price growth exhibits greater persistence during boom phases ($\beta_1 = 1.01, (0.30)$) compared to bust phases ($\beta_2 = 0.88, (0.09)$). This result is consistent with the findings from the estimations of expectations formation. Moreover, the perceived persistence parameter governing the law of motion of house prices (ϱ) is determined endogenously through the Kalman filter:

$$g = \frac{\varrho^2\sigma^2 + \sigma_v^2}{\varrho^2\sigma^2 + \sigma_v^2 + \sigma_e^2}$$

$$\sigma^2 = \frac{1}{2\varrho^2}\left[-(\sigma_v^2 + (1-\varrho^2)\sigma_e^2) + \sqrt{(\sigma_v^2 + (1-\varrho^2)\sigma_e^2)^2 + 4\varrho^2\sigma_v^2\sigma_e^2}\right]$$

One can show that $\frac{\partial g}{\partial \varrho} < 0$ if $2 \times \sigma^2 + \varrho\frac{\partial \sigma^2}{\partial \varrho} < 0$. Given that σ^2 is strictly positive, it follows that $\frac{\partial g}{\partial \varrho} < 0$ holds whenever $\frac{\partial \sigma^2}{\partial \varrho}$ is sufficiently negative. Given that σ^2 is highly non-linear, we rely on numerical simulations to infer when $\frac{\partial g}{\partial \varrho} < 0$ holds. To do so, we fix σ_v, solve for σ_e such that we either arrive at g^h or g^l, and then compute the $\frac{\partial g}{\partial \varrho}$ at this point. Figure (C.4) plots the results. one can see that for a small enough σ_v, a drop of ϱ would result in an increase in g. Hence, the decrease in the perceived persistence, which aliens with the actual persistence, could result in stronger updating without a shift in σ_v and σ_e.

Figure C.4: $\frac{\partial g}{\partial \varrho}$ in booms and busts



Notes: Left: $\frac{\partial g}{\partial \varrho}$ as function of σ_v resulting in g^l or g^h. Right: corresponding σ_e as a function of σ_e.

As a second exercise, we can directly estimate equation (14) using actual house price realizations instead of expectations. Hence, we estimate the following equation using the same procedure as before:

$$\Delta q_{t+1} = \mathbb{1}(q_{t-1} > \bar{\omega}) \left(\bar{c}^h + \bar{\varrho}^h (1 - \bar{g}^h) \times \Delta q_t + \bar{\varrho}^h \bar{g}^h \times \Delta q_{t-1} \right) + \\ \mathbb{1}(\Delta q_{t-1} < \bar{\omega}) \left(\bar{c}^l + \bar{\varrho}^l (1 - \bar{g}^l) \times \Delta q_t + \bar{\varrho}^l \bar{g}^l \times \Delta q_{t-1} \right) \quad \text{(C.2)}$$

We estimate the following parameter values: $\bar{\varrho}^h = 1.00$, $\bar{g}^h = 0.66$, $\bar{\varrho}^l = 0.54$, $\bar{g}^l = 0.99$, and $\bar{\omega} = 0.10$. Consistent with the results from the expectations formation exercise, we find a higher degree of persistence and a lower degree of updating in the boom regime. Moreover, the estimated threshold is close to zero. We therefore conclude that the perceived law of motion mirrors the properties of the actual law of motion.

Rational Inattention. We assume that households believe that house prices follow

$$q' = \rho q + \nu, \quad \rho \in [0, 1], \quad \nu \sim i.i.N.(0, \sigma_\nu^2)$$

Following Pfäuti (2025) we make the following assumptions. Households want to minimize expected forecast errors, but house prices today are unobservable, and acquiring/processing information is costly. Further, the loss from mistakes is quadratic with scaling factor r, and the cost of information is linear with scaling factor λ. The agents solve the following problem:

$$\max_f \int U(q^e, q) f(q^e, q) dq dq^e - C(f)$$

$$s.t. \quad \int f(q^e, q) = g(q), \forall q$$

$g(q)$ is the prior assumed to be Gaussian $q \sim N(\widehat{q}, \sigma_q^2)$. q^e is the inflation forecast, $f(q^e, q)$ the joint distribution, $U(q^e, q)$ the negative loss, and $C(.)$ the cost of information which are given by

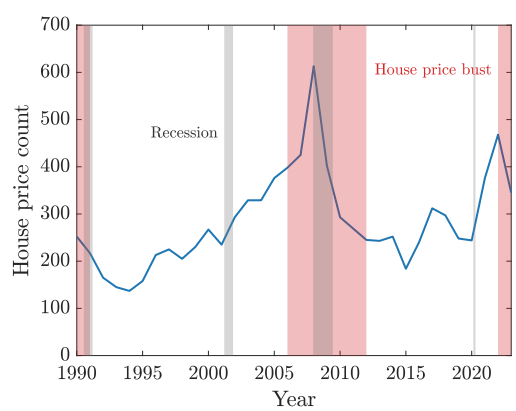
$$U(q^e, q) = -r(\rho q - q^e)^2$$
$$C(f) = \lambda I(q; q^e) = \lambda(H(q) - E[H(q|q^e)])$$

$H(q) = -\int f(q) log(f(q)) dq$ is the entropy. It can then be shown that the Kalman gain, hence the optimal level of attention, is given by:

$$g = max(0, 1 - \frac{\lambda}{2r\rho^2\sigma_q^2})$$

The expected law of motion of house prices is equivalent to Lemma (1). The derivations suggest that the Kalman gain can also be interpreted as the level of attention to a certain variable. In figure (C.5), we counted the number of times the word "house price" was mentioned in the New York Times. It becomes clear that the word occurs more frequently in house price busts, suggesting a higher level of attention.

Figure C.5: Occurrence of "House Price" in New York Times



Number of times the New York Times mentioned the word "house price" in a given year.

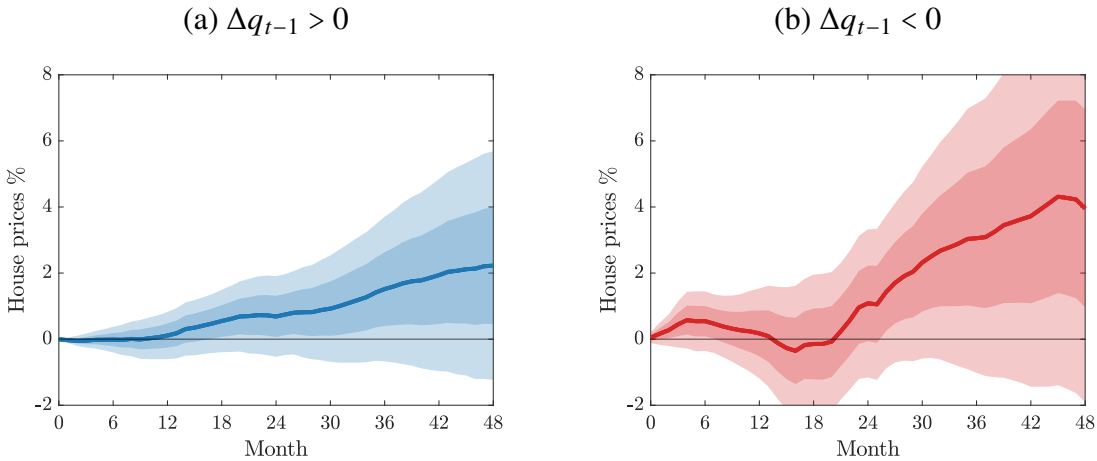
C.5 House price response to monetary policy shocks in booms and busts

To test whether house prices are more responsive in busts, we run local projections on house prices responding to a monetary policy shock, conditioning on booms and busts. The estimation equation is given by:

$$q_{t+h} = \alpha^h + \mathbb{1}(\Delta q_{t-1} > 0) \times \beta_1 \epsilon_t^{MP} + \mathbb{1}(\Delta q_{t-1} < 0) \times \beta_2 \epsilon_t^{MP} + x_t + u_{t+h} \tag{C.3}$$

The monetary policy shock, ϵ_t^{MP}, is the high frequency identified and orthogonalized shock from Degasperi and Ricco (2021). The left-hand side variable, q_{t+h}, is the log house price. The sample runs from 1991-2019 and is in monthly frequency. The controls, x_t, contain 6 lags of the left-hand side variable, log of industrial production, the log of CPI, the FFR, and the shocks. We will focus on expansionary monetary policy shocks throughout the whole empirical analysis. Figure (C.6) plots the results. The blue line shows the response if house prices were increasing in the past, and the red line if they were decreasing. We find that house prices are notably more responsive to monetary policy shocks in times when they have been decreasing. The peak response of the point estimates doubles. Quantitatively, the response seems sizable in both cases; however, asset prices, including house prices, generally exhibit strong responsiveness to variations in interest rates. Further, the estimates exhibit large estimation uncertainty.

Figure C.6: House price response to monetary policy shock, boom-bust



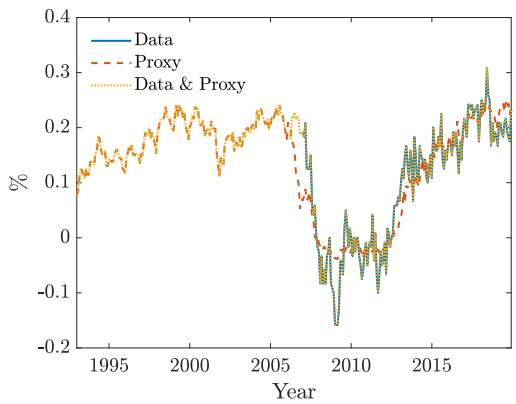
Notes: Responses to expansionary MP shock (25 bps); Confidence Intervals: 68% and 95% (Newey-West).

C.6 House price expectations: extended dataset

Jacobson (2022) proposes to backcast by regression expected house price changes on perceived buying conditions taken from the Michigan Survey. As the survey question about buying condition is goes back to 1993, this relation allows a backcast. We follow her proposal and combine the backcast with the actual house price expectations. However, the backcast leads the actual house price expectations by roughly one year at the beginning of the housing bust in 2007. We therefore use the proxy up until December 2005 and the actual data starting in January 2007. To bridge the gap, we regress house price expectations on three lags of past house price growth and compute the

gap using the estimated coefficients. Figure (C.7) shows the results: one can clearly see that the proxy leads at the beginning of the crisis. Our approach corrects for this. For the remainder of the sample, the proxy does rather well.

Figure C.7: House price expectations proxy



Data is the actual house price expectations series from the Michigan Survey. Proxy is the approximation as suggested by Jacobson (2022). Data & proxy combine the two after accounting for the adjustment described in the text.

Using this series, we re-estimate the equation (14). We also directly control for booms, bust and normal times. We define a bust in the same manner as before, while a boom is a time when house price growth is twice as high as the long-run mean. Table (C.2) shows the results. First, for the baseline estimation, we find that qualitatively, our results remain valid. Even the estimated threshold is close to our value presented in the main text. Second, our results also remain robust if we specifically control for boom periods.

Table C.2: House price belief model: parameters for proxy

Specification	ϱ^{normal}	ϱ^{h}	ϱ^{l}	g^{normal}	g^{h}	g^{l}	*Threshold*	MSE^{i}/MSE^{lin}
Baseline		0.95	0.91		0.003	0.015	0.002	0.48
Boom/Bust/Normal	0.95	0.96	0.92	0.012	0.013	0.016		0.45

Notes: Baseline: estimated parameters from (14) using the proxy data series; Boom/Bust/Normal: Busts are defined by negative house price growth in the previous month, Booms are defined by house price growth being twice as high as the long-run mean. The share in the MSE is computed relative to using a linear model for the same sample.

Finally, we re-run the regressions proposed by Case et al. (2012) using the extent expectations series. Table (C.3). As before, the pass-through from realized house prices to expectations is larger in a bust. Furthermore, the previously insignificant negative coefficient for the boom becomes positive and significant. Finally, we can still reject the null of both coefficients being equal.

Table C.3: Pass-through from lagged prices to expectations, proxy

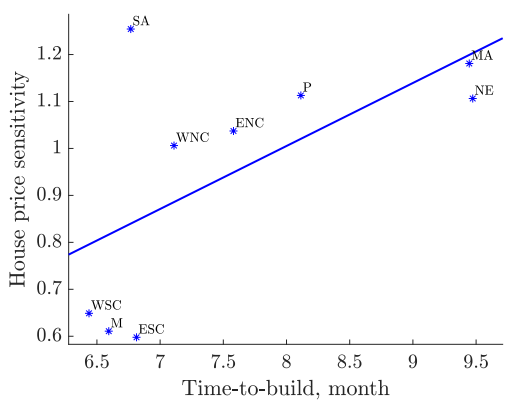
b^{boom}	b^{bust}	R^2	N	$p-val.\ H_0: b^{boom} = b^{bust}$
0.06^{***}	0.13^{***}	0.59	323	0.06
(0.02)	(0.03)			

Notes: Estimation of the Case et al. (2012) regression for house prices, conditioning on boom and bust regimes. Reported standard errors are Newey-West. Significant levels: $***p < 0.01, **p < 0.05, *p < 0.1$. Column 5 shows the p-value for the null hypothesis that both coefficients are equal.

C.7 Time-to-build in the U.S.

The time it takes to obtain a building permit or, generally, to build a house is not available for the U.S. on a state level. However, the Census Bureau documents construction times for homes on a census division level. Below, we aggregate the housing sensitivity indicator from Guren et al. (2021) to a census division level and correlate them with the housing construction time measured from the date of authorization to completion. One can see in figure (C.8) that these are clearly positively correlated.

Figure C.8: Correlation of house price sensitivity indicator with time-to-build



Correlation of house price sensitivity indicator from Guren et al. (2021) with time-to-build a house across census divisions. Time-to-build is measured from the date of the authorization until the completion of the building activities.

Table C.4: Regional housing supply side heterogeneities

City	Elasticity	Time-to-build	City	Elasticity	Time-to-build
Atlanta (Atl)	1.05	7.2	Minneapolis (Min)	1.3	8.5
Charlotte (Cha)	0.65	4.0	Miami (Mia)	2.4	17.1
Dallas (Dal)	0.65	4.2	New York (NYC)	1.4	9.7
Denver (Den)	0.41	2.4	Phoenix (Pho)	0.95	6.4
Detroit (Det)	1.50	10.5	Portland (Pho)	0.61	3.9
Boston (Bos)	0.98	6.7	San Diego (SD)	1.1	7.7
Chicago (Chi)	1.13	7.8	San Francisco (SF)	1.2	8.1
Cleveland (Cle)	0.94	6.4	Seattle (Sea)	0.68	4.4
Las Vegas (LV)	1.43	10.0	Tampa (Tam)	2.0	14.6
Los Angeles (LA)	1.2	8.2	Washington D.C. (DC)	1.1	7.6

Notes: The elasticity data is taken from Guren et al. (2021). The time-to-build is in monthly frequency. Since the model is in quarterly frequency, we will round the time-to-build to the nearest quarter.

C.8 Model: steady-state values

Table C.5: Steady-state values

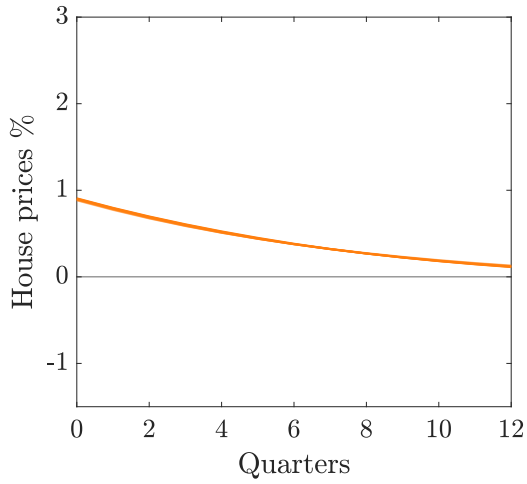
Variable	Value	
$\xi_{x,ss}$	1.46	housing productivity shifter
c_{ss}	0.98	consumption
x_{ss}	0.06	housing investment
h_{ss}	10.09	housing
y_{ss}	1.04	output
b_{ss}	0.00	bond holdings
w_{ss}	1.00	wage

Notes: All steady states except the house price are symmetric across countries. Time-to-build and the steady-state housing preference shifter are not symmetric across countries. All elements of ξ_{ss} not explicitly mentioned assume the value 1.

D. Quantitative model

D.1 Further results

Figure D.1: Regional house price and output response under rational expectations



Notes: Responses to an expansionary monetary policy shock (25 bp).

Table D.1: Aggregate standard deviations across policy rules

			Boom			*Bust*		
Model	ϕ^{π}	ϕ^{q}	π	y	q	π	y	q
Rational Expectations	1.5	0.00	0.05	0.14	0.17	0.05	0.14	0.17
	1.5	0.125	0.07	0.09	0.11	0.07	0.09	0.11
Subjective Expectations	1.5	0.00	0.01	0.34	0.70	0.02	0.37	0.78
	1.5	0.125	0.06	0.10	0.20	0.06	0.10	0.20

Notes: The table shows standard deviations of several variables in response to an expansionary productivity shock (100 bp); The standard deviations are computed over the first 24 quarters.

D.2 Loss function

We assume that the policy-maker's objective is the utilitarian average welfare in the economy, evaluated under rational expectations. This definition leaves open the question of what information the policy-maker possesses. Conditional welfare, $\mathbb{W}_{-1}$, is the average expected welfare in the union, conditional on (i) the economy being in the steady state in the current and all past periods $t \leq 0$,

Table D.2: Cross-region standard deviations across policy rules, Subjective expectations

			Boom			*Bust*		
	ϕ^π	ϕ^q	π	y	q	π	y	q
Subjective Expectations	1.5	0.00	0.00	0.05	0.17	0.00	0.09	0.28
	1.5	0.125	0.00	0.02	0.05	0.00	0.02	0.06

Notes: The table shows standard deviations of several variables in response to an expansionary productivity shock (100 bp) under subjective expectations; the standard deviations are computed over the first 24 quarters.

and on (ii) the policy-maker knowing that in $t = 0$ a set of shocks $(\xi_t)_{t\geq 0} \in (\mathbb{R}^n)^\infty$ will realize with a Gaussian probability distribution but not knowing about the exact realization of the shocks. The experiment here is that the policy-maker decides about about their instruments at the end of period $t = -1$, maximizing the average utility in the union under the expectation that a set of surprise shocks might materialize in the next period. According to a known distribution—let $\mathbb{E}_{-1}$ denote the operator drawing the expectation with respect to the rational expectations measure and conditional on (i) and (ii). Specifically,

$$\mathbb{W}_{-1} := \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \gamma \sum_{i=1}^{N} u_{i,t}$$
$$\mathbb{W} := \mathbb{E}\gamma \sum_{i=1}^{N} u_{i,t}$$

where $u_{i,t}$ is the utility function of region i.[57] We approximate both criteria to the second order around the non-stochastic steady state: Assuming that the allocation in the non-stochastic steady state is symmetric, we have that

$$\mathbb{W}_{-1} = \mathbb{E}_{-1} \sum_{t=0}^{\infty} \beta^t \gamma \sum_{i=1}^{N} \Big[-\frac{c^{-\sigma}}{2} (\kappa y \pi_{iit}^2 + \sigma c \widehat{c}_{it}^2 + \xi_{hi} h^{1-\nu} c^\sigma \widehat{h}_{it}^2 + \varphi n^{1+\varphi} c^\sigma \widehat{n}_{it}^2 + x(1-\eta)\widehat{x}_{it}^2 + \psi y^2 \widehat{b}_{it}^2)$$
$$+ n^{1+\varphi} \widehat{\xi}_{at} \widehat{n}_{it} + c^{-\sigma} \bar{s}_3 \sum_{j\neq i}^{N} \widehat{s}_{ijt}^2 \Big] + O(3) + tips$$

where $\bar{s}_3$ consists only of model parameters, $O(3)$ are terms of the third or higher order and "t.i.p." stands for the "terms independent of the policy," that is, the variables that are beyond the policy-maker's control and, thus, are irrelevant for the policy selection. Re-writing the loss function using

[57] Note that it holds that $\mathbb{E}\mathbb{W}_{-1} = \mathbb{W}$.

market-clearing conditions gives:

$$\frac{\mathbb{W}_{-1}}{c_{ss}} = -c_{ss}^{\sigma}\mathbb{E}_{-1}\sum_{t=0}^{\infty}\beta^{t}\sum_{i=1}^{N}\left[\frac{\gamma}{2}\Omega_{i,t} + \frac{1}{2}\Gamma^{s}\sum_{j\neq i}^{N}\widehat{s}_{ijt}^{2} + O(3) + \text{t.i.p.}\right] \tag{D.1}$$

$\Omega_{i,t}$ collects the welfare-relevant choices and is defined as

$$\Omega_{i,t} = \Gamma^{\pi}\widehat{\pi}_{ii,t}^{2} + \Gamma^{c}\widehat{c}_{i,t}^{2} + \Gamma^{h}\widehat{h}_{i,t}^{2} + \Gamma^{n}\widehat{n}_{i,t}^{2} + \Gamma^{x}\widehat{x}_{i,t}^{2} + \Gamma^{b}\widehat{b}_{i,t}^{2} - \Gamma^{\xi_a}\widehat{\xi}_{a,t}\widehat{n}_{i,t}$$

This can be easily rewritten in the form presented in the main text.